

The Far Side of the Constraint

*A Civilizational Thesis on Energy Abundance,
Physical Automation, and the Distribution of Ownership*

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Abstract

This thesis is written for sovereign wealth funds, venture capital, other investors, and founders building in deep tech. It is a hypothesis that analyzes, conceives and designs outside the present constraint rather than inside it, and it is meant to serve, as a hypothesis, as a compass for investment.

Suppose a future society in which energy is abundant, in which artificial intelligence and physical AI autonomously raise the productivity of every process, and in which institutions function. The constraint that has bound our capacity to conceive large-scale investment comes off. People are allotted capital before they ever work, and can devote themselves to the creation of value in its proper sense rather than to productivity. In that world the macro problems we now fear, population collapse among them, dissolve as a by-product of two constraints being lifted, energy and labor input, together with the distribution of wealth. They do not wait on symptomatic treatment.

The argument of this thesis is that contemporary civilization faces one structural constraint, a stalled capacity to build, defined by three deficits: (i) the absence of effective energy abundance, (ii) a lag in automation, and (iii) the absence of institutions. Economic stagnation and population collapse are not the cause. They are the symptoms the constraint produced.

The argument begins with an interpretation of the past industrial revolutions and a diagnosis of the present.

1. The industrial revolutions are re-read on two axes: the efficiency of placing matter and the efficiency of prediction.
2. The supply constraint is argued to be suppressing, at the macro layer, our capacity to conceive large new investment.
3. The supply constraint is argued to be suppressing population as a second Malthusian trap, with the rising prices of exactly those goods automation never reached striking households directly.

From there the thesis sets out how the accelerating arrival of effective energy abundance and of automation restores that capacity to conceive and expands demand. In place of allocating income by contribution to tasks, it proposes a model of distributed ownership built on capital accounts at birth. And as a departure from GDP, it proposes a model of value creation and measurement centered on use-value, applying the principles of service-dominant logic.

Finally it examines, region by region, the subcategories bearing on effective energy abundance, automation and institutions, and forecasts the composition of the world at the level of clusters. It identifies Texas and the UAE as the candidates for long and sustained success, the San Francisco Bay Area, the Yangtze Delta and the Pearl River Delta as likely to keep growing, and Japan-Korea-Taiwan as consolidating into technological chokepoints.

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Premise

“Wealth consists of physical energy (as matter or radiation) combined with metaphysical know-what and know-how.”

R. Buckminster Fuller, *Critical Path* (1981), p.198

You can see the future first in San Francisco, wrote Leopold Aschenbrenner.¹ Half of that is right. The future of computation can indeed be seen from San Francisco. But the future of civilization, which is to say the future of energy, population, and distribution, is visible first from the United Arab Emirates, where I live.

This thesis is that report. There is only one claim in it. Economic stagnation, collapsing birth rates, and wealth that no longer reaches people are not separate problems. Put them in the order they happened and they are one sequence. It begins with energy, which stopped getting cheaper. Then automation, which never reached the things a household has to buy, its housing and childcare and schooling and medicine. Then institutions, which were never built to pass what energy and automation do produce through to the household. So the advanced economies lost the capacity to build at scale, households could no longer afford children, and the wealth that was created stopped reaching people. Energy, automation, institutions. All three are moving at once now, for the first time in half a century. What happens when they line up, and where to be standing when they do, is what follows.

Where the given and the chosen change places

The first thing living in the Gulf does to you is break your sense of what can be moved and what cannot. What the advanced economies treat as fixed, and what they treat as theirs to design, are swapped here.

First, energy. In the advanced economies energy is a cost, and supply is planned against demand. Here the order runs the other way. Neither Barakah’s 5.6 gigawatts of nuclear nor Al Dhafra’s 2 gigawatts of solar was built by waiting for demand to accumulate.² You lay the power down first and decide afterwards what to build on it. That changes the question people ask. Not what a kilowatt-hour costs, but how many gigawatts you can use, for how many years, without having to count the cost. The two questions look alike and are not alike, and most of this thesis stands on the difference.

Second, fertility. Fertility among UAE nationals and among all residents differs by nearly a factor of three.³ Same city, same streets, same shops, same summer. What differs is who is staying. One group is settled, and inside the arrangements that share out the country’s wealth. The other holds a permit tied to an employer and goes home when it ends, and people who come to earn for a few years do not have children where they are earning. What explains that gap is not culture. It is whether people are staying, and how the wealth is shared.

Third, talent. Energy came first, then capital, then infrastructure. All three arrive if you decide they will. The fourth does not: the person who makes one out of zero.

¹ San Francisco. Leopold Aschenbrenner, “Introduction,” *Situational Awareness: The Decade Ahead*, June 2024.

<https://situational-awareness.ai/> The opening sentence reads “You can see the future first in San Francisco.”

² Energy in the UAE. The Barakah nuclear plant comprises four APR1400 units for a combined 5.6 GW, covering up to 25 percent of national electricity demand at full output (Emirates Nuclear Energy Corporation). Al Dhafra solar is 2 GW, and its 2020 award price of 1.35 US cents per kilowatt-hour was the lowest in the world at the time (Emirates Water and Electricity Company, 2020). What the auction settled was the developer and the price; the procured capacity itself was fixed in advance. The fuller picture, including historically subsidized domestic gas, is in the notes to Chapter 1.

³ The UAE’s two fertility rates. The total fertility rate of Emirati women fell from 6.7 in the 1970s to 3.7 in 2017 (Shukla,

They choose where to move by density rather than by tax rate, by the chance of running into someone smarter than themselves every week.

Put energy, fertility and talent side by side and the inversion is plain. Energy and population, which the advanced economies hold fixed, are variables here. Talent, which they treat as a variable that money sets, is the most fixed of the three. That inversion is where this thesis began. If energy is chosen rather than given, then the stagnation stacked on top of it, and the falling births, and the wealth that stops short of people, were chosen too, and can be unchosen.

Two puzzles

The future is unreadable if you watch what people say they want. It becomes readable if you watch what they want it for. Two puzzles, a century and a half apart, make the point.

At the start of the industrial revolution, if you had asked people what they wanted, they would have said a faster carriage. What arrived was the railway, and then the automobile. Something nobody ordered delivered what everybody wanted.⁴

The other is the humanoid robot that appeared in almost every account of the future written in the 1980s. Humanoids are arriving now, but not for the shopping errand. The value the shopping robot was supposed to deliver, which is having what you want appear at your door, came instead from Amazon, from rearranging warehousing, logistics, and the act of ordering. The imagined form never arrived. The value did.

The two puzzles are one puzzle. In both, the value stayed and the thing that had been delivering it was swapped out. What people wanted was not a carriage but to go farther, faster. Not a robot, but to have what they wanted appear at the door. A person's goal always sits on the side of the value. It never sits on the side of the form. And the form is the part that gets replaced. Hold value and form apart and most investment questions settle themselves, because what you can see, and price, and buy is always the form. Follow the same separation all the way down and it runs in one line from the causes of falling fertility through the structural limits of renewables, the trajectory of the Chinese economy, the next form of capitalism, and the end of money. This thesis is an attempt to draw that whole line.

The energy question is older than AI

Energy abundance is not an AI story. It is a two-century story, and AI is a newcomer to it.

In the discourse of 2026 the phrase circulates almost entirely inside the AI context: how to supply data centers with power, how to thicken the bottom layer of what Jensen Huang calls the five-layer cake. That the bottom layer is energy is a point I would make too.⁵ The data center is only the newest thing to run into the wall of energy.

What this thesis treats is the wall itself. The abundance of energy has governed the skeleton of civilization since the industrial revolution. What can be built from

Karabchuk & Al Neyadi, *Reproductive Health* 20:138, 2023; underlying figures from the UAE Federal Competitiveness and Statistics Centre). The source says “the 1970s,” not a single year of 1970. For the same year of 2017 the all-resident figure is 1.35, and the most recent figure for 2024 is 1.21 (World Bank). Detail in the notes to Chapter 1.

⁴ From carriage to railway and automobile. The question in the body is a thought experiment, not a quotation. The well-known “faster horse” line attributed to Henry Ford has no source, and there is no passage about horses or carriages in Peter Thiel's *Zero to One*. The scale of the substitution itself is, however, documented. Horses outside American farms fell from 3.45 million in 1910 to 380,000 in 1925, an 89 percent decline in fifteen years (Olmstead & Rhode, “The Diffusion of the Tractor in American Agriculture: 1910–60,” NBER Working Paper 7947, 2000), while motor vehicle registrations rose from 468,500 in 1910 to 26,749,853 in 1930 (Federal Highway Administration, *Highway Statistics Summary to 1995*, Table MV-200, 1997). A New York traffic count first recorded automobiles outnumbering horses in 1912 (Eric Morris, “From Horse Power to Horsepower,” *ACCESS* No.30, 2007, 2–9).

⁵ The five-layer cake. Jensen Huang, “AI Is a 5-Layer Cake,” NVIDIA Blog, March 10, 2026. <https://blogs.nvidia.com/blog/ai-5-layer-cake/>

nothing, whether population grows, how wealth is distributed, what value is measured by. AI is the current name of that two-century trend.

Physics runs ahead of institutions

You do not win the argument about institutions. You replace the premises the argument is standing on. Two cases: Britain, and my own family.

In 1798 Malthus wrote that there is a ceiling on the number of people this island can feed. For half a century afterward Britain fought over that ceiling. The Corn Laws, the Poor Laws, rent and wages. The argument reached a political settlement in 1846, but what moved the height of the ceiling itself was not the argument. It was coal. To get, from firewood cut at a rate the woods could sustain, the heat that England was digging out of the ground in 1850 would have required 48.1 million acres. One and a half times the surface of the country.⁶

Malthus's logic was correct about the world he was describing. In an economy running on nothing but the sunlight that arrived that year, output is bound to land. What broke was not the logic. It was the premise. Coal is the savings of past sunlight, and drawing on it removed the constraint of land on which the entire argument had been resting.

The same replacement happened in Japan, and my own family was inside it. In the last years of the Tokugawa shogunate, while the argument over expelling the foreigners or opening the country was still running, my great-great-grandfather's generation gave practical priority to installing the most advanced systems of the day, which were the steam engine and the telegraph. What they learned was that once the machines are in, the argument itself changes under you. That dispute did not get resolved. It disappeared as a question, somewhere past the point where the foreign steamships' engines had been taken apart and copied, the telegraph lines strung, and the railways laid.

That order is the method running through this thesis, and it is what is happening now in the Gulf.

On the three variables

Falling fertility is treated as culture, AI as technology, inequality as politics. Seen through three separate windows, the prescriptions contradict each other. Raising the birth rate calls for fiscal spending, AI policy calls for attracting investment, and inequality calls for taxing capital. Do all three at once and the capital that dislikes being taxed leaves the country, and none of the three works. That is what is happening.

What is needed is not a fourth prescription but putting fertility, AI and inequality on one board. Most of what can be observed about a civilization is explained by three variables and no more: institutions, the effective abundance of energy, and labor input. Effective abundance is not the same thing as endowment. The same reserves become almost free energy in one society and stay in the ground in another.

⁶ Malthus and Britain's ceiling. T. R. Malthus, *An Essay on the Principle of Population, as it affects the Future Improvement of Society*, London: J. Johnson, 1798, ch.1. That chapter takes "this island" as its named example and argues that subsistence can increase only arithmetically. Malthus did not predict a future collision; he argued that the constraint was already operating continuously in the present. The woodland equivalence is from E. A. Wrigley, *Energy and the English Industrial Revolution*, Cambridge University Press, 2010: the woodland area required to obtain the same heat by sustainable harvest was 4.3 million acres in 1750, 11.2 million in 1800 (35 percent of the land surface), and 48.1 million in 1850 (150 percent). The 1846 repeal of the Corn Laws was a real political event, and Peel paid for it with his cabinet and the split of his party. This thesis does not claim physics substituted for that. It claims the premises were replaced. How much weight to give coal remains contested; Clark & Jacks (*European Review of Economic History* 11(1), 2007) and Tepper & Borowiecki (FRBNY Staff Report 639, 2013) estimate coal's contribution as smaller. And the escape was not uniform. In Ireland, within the same United Kingdom, the Malthusian outcome arrived on schedule in the 1840s.

Placed on one board, two things change. The order of cause and symptom inverts, and fertility becomes not a cause but the slowest-moving symptom. And the order of action follows. Rather than lining up one remedy per symptom, you fill in whichever of the three variables is missing.

Cutting it to three variables is crude. But a way of laying things out is never judged right or wrong, only by what it explains and what it predicts. The chapters that follow are that test. For an investor the frame comes down to one question, asked before any judgment about a particular security or a particular technology: which of the three is moving now?

On a hypothesis that cannot be proved

Consider a society in which energy is supplied so cheaply that its price need not be counted, in which all processes are automated, and in which the way the wealth gets shared has been designed as well. The constraint that has bound what can be made into a business, and where large capital can be put to work, comes off. The wealth that is created is distributed to everyone as a stake held from birth, and by the age of eighteen a person is a capitalist without ever having worked.

At that point what a person does with their time changes. Productivity stops being the aim, because machines supply it. What is left in its place is the thing productivity was only ever a stand-in for, which is applying one's own competence for the benefit of others. That also becomes where income comes from, because by then the fact that someone received a benefit can be recorded directly, without a price standing in as the proxy.

And the problems governments now treat as intractable come from that same constraint: the economy that stopped growing, the social insurance bill that keeps swelling, the gap that keeps widening, the collapse of population. They begin to resolve when it comes off. Increasing population is not the objective. People stopped having children under a larger structural constraint. That is all it was, and when the constraint comes off, births return. Humanity goes farther, production and transport move off the planet, and the world of bits, refined to its limit, becomes nearly indistinguishable from magic.

This is not a proof. It is a hypothesis, and I cannot prove it. Statements about the future are like that in principle.

But being unprovable and being groundless are different things. Having analyzed and diagnosed civilization, one can say that it looks as though it will go this way. Sharpening that "as though" until it is precise enough to commit capital to is the work of this thesis. That is why it sets out eleven predictions and, for each, what would falsify it.

So we begin with the analysis of civilization.

Chapter 1. On Building

1.1 Everything is a conversion of energy

Everything in this world is a conversion of energy. Cars, calculators, computers, aircraft, VR headsets, and the computing systems that drive them. All of it rests on heat turned into electricity, on electricity turned into hydrogen, or on fossil fuels, which are plant carbon formed by photosynthesis and buried. Buckminster Fuller, in *Operating Manual for Spaceship Earth* (1969), treated fossil fuels as a store loaded aboard the ship and warned against a civilization that keeps funding its daily voyage out of that savings account. Sunlight, and the wind that arises from the heat differentials sunlight creates, are income arriving continuously from outside the ship. What I want from Fuller is not the warning but a second implication of the metaphor: that the whole activity of a civilization sits on the ship's energy balance sheet.¹

Nor is this only about appliances. The desk in front of you, the chair, everything, is accumulated energy input. The sum of what went into felling, milling, transporting, drying, and assembling is fixed there now in the shape of a desk. Industrial ecology calls this embodied energy, the total energy invested across a good's entire supply chain. Where a human made it by hand, the muscle and the running of the brain were energy all the same. An adult uses about 2,000 kilocalories a day, which averaged out is a heat engine of roughly 100 watts, and the brain is a computer drawing about 20 watts.² Twenty watts is one phone charger. Whether or not there is a computer inside, energy conversion is energy conversion.

Thermodynamics is the higher law over all economic activity. What economics calls production appears to physics as a process that takes in energy and matter still in a usable, ordered form, which is to say low entropy, converts them into ordered structure such as goods, services and information, and expels waste heat that can no longer do any work, which is to say high entropy. As Nicholas Georgescu-Roegen formalized in *The Entropy Law and the Economic Process* (1971), no economic activity is free of that constraint.³ Mainstream economics has long treated the point as a philosophical footnote. This thesis puts it in Chapter 1 instead.

Widen the same conversion to the level of society and twelve thousand years change the order of magnitude. A Palaeolithic hunter-gatherer moved roughly 300 watts continuously, fire included. In the twentieth century alone, world average primary energy consumption per capita more than doubled, from 28 to 65 gigajoules. Sixty-five gigajoules a year, divided by the seconds in a year, is a continuous output of about 2,060 watts. Divide the United States the same way and you get roughly 8,800 watts.⁴

For the United States, that is eighty-eight people at 100 watts each. Against the Palaeolithic 300 watts it is thirty times. A wealthy industrial civilization is one in which everybody employs something close to a hundred invisible servants.

What grew over twelve thousand years is not the amount a human eats but the amount of work done outside the human.

¹ Spaceship Earth. R. Buckminster Fuller, *Operating Manual for Spaceship Earth*, Southern Illinois University Press, 1969.

² The wattage of a human body. Two thousand kilocalories a day averages out to about 97 watts. This is total energy expenditure, not basal metabolic rate (FAO/WHO/UNU, *Human Energy Requirements*, 2004). The brain is about 2 percent of body mass and accounts for 20 percent of whole-body oxygen consumption at rest, giving a metabolic rate of roughly 20 watts (Clarke & Sokoloff, "Circulation and Energy Metabolism of the Brain," in *Basic Neurochemistry*, 6th ed., 1999).

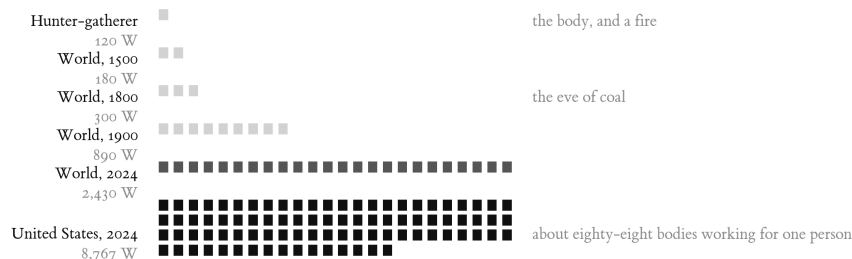
³ The entropy law and the economic process. Nicholas Georgescu-Roegen, *The Entropy Law and the Economic Process*, Harvard University Press, 1971.

⁴ Energy moved per person. The comparison is not strictly measuring the same thing: the numerator is the primary energy a society mobilizes, the denominator the food an individual takes in through the mouth. Palaeolithic food intake of roughly 10 megajoules per day is from Vaclav Smil, *Energy and Civilization: A History*, MIT Press, 2017. The twentieth-century rise in the world average from 28 to 65 gigajoules is

Energy means a way of getting work done without spending human hands. Buying that with coal and buying it with machines are the same purchase. But a machine without electricity does not run, and energy without machines still needs hands, so the two cannot be counted as one.

Figure 1-1 Energy commanded per person, in human-body equivalents

one mark = 100 watts of continuous power, roughly one human body at rest; each row is 24 marks



Twelve thousand years separate the top row from the bottom, and almost all of the difference arrives after 1800. Values before that date are order-of-magnitude estimates.

Source: Our World in Data (CC BY 4.0); Smil, *Energy and Civilization* (2017).

Smil's own series on a total primary energy supply basis, in "Science, Energy, Ethics, and Civilization." The same author's "World History and Energy" gives 60 gigajoules on a commercial-energy basis; the two series must not be conflated. The figure of roughly 300 watts including fire is my own estimate, adding to food intake the burning of one kilogram of wood per person per day at 16 megajoules per kilogram; it is not Smil's number. The 8,767 watts for the United States (2024) and the series in Figure 1-1 are from Our World in Data (CC BY 4.0). US Energy Information Administration figures give roughly 9,300 watts. The difference is one of accounting scope rather than conversion method, and this thesis keeps figure and text on the same series.

1.2 The three terms of competitiveness

$$\text{Civilizational competitiveness } C \approx \text{Institutions } I \times \text{Effective energy abundance } E \times \text{Labor input (automation) } L$$

If any one of the three is near zero, the output is near zero however high the others are.

What this thesis treats is the competitiveness of a civilization. Competitiveness here means how fast it can build the next physical substrate and how much new investment at scale it can execute. Quality, meaning the goodness of life for the people living there, is not treated.

The three terms and their decomposition

Institutions I. That returns are preserved by private property rights, investor protection, tax treatment, and the speed of permitting. That calls in capital. And an environment in which failure is permitted and in which there is social inclusion. That calls in people. Neither is competitiveness in itself. Both set what the other terms get multiplied by. What does the building is the capital and the people who arrive.

Effective energy abundance E. Whether a society has access to a source of energy that is effectively free, of the kind British coal was in the 1840s. And whether that cheapness is believed to hold for a good part of the life of whatever is about to be built. The question is not how many percentage points of competitiveness a twenty percent fall in electricity prices buys.

Labor input L. The power to actually get the work done, which in the present era approximates to automation. Better conversion of energy, meaning machines taking

the place of human hands. And better conversion of information, meaning AI and computing capacity speeding up judgment and design.

$$L \approx \text{Deep tech } T \times \text{Large capital } K$$

Deep tech T. Whether the chokepoint technologies, the ones that stop everyone else the moment they are withheld, are held, or whoever holds them is within reach.

Large capital K. Installing autonomy and computing substrate demands capital expenditure of another order. The question is whether the assets already accumulated can be moved at that scale.

Japan-Korea-Taiwan, holding between them the depth of an advanced-industrial supply chain, sit upstream on T. That is why they cannot be substituted for. The United States holds much of T and holds K, and what it lacks arrives through alliance. China holds much of T and much of K but sits outside the alliance. For a country with neither, two paths remain: accept enormous investment through alliance, or receive a transfer of technology. L can be obtained without building it yourself. It just requires somebody's permission.

The constraint binds what can be conceived

What this equation settles is not the wealth a country already holds but the amount it can still build. The set of countries that are wealthy now and the set that can build now are different sets. The first is running on inventory laid down in a period when all three were present.

What the absence of abundance takes away is not prosperity in general but the option to build. Steel mills, chemical plants, gigawatt-scale computing substrate, and cities themselves fall out of the set of things that society can choose. And for the coming decades what the world needs is the building, not the inventory.

Options are not considered and rejected. They stop being proposed. Where a premise is known to be fixed rather than variable, no business plan is written on that premise, nothing reaches an investment committee, and nothing is conceived in the first place. Let a constraint persist long enough and people take the inside of it for the whole world. The range of ventures we currently feel to be realistic, and our common sense about the scale of investment, were both built on the assumption that the three terms sit where they now sit. What this equation binds is not only what can be built. It binds what can be conceived.

1.3 What a civilization can build from nothing

Only a society with all three at once can build a rail network, a future city, or gigawatt-scale computing substrate from nothing. The argument about electricity and gas bills asks whether an existing factory is in the black. The abundance argument asks who can build the next civilization's stock. They are arguments of different dimensions.

Windows that open and windows that close

Two societies have shown that capacity in extreme form: Britain in the 1840s and the UAE today. On the face of it they share nothing. Look at the structure and the same three are present. Cheap energy with no realistic prospect of exhaustion. Cheap labor input. And institutions that preserve returns and call in people and capital.

Britain in the 1840s. British coal output rose twentyfold over the nineteenth century, and one country was digging roughly two thirds of the world's coal.⁵ Seen at the mouth of the mine, coal was a resource with no prospect of running out.⁵

Labor was abnormally cheap as well. In 1841 roughly 216,000 people worked in the mines, and among them were children who from the age of five or six opened and closed ventilation doors underground.⁶ Part of the cost of extraction was covered, literally, by the labor of children. But structurally this was a society in which the cost of adding one more unit had hit bottom for both energy and labor.

The institutional side was exceptional for its time as well. The Joint Stock Companies Act of 1844 permitted incorporation by registration alone. The Continent did not arrive at the same principle for more than twenty years. Railways were excluded from that act and required Parliament to pass a separate act for each line, but in the frenzy of 1845 and 1846 Parliament processed more than seven hundred railway bills across two sessions.⁷ And the Lands Clauses Consolidation Act of 1845 standardized the assessment and payment of compensation for the compulsory purchase those separate acts conferred.⁷ The machinery for assembling capital and the machinery for assembling land were in place at the same time.

The consequence was the Railway Mania, the investment frenzy of those years. Actual investment in railway construction reached 44 million pounds in 1847, just under 8 percent of British GDP that year and twice that year's military budget. The network more than quadrupled in a decade.⁸ This was not a state plan but private capital. One new industry, on private judgment alone, remade the country.

Britain in the 2020s. Now the same country in the present. The 134 miles of HS2's first phase, from London to Birmingham, run to roughly 396 million pounds per mile, the highest price for a completed surface railway anywhere in the world.⁹ In nuclear, Hinkley Point C has swollen to more than double its original estimate, with completion pushed back to 2030.¹⁰ Industrial electricity prices are the highest among 28 compared countries, roughly four times those of the United States.¹¹ The country that laid thousands of miles in a decade on private capital alone has now spent seventeen years and 46.8 billion pounds without opening a single mile of high-speed line.⁹ Raising a future city out of open country is out of the question. What changed? Not vision, and not national character. The structure of abundance disappeared.

The UAE today. And the same three conditions are now assembled somewhere else. Energy is cheap three times over: historically subsidized domestic gas, the 5.6 gigawatts of the Barakah nuclear plant, and Al Dhafra solar, which set what was then a world record price in 2020.¹² Labor is carried by migrants who make up more than eighty percent of the resident population. Land, being desert, was effectively unlimited. And the institutional side is moving as well. As a place where

⁵ British coal. Output rose from roughly 8.5 million tons in 1800 to roughly 224 million tons in 1900 (B. R. Mitchell, *Economic Development of the British Coal Industry 1800–1914*, Cambridge University Press, 1984). “Roughly two thirds of the world” is a generalization about the nineteenth century as a whole; by 1913 the British share had fallen to about 24 percent. The quotation is from Ticquet (1738), but by way of Robert C. Allen, *The British Industrial Revolution in Global Perspective* (Cambridge University Press, 2009), p.80; the original has not been consulted.

⁶ Labor in the mines. For the roughly 216,000 mine workers in 1841 and the children of five and six operating ventilation doors, see Children's Employment Commission, *First Report of the Commissioners (Mines)*, 1842 (British Parliamentary Papers). The Mines Act of 1842 (5 & 6 Vict. c. 99, royal assent 10 August) prohibited underground work by women of any age in section 1 and by boys under ten in section 2.

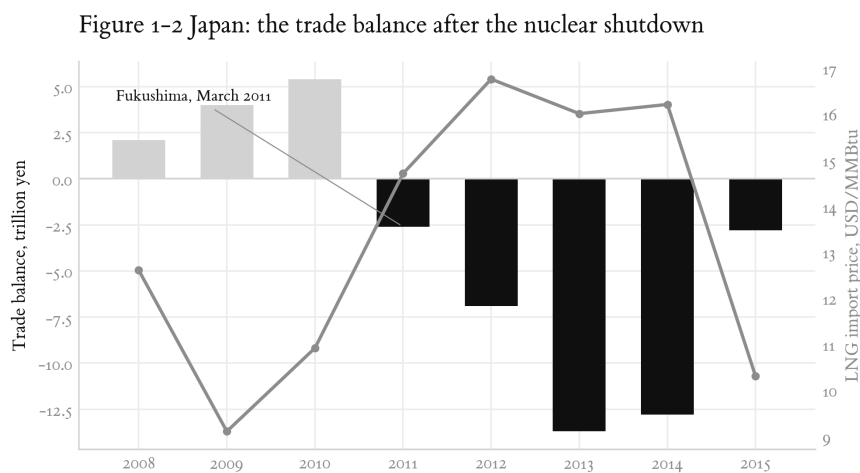
⁷ The Companies Act and the railway bills. The Joint Stock Companies Act of 1844 (7 & 8 Vict. c. 110) permitted incorporation by registration alone, but undertakings requiring parliamentary authorization, meaning bridges, roads, canals, and railways, were excluded. France arrived at full general incorporation with the law of 24 July 1867, the North German Confederation with that of 11 June 1870 (Guinnane, Harris, Lamoreaux & Rosenthal, “Putting the Corporation in its Place,” *Enterprise & Society* 8(3), 2007). Railway bill applications numbered 210 in the 1845 session and 508 in 1846, roughly 720 in total, of which 121 and 272 passed, 393 in all (Esteves & Geisler Mesevage, *Journal of Economic History* 81(4), 2021; primary sources British Parliamentary Papers 1845 (620) and 1846 (723–II)). The Lands Clauses Consolidation Act 1845 is 8 & 9 Vict. c. 18.

⁸ The scale of the railway mania. Actual investment in railway construction was 13 million pounds in 1845, 30 million in 1846, and peaked at 44 million in 1847, which is

returns are preserved, tax burdens are light, and the barrier to starting a business is low, frontier researchers and founders are beginning, slowly, to gather. The consequence is a set of future cities raised from desert in thirty years, and now a shift of investment from concrete to computation. Abu Dhabi's 5-gigawatt AI campus, and a single fund that gathered 49 billion dollars, are new investments at a scale that would not even be conceived except on a structure of abundance.¹³ What nineteenth-century Britain poured into railways, the twenty-first-century UAE is pouring into computing substrate.

Windows do not only open. They close.

Japan and Germany, where they closed. Japan's postwar high growth stood on cheap Middle Eastern crude and on cheap labor coming out of the countryside. The window closed in 1973. Real growth halved over that period, from an annual 9.1 percent to 4.2.¹⁴ The nuclear shutdown of 2011 was a second shock on top of that, and the trade balance swung from surplus to deficit.¹⁴ Germany closed its own. In 2023 it stopped its last three reactors, lost Russian pipeline gas at almost the same moment, and industrial electricity moved to two or three times the American level.¹⁵ The word *Deindustrialisierung* became central to its economic debate after that.



A swing of roughly 20 trillion yen in three years, paid to foreign energy suppliers.
Source: Ministry of Finance trade statistics; IEEJ.

The United States, where it was reopened. Shale held natural gas at a third of European and Asian prices and kept it there, and cheap gas became cheap electricity and cheap feedstock directly.¹⁶ The return of manufacturing is narrated as the achievement of policy, but the policy only became possible on top of that physics. Policy can only choose from within the set of options the energy structure permits.

Buying abundance back with machines

The abundance of 1840s Britain rested in part on child labor and colonies, human costs that must never be repeated. The labor cost of the present-day UAE stands on a global wage gap. But the combination of energy abundance, fully automated manufacture and transport, and correct institutions makes those three conditions obtainable, for the first time in history, without human sacrifice, anywhere. Cheap

just under 8 percent of GDP that year and twice the military budget (Andrew Odlyzko, "The collapse of the Railway Mania...", revised June 25, 2011). Route mileage was 1,498 in 1840 and 6,621 in 1850 (B. R. Mitchell, *British Historical Statistics*, Cambridge University Press, 1988). That series appears to be on a United Kingdom basis including Ireland.

⁹ HS2. Phase one, 134 miles, at a total of 53.1 billion pounds, is roughly 396 million pounds per mile. The French line from Tours to Bordeaux is roughly 46 million pounds per mile (Britain Remade, 2023). The UK Department for Transport's official figure uses a different base year. Cumulative spend was 46.8 billion pounds at the end of March 2026, in nominal terms, including spend on the canceled phase two (National Audit Office, *High Speed Two reset*, HC 52, 2026). The same report gives an opening range for Old Oak Common to Birmingham of May 2036 to October 2039. HS2 Ltd began operating in January 2009.

¹⁰ Hinkley Point C. The original construction estimate was 18 billion pounds, in 2016 prices (National Audit Office, *Hinkley Point C*, HC 40, 2017). EDF's estimate as of February 2026 is 35 billion pounds in 2015 prices, 48 to 49 billion in today's prices, with completion pushed back to 2030 (*The Guardian*, 20 February 2026, reporting on EDF's announcement). The original and updated figures are not on the same deflator basis, so the multiple must be read as a range.

¹¹ UK industrial electricity prices. 25.85 pence per kilowatt-hour in 2023, the highest of 28 compared countries and roughly four times the United States (UK Department for Energy Security and Net Zero, *International non-domestic energy prices*). This is a tax-inclusive price.

¹² Energy in the UAE. Energy subsidies were 5.7 percent of GDP in 2011 and 4.6 percent in 2015 on the IMF's pre-tax estimate (IMF Country Report No. 15/220, 2015). The Barakah nuclear plant comprises four APR1400 units for a combined 5.6 gigawatts, covering up to 25 percent of national electricity demand at full output (Emirates Nuclear Energy Corporation). The 2-gigawatt Al Dhafra solar project was awarded at 1.35 US cents per kilowatt-hour, the lowest price in the world at the time (Emirates Water and Electricity Company, 2020).

¹³ Labor and capital in the UAE. Migrant workers number roughly 8.7 million, more

energy is supplied by next-generation energy technology, fusion among it; cheap labor by the next generation of machines. At that point a wave of private investment absorbing just under 8 percent of GDP could rise again on the same English plains where no future city could be built in the 2020s. The abundance the nineteenth century bought with human pain, the twenty-first buys back with machines. That is the civilizational meaning of committing capital to physical AI and energy.

Abundance reproduces population too

What drove the surge in English population after 1750 was not falling mortality but rising fertility, and what pushed fertility up was earlier marriage brought on by rising real incomes. The postwar American baby boom, in the golden age of cheap energy, and the high fertility of the Gulf during the oil boom, stand on the same structure.¹⁷ And in the UAE, in the same city, the same climate, the same prices, a factor of three in fertility has opened between a population that settles and one that returns home in a few years.¹⁸ That is not a gap culture can explain. Where distribution works inside abundance, people have children without worrying about their careers. Where distribution fails, scarcity throttles births through the price system.

1.4 One constraint, many symptoms

What a civilization can newly build in the physical world is settled by the product of three terms: institutions, the effective abundance of energy, and labor input. Britain in the 1840s and the UAE today both had the three at once. In Japan and Germany what had been present was lost; in the United States what had been lost came back.

But what the capacity to build settles is not only what gets built. What can be built determines what becomes cheap. The price of what did not become cheap becomes the household's constraint.

Stagnation, rising prices, the failure of distribution, and falling births have each been argued as separate problems. This thesis reads them as multiple symptoms produced by one constraint. The prescription is therefore one in total: fill whichever of the three is missing. Symptom-by-symptom measures did not fail because they were badly designed. They failed because we were firing at symptoms in the belief that they were causes.

Births, raised in the previous section, are the slowest-moving of those symptoms. That is why they bite last and are hardest to take back. This thesis gives population a large place not because raising it is the objective, but because whether the constraint has come off shows up there last of all.

than 80 percent of the resident population (International Labour Organization). The Abu Dhabi plan is a 5-gigawatt UAE-US AI campus, inside which Stargate UAE is a 1-gigawatt cluster. The 200-megawatt first phase targets operation in 2026 (G42, 22 May 2025). MGX closed its first fund at 49 billion dollars (MGX, 1 July 2026). That is the size of the fund, not assets under management, and the widely circulated "100 billion dollar AUM target" is not a figure MGX has published.

¹⁴ Japan. The standard account attributes the main cause of the halving to the end of catch-up, with energy a contributing factor layered on top. Real growth averaged 9.1 percent from fiscal 1956 to 1973 and 4.2 percent from fiscal 1974 to 1990, on a fiscal-year basis (Cabinet Office, Long-Term Economic Statistics, National Accounts). The series is spliced across the 68SNA and 93SNA. The trade balance was a surplus of 6.6347 trillion yen in 2010 and a deficit of 11.4684 trillion yen in 2013, on a customs-clearance basis (Ministry of Finance Trade Statistics). On a balance-of-payments basis the same two years are +9.5 trillion and -8.8 trillion yen; the endpoints do not coincide.

¹⁵ Germany. Isar 2, Emsland, and Neckarwestheim 2 were shut down on 15 April 2023 (Federal Office for the Safety of Nuclear Waste Management). EU electricity prices for energy-intensive industry in 2025 were more than double those of the United States (IEA, *Electricity 2026*). For Germany alone the figure varies by consumption band: roughly three times the US industrial average in the 500 to 2,000 megawatt-hour band, and roughly 1.9 times for the largest consumers above 70,000 megawatt-hours.

¹⁶ American gas. The 2025 average was 3.6 dollars per million Btu at Henry Hub, just under 12 dollars for European TTF, and just over 12 for Asian JKM (IEA, *Gas Market Report, Q1-2026*). Henry Hub was 2.53 and 2.19 dollars in 2023 and 2024, when the ratio was wider still. 2025 is used because it is the conservative choice.

¹⁷ Abundance and births. For the period 1757 to 1871, the effect of variation in fertility on population growth was roughly twice that of mortality, and about half of the rise in the gross reproduction rate came from a fall in women's age at first marriage, from 26.5 in the 1650s to 23.4 in the early 1800s (Wrigley & Schofield, *The Population History of England 1541-1871*, Cambridge University Press, 1981). The American total fertility rate peaked at 3.77 in 1957 (National Center for Health Statistics). In the 1960 Gulf,

Chapter 2. On Convergence

2.1 One substrate, two directions

The progress of civilization is made of one substrate and two directions of development. The substrate is energy, and the two directions ride on top of it. Not two axes placed side by side, but two layers stacked.

Direction 1 is productivity in the physical world. Turning energy into momentum, building machines, transporting, constructing, traveling farther. The goal of moving matter with less effort and at lower cost has not changed since the industrial revolution. Jensen Huang divides AI into five layers: energy, chips, infrastructure, models, and applications.¹ Placing energy at the bottom is right. But seen from the side of civilization, two layers suffice: energy, and the applications that run on it. The layers in between rise and fall as means to that goal.

Direction 2 is productivity in the world of bits. A lineage that begins with language and continues through writing, printing, the telegraph, and the telephone: development on the side where almost no mass has to be moved. Solutions here do not necessarily take physical form. People once imagined a future in which a shopping robot took our order and came back with the goods; what actually delivered it was Amazon, a system. At the far end of this direction lies what Yoichi Ochiai calls technology becoming magic, which is to say technology that dissolves into the environment and disappears from view.

2.2 The two endgoals

Each of the two directions has an endpoint. For the physical side it is driving to zero the cost of reaching the configuration of matter you were aiming at, the convergence cost of matter configuration. For the virtual side it is driving to zero the cost of reaching a predictive model of the accuracy you need, the convergence cost of predictive models. The entire history of civilization along the axis of efficiency is those two falling toward zero.

The clue lies in the distinction between meaning and symbol. Language and writing are only vehicles for conveying meaning. And what one notices is that what the telegraph, the telephone, the internet, and Zoom have overwhelmingly improved over the past century is not the conveyance of meaning but the fidelity of symbol reproduction.

That intuition coincides exactly with the definition given by the father of information theory. Claude Shannon opened “A Mathematical Theory of Communication” by defining the fundamental problem of communication as “reproducing at one point either exactly or approximately a message selected at another point,” and added, deliberately: “Frequently the messages have meaning. (...) These semantic aspects of communication are irrelevant to the engineering problem”.² The past century of information technology has been a century of solving exactly the problem Shannon defined, faster, more accurately, and more cheaply. The telegraph reproduces letters, the telephone reproduces sound

Saudi Arabia was 7.63, Oman 7.28, and Kuwait 7.22. Oman rose further, to 7.89 in 1980 (World Bank SP.DYN.TFRT.IN).

¹⁸ The UAE’s two fertility rates. The total fertility rate of Emirati women fell from 6.7 in the 1970s to 3.7 in 2017 (Shukla, Karabchuk & Al Neyadi, *Reproductive Health* 20:138, 2023; underlying figures from the UAE Federal Competitiveness and Statistics Centre). The source says “the 1970s,” not a single year of 1970. The all-resident figure for the same year of 2017 is 1.35, and the most recent figure for 2024 is 1.21 (World Bank). The all-resident figure is lower because it is diluted by a migrant population that does not have children there.

¹ The five-layer cake. Jensen Huang, “AI Is a 5-Layer Cake,” NVIDIA Blog, March 10, 2026. <https://blogs.nvidia.com/blog/ai-5-layer-cake/> The five layers stack in the order of energy, chips, infrastructure, models, and applications.

² Shannon’s definition. Claude E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal* 27 (1948), 379–423, 623–656. The

waveforms, television reproduces images, the internet reproduces arbitrary bit strings. Generative AI is the point at which that lineage changes in kind: what it reproduces is not symbols but the probability distribution that generates symbols, which is to say a model of the world itself.

quotations are the opening two sentences: the definition of the fundamental problem of communication, and the proviso separating semantic aspects from the engineering problem.

The convergence cost of matter configuration (C_{config}) is the total cost of getting arbitrary matter to an arbitrary place in an arbitrary configuration. Transport, manufacture, construction, agriculture, and the opening of space are all special cases of convergence on a desired configuration of matter, and thermodynamics sets the ceiling on their efficiency. The convergence cost of predictive models (C_{pred}) is the total cost of having on hand, at the required accuracy, a model of the state and behavior of some system: another person's mind, a market, a protein, a city. Communication, computation, science, education, and AI are all special cases of convergence on a sufficiently good predictive model. The wheel, the sailing ship, the steam engine, the container, and reusable rockets are landmarks of the first; language, writing, printing, the telegraph, the internet, and deep learning are landmarks of the second.

2.3 The law of competition

The two directions compete over the same problem. And the side with the lower convergence cost wins.

The law of competition: for any given problem, a direction-1 solution and a direction-2 solution are potentially in competition, and the one with the lower convergence cost wins. The more the essence of the problem is information processing, the more direction 2 wins; the more matter configuration is unavoidable, the more direction 1 remains.

The two puzzles of the Premise are its first application. The shopping robot was a direction-1 solution to the problem of shopping: a humanoid machine physically rearranges the configuration of matter. Amazon is a direction-2 solution to the same problem. Most of shopping, meaning search, comparison, ordering, and payment, is not matter configuration but information processing, and Amazon solved that part in virtual space and pushed only the residual matter configuration through the existing logistics network. What decided the outcome was not the superiority of one vision but cost structure. A direction-2 solution replicates at nearly zero marginal cost and is less bound by physical law and safety regulation.

To make this law a tool, one has to be able to measure, before seeing who wins, how much of a problem belongs to direction 1 and how much to direction 2. There is one way to measure it. Decompose the problem into steps and count the mass that must be moved at each step. For shopping, search, comparison, ordering, and payment all move zero mass. The only step that does not is the last leg, from warehouse to doorstep. So this problem could be read in advance as one that would mostly move to direction 2. Apply the same question elsewhere: what Zoom substituted for was the leg of a business trip that carries a human body, and what Uber Eats did was move cooking, the step that carries mass, out of the household

and into somebody else's kitchen. Steps of zero mass migrate to direction 2 sooner or later.

There is, however, a domain in which the two compose rather than compete. Contemporary drug discovery is the type case. Structure prediction,³ of which AlphaFold is the representative, builds a predictive model of how a protein folds entirely inside computation, and only the narrowed candidates are sent onward to the matter-configuration process of synthesis and trials. As much of the predictive-model component of the problem as possible is moved to direction 2, and only the matter-configuration component is left to direction 1. What is needed is not replacement but a shift in the mixing ratio. The same structure repeats in the training of autonomous vehicles, in sim-to-real in robotics, meaning training in simulation and then transferring to the physical machine, in materials science, and in the design of fusion reactors. Direction 2 has become the lever that collapses the development cost of direction 1.

2.4 Thirty years of asymmetry

Capital and talent flowed one-sidedly into direction 2 for thirty years because that side needed less energy. And in the 2020s direction 2 collided head-on with the wall of energy.

As Marc Andreessen declared, software ate the world. The well-known Founders Fund line, "we wanted flying cars, instead we got 140 characters," is correct as an observation of the asymmetry.⁴ As an explanation of the cause it is insufficient. Direction 2 has three properties: replication at zero marginal cost, low friction, and a small energy requirement. Only the last varies with the era. In a constrained world capital flows toward the looser constraint.

Then why could improved efficiency not avert the wall? Should consumption not fall as efficiency rises? It does not fall. That improved efficiency instead increases total consumption is the paradox known since William Stanley Jevons pointed it out in *The Coal Question*.⁵ In the framework of this thesis it is not even a paradox. A fall in convergence cost unlocks applications that were not previously viable, so demand grows by more than the efficiency gain. On the International Energy Agency's estimate, world data center electricity demand rose 17 percent year on year in 2025 and will roughly double by 2030; and AI-specific facilities alone rose 50 percent in 2025.⁶ Over that same period, the electricity consumed per AI task has been falling rapidly. Cheaper computation was not saved. It was used more widely, more often, and for heavier purposes.

So the wall came closer. The doubled figure is slightly above Japan's entire electricity consumption today. Goldman Sachs put the increase at up to 165 percent by 2030 and later revised it upward to 175.⁷ However magical direction 2 becomes, somebody has to turn a turbine. The material substrate of civilization, meaning cities, transport, housing, and food, does not move by a single gram in direction 2.

The movement of capital speaks more plainly than forecasts. On 25 August 2025 the Palisades plant in Michigan became the first reactor in American history to return from the decommissioning process to operating status.⁸ What this thesis depends on is a single point: the structural change by which computing demand

³ Structure prediction. John Jumper et al., "Highly accurate protein structure prediction with AlphaFold," *Nature* 596 (2021), 583–589. The barrier against excessive expectation is Thomas C. Terwilliger et al., "AlphaFold predictions are valuable hypotheses and accelerate but do not replace experimental structure determination," *Nature Methods* 21 (2024), 110–116. This thesis does not claim replacement.

⁴ Software and flying cars. Marc Andreessen, "Why Software Is Eating the World," *The Wall Street Journal*, August 20, 2011. The line "we wanted flying cars, instead we got 140 characters" is from Bruce Gibney, "What Happened to the Future?," Founders Fund, 2011. <https://foundersfund.com/the-future/> The author is Gibney, not Peter Thiel. The original reads on its cover, in the present tense, "We Wanted Flying Cars, Instead We Get 140 Characters," and the line does not appear in the body text. It circulates as Thiel's because he spread it in talks.

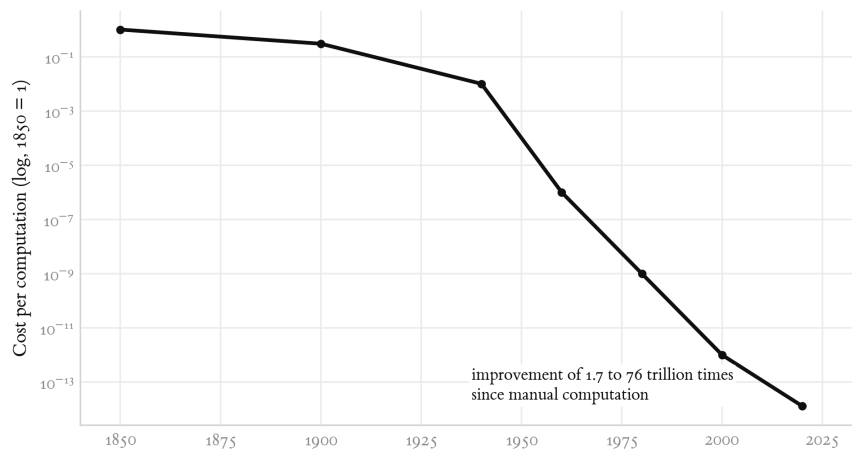
⁵ Jevons's paradox. William Stanley Jevons, *The Coal Question*, Macmillan, 1865.

⁶ Data center electricity demand. IEA, *Key Questions on Energy and AI*, July 2026 (data center electricity demand up 17 percent year on year in 2025, AI-specific facilities up 50 percent in the same year; 485 TWh in 2025 to 950 TWh in 2030). The IEA states that energy per AI task falls by at least an order of magnitude a year but publishes no series of levels, so the right-hand side of Figure 2-2 is drawn as a band rather than as a series.

⁷ IEA and Goldman Sachs projections. IEA, *Energy and AI*, Special Report, April 2025 (from roughly 415 TWh in 2024 to roughly 945 TWh in 2030). Goldman Sachs

made investment in energy supply politically unavoidable. A reactor pulled back out of the decommissioning queue shows that this has already happened.

Figure 2-1 Two centuries of the price of computing

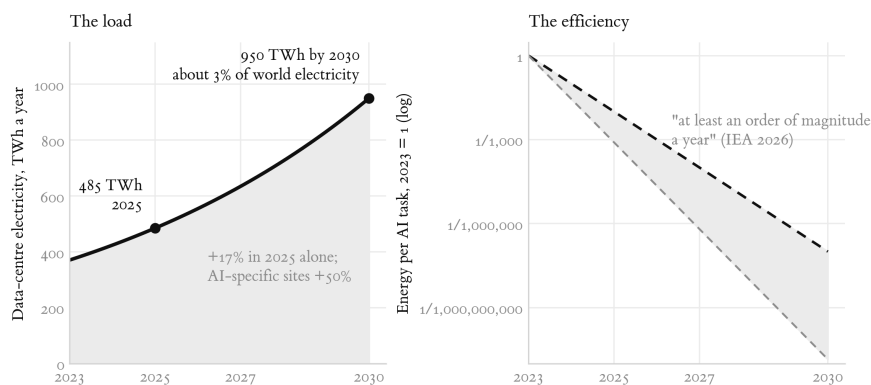


Source: Nordhaus (2007), *Journal of Economic History* 67(1).

Research, “AI to drive 165% increase in data center power demand by 2030,” February 4, 2025, revised upward to 175 percent in the follow-up GS SUSTAIN “6 Ps” report. Because the base year has been updated, this thesis uses the newer IEA series from section 2.4 onward. These are projections and not observations, and the two houses’ definitions of the data center perimeter differ.

⁸ The Palisades plant. Utility Dive, “Palisades becomes first decommissioned US nuclear plant to reach ‘operations’ status,” August 27, 2025 (the event was 25 August). “Operations” status here is a regulatory category, not the resumption of generation. Fuel loading began on 31 August 2026, and the reactor has not yet gone critical (American Nuclear Society, Nuclear Newswire, July 8, 2026; Holtec International, 31 August 2026). What this thesis uses is only the fact that it was pulled back out of the decommissioning queue.

Figure 2-2 Jevons, measured: the load rose while the task got cheaper



Left: the two figures the IEA states, on a constant-growth path between them. Right: a band, not a series. The IEA reports the fall in energy per AI task without giving a level, so only its stated rate is drawn.

Source: IEA, *Key Questions on Energy and AI* (July 2026); IEA, *Energy and AI* (2025).

The commitment of capital to physical AI and energy is not “the next trend” but capital’s historical return, of the two directions of the last twelve thousand years, to the side that was left behind.

2.5 The industrial revolution never ended

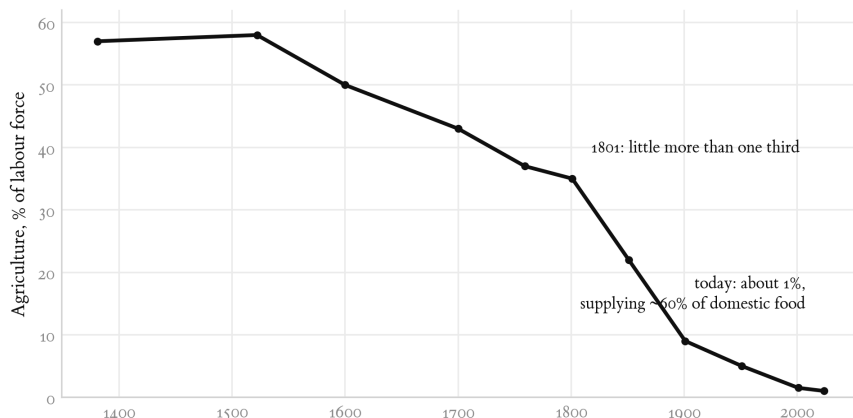
The word used was “return,” but only one side had stopped. The pressure toward efficiency itself has not broken once in two hundred years.

What speaks to this most plainly is a completed experiment with two centuries of data behind it: British agriculture. On Wrigley’s account, roughly three quarters of the labor force was in agriculture under the Tudors. By the early eighteenth century it was about half, and by 1800 under two fifths. Today agriculture is under 1 percent of the British labor force. And with that under 1 percent Britain produces

⁹ Figures 2-1 and 2-2. Figure 2-1 redraws the series in William D. Nordhaus, “Two Centuries of Productivity Growth in Computing,” *Journal of Economic History* 67(1), 2007, 128–159. Figure 2-2 is from IEA, *Key Questions on Energy and AI* (July 2026) and *Energy and AI* (2025). The right-hand panel is a band indicating a direction, not a data series, and should not be read as one.

the equivalent of roughly 60 percent of the food it consumes.¹⁰ On the same island, under a fortieth of the hands now grows more.

Figure 2-3 Britain: agriculture's share of the labour force, 1381-2024



The industrial revolution never ended; it merely changed sectors.

Source: To 1851: Broadberry, Campbell & van Leeuwen (2013); Broadberry et al. (2015). Later: ONS and the ILO modelled estimate (World Bank SL.AGR.EMPL.ZS). Series spliced.

What matters is that nowhere in that process is there an end of the revolution. From enclosure through crop rotation, seed improvement, mechanization, Haber-Bosch fertilizer, the tractor, the combine, and now precision agriculture and automated glasshouses, one and the same pressure, to produce with less effort and at lower cost, worked without interruption for two centuries. The habit of numbering industrial revolutions first, second, third, and fourth is a set of graduations laid retrospectively over that continuum. The graduations change; the direction of that pressure has not changed once.

The size of the current AI boom, too, is measured on this continuum. AI is the present form of that continuum, not a rupture. Discourse that narrates it as a rupture stirs excitement and fear while blurring the practical question. What sets the speed of progress in AI? For two hundred years what has set the speed of automation is energy and capital goods, and this time is the same. There is one question worth asking: does it run inside the energy constraint, or outside it?

2.6 The measure of human worth has changed twice

The measure of human worth has changed twice in the past two hundred years. Both times it changed immediately after an externalization.

The first change, early nineteenth century. The externalization of muscle. Until then human worth was measured mainly by physical capacity in agriculture and manufacture. Laborers, sailors, farmhands. Wage structures and conscription examinations alike were measuring the body. Steam and mechanization replaced most muscular work, and as a result the computational side of the human came to the front. From the 1820s, society began to look at the human brain with the gaze it had used on muscle.

¹⁰ British agriculture. The 40 percent not covered by domestic food supply is externalized to the labor and land of other countries, so the system does not close within Britain alone. The fall in labor share is also mixed with a denominator effect, as other sectors swelled. Wrigley's account is from E. A. Wrigley, *Energy and the English Industrial Revolution*, Cambridge University Press, 2010, pp.33, 196: roughly three quarters of the labor force in agriculture under the Tudors, about half by the early eighteenth century, under two fifths by 1800. The 1759 figure of about 37 percent is from Broadberry, Campbell & van Leeuwen, *Explorations in Economic History* 50(1), 2013, and Broadberry et al., *British Economic Growth, 1270-1870*, Cambridge University Press, 2015. Current agricultural employment is 0.85 percent (2025, World Bank SL.AGR.EMPL.ZS, ILO modelled estimate). The 60 percent for food is the production-to-supply ratio of the UK Department for Environment, Food and Rural Affairs (2025, 60 percent across all food, 72 percent for items that can be produced domestically), and it is a United Kingdom figure rather than an England one. This change cannot be narrated through labor productivity. On Gregory Clark's estimates, output per agricultural worker grew only about 19 percent between 1700 and 1860, and almost all of the change came after mechanization (Clark, "The Agricultural Revolution and the Industrial Revolution: England, 1500-1912," unpublished draft, 2002, Table 5).

¹¹ Figure 2-3. Points to 1851 are Broadberry, Campbell & van Leeuwen (2013) and Broadberry et al. (2015); later points are the UK Office for National Statistics and the ILO modelled estimate (World Bank SL.AGR.EMPL.ZS). The two are spliced, and geography, method and the definition of agricultural employment all change across the join.

The second change, from the 1950s. The externalization of the brain. Until the 1940s the world's total computing capacity depended almost entirely on the sum of human brains. This is not a metaphor. The English word "computer" was, until the middle of the twentieth century, a job title for a person who computed.¹² Those calculating ballistic tables, almanacs, and actuarial values were human computers at rows of desks. After the arrival of the electronic computer, the share of computation carried by human brains fell sharply. On William Nordhaus's measurement, the price-performance of computation has improved by between 1.7 trillion and 76 trillion times since the era of hand cranking.¹³ No other good in human history has seen its price collapse at that speed.

The human brain is not an organ designed to maximize computational capacity but one optimized across multiple purposes: survival, emotion, sociality, intuition. And yet, because muscle was externalized, society placed at the center of value the work of the head, and specifically only its computational aspect. Which is to say that for two hundred years we have been required, as the macro structure of the economy dictated, to maximize performance on a dimension other than the one we were designed for.

There is corroboration from robotics. Hans Moravec wrote in *Mind Children* (1988) that it is comparatively easy to make a computer perform at adult level on intelligence tests, and difficult or impossible to give it the perceptual and motor skills of a one-year-old. The reason lies in evolutionary time. "The large, highly evolved sensory and motor portions of the human brain encode a billion years of experience. (...) The deliberate process we call reasoning is, I believe, the thinnest veneer of human thought".¹⁴ The present sight of white-collar intellectual work being eroded by AI ahead of skilled blue-collar physical work is consistent with that view. Consistent, and no more than that.

And in a future where computation has been fully externalized, on the ruler of computational volume, differences between humans vanish. That is not the same as the differences themselves vanishing. Judgment remains, relationships remain, the body remains. What retires is the single ruler that stood in front of them for two hundred years.

2.7 Rewriting the equation of national power

The contents of the production function are replaced as automation advances. Production could once be written as the product of productivity and the working-age population. As the human input thins, the productivity term is absorbed into the penetration of automation, and the production function moves toward the product of institutions, effective energy abundance, and labor input. There is no need to erect a new production function. The equation of Chapter 1 is itself the answer.

Total factor productivity is a name given to the place where explanation was abandoned. Classical growth accounting is written $Y = A \cdot f(K, L)$, naming whatever capital and labor cannot explain total factor productivity A and pushing it in there. The share of that A explicable by thermodynamic efficiency improvement is, on the

¹² Human computers. David Alan Grier, *When Computers Were Human*, Princeton University Press, 2005.

¹³ The price-performance of computation. Nordhaus (2007), cited above. Improvement of between 1.7 trillion and 76 trillion times since the era of hand cranking, depending on the measure used, with improvement at roughly 55 percent a year between 1940 and 2001.

¹⁴ Moravec's paradox. Hans Moravec, *Mind Children: The Future of Robot and Human Intelligence*, Harvard University Press, 1988. The quotation is from two places in that book. The point that it has never been tested is Arvind Narayanan, "Fact checking Moravec's paradox," *AI as Normal Technology*, January 29, 2026, <https://www.normaltech.ai/p/fact-checking-moravecs-paradox>. His claim is not that the proposition has been empirically refuted but that it has never been empirically tested, and that it reads more accurately as a sociological observation about what AI research has found interesting.

most rigorous attempt to reproduce the result, about a quarter.¹⁵ But if a quarter of a nameless residual has now been given a name, the rest may yet get names too.

The measure of national power therefore converges on the same equation. The twentieth-century equation of national power was “population times productivity.” The twenty-first-century one is institutions times effective energy abundance times labor input. What can be built, how much can be produced, how strong a country is. The three look like separate questions and are the same product seen from different angles. This is why there is so little point in staring at GDP per capita and lamenting. That is a report card, not a cause.

Three things follow from this replacement.

One: this is an equation of supply

Of the three terms, effective energy abundance and labor input are both quantities on the supply side. Institutions, too, act on the supply side, in the sense of calling in capital and talent. Demand appears nowhere.

Bring about a supply revolution and demand follows. Demand is contracting now not because people have lost things they want, but because population, the substance of demand, is contracting under the supply constraint. The prices households face are a function of supply, and births are a function of those prices. The order runs from supply to demand, and it does not run in reverse.

So if the three terms rise, capital arrives afterward. And if the wealth created is properly distributed, population returns and demand returns. A rotation that began on the supply side comes round and reaches demand. There is a cost, however. In a world where supply is automated, what becomes scarce is not productive capacity but demand. Productive capacity nobody can sell to is a liability, not an asset.

Two: the contents of labor input have been swapped

The method of throwing in large volumes of hands to make things cheaply ended in China in the 2010s. From the 2020s onward, labor input is another name for automation, and its substance is deep tech and large capital. Possessing cheap labor is no longer an entry ticket. If a society sets out to climb the same road now, automation will have gone farther before it catches up. What is asked of this term, therefore, is not population but whether chokepoint technologies are within reach and whether capital expenditure of another order of magnitude can be moved.

For the same reason the meaning of talent changes. Talent is no longer a quantity poured into production. It moves to the side that decides what gets made. Once machines take over what to make and how much of it, human work becomes co-creation with machines: deciding what ought to be made, and seeing possibilities nobody is looking at. So what is asked is how large a pool such people emerge from, and whether that pool can be attracted.

Three: GDP moves away from value

As supply is automated, GDP moves away from the concept of the sum of value created. There are two reasons. First, wealth produced by machines is counted

¹⁵ A quarter of the residual. Kento Sakai, Paul Brockway, John Barrett & Peter Taylor, “Thermodynamic Efficiency Gains and their Role as a Key ‘Engine of Economic Growth,’” *Energies* 12(1), 2019, 110, giving roughly 25 percent of the increase in UK GDP over 1971–2013. See “Objections and Limits” at the end of this chapter.

regardless of whom it was useful to. The idea that only what sells counts as value was an approximation for an era when there were humans on the supply side. Second, because those creating value stop being conscious of productivity. Once human time is released from the direction of productivity, the value that time generates is, by definition, not recorded on an index of productivity. GDP moves away from value not because GDP breaks but because human activity moves outside the range GDP counts.

Chapter 3. On Scarcity

3.1 The 1970s as a fault line

The slowing of growth across the advanced economies over this past half-century is not the middle-income trap that stalls developing countries but something better called an advanced-country trap. The factors are various, but a considerable part of the slowdown comes from energy supply that stopped growing.

Coal, then oil, then natural gas: each time a new and more efficient source of energy was found and its cost brought down, the advanced countries turned round and set limits on their own use of it. The period from the beginning of the energy transition in the 1970s to the present may well have been a kind of transitional age. The move off oil after the oil shock, environmental regulation, the shift to renewables, nuclear phase-out in some countries, price management by OPEC+. These overlapped, and the balance of energy supply and demand broke down in ways nobody intended. Energy prices then ran for decades above the level that the available technology and the available resource base would otherwise have implied. In most advanced importing countries the cost of the energy transition ended up carried by households over a long horizon, the cost of living crept up faster than real wages, and disposable income was shaved away.

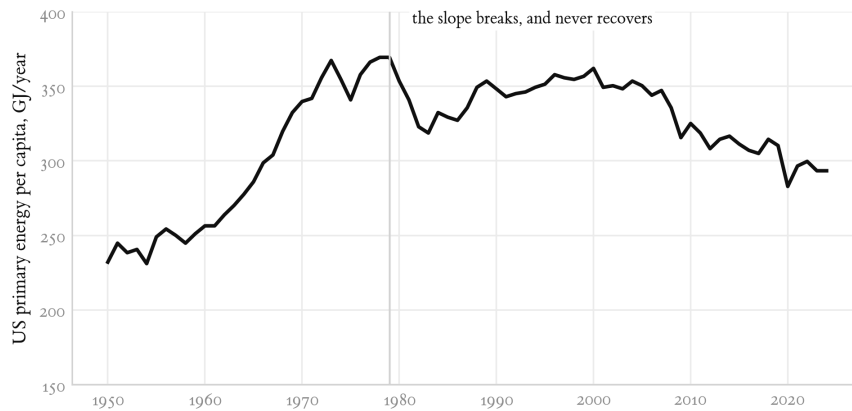
The problem is not the quantity of energy as such. It is what the effective price of energy permits a household to do and what it forbids. Fertility, housing and education all sit on that list of prohibitions.

Three exhibits for the fault line

First, energy use per person stopped rising. Primary energy consumption per capita in the United States climbed almost without interruption from the founding of the republic, topped out at the end of the 1970s, and has run flat to slightly downward for the half-century since.¹ The slope of a curve a century and a half long broke in that one decade. Some part of this is explained by efficiency gains, but set it against Jevons's law, that efficiency gains have throughout history always raised total consumption, and it reads not as the fruit of efficiency but as the halt of expansion.

¹ American energy consumption per capita, and Figure 3-1. U.S. Energy Information Administration, *Monthly Energy Review*, Table 1.7 (Primary Energy Consumption per Capita).

Figure 3-1 A century and a half of growth, stopped in one decade



Source: U.S. EIA, Monthly Energy Review, Table 1.7 (Million Btu per capita, converted at 1.055056 GJ/MMBtu).

Second, productivity decelerated. In Robert Gordon's monumental aggregation, total factor productivity in the United States grew at roughly 1.9 percent a year from 1920 to 1970, and after 1970 fell to about a third of that rate, the IT phase of the late 1990s excepted, with no recovery since.² And in the model with which Ayres and Warr asked how much of growth the useful work extracted from energy can explain, the point at which the unexplained remainder reappears is 1975.³ Two independent measurements point at the same decade.

Third, nuclear power was frozen. On the IAEA's record, all 41 reactors ordered in the United States after 1973 were canceled, and more than two thirds of all reactors ordered after January 1970 were canceled in the end.⁴ New construction stopped in practice for decades. What matters is that this was not a consequence of physical law. As the international comparison by Lovering and colleagues showed, the escalation of reactor construction costs is not a universal phenomenon.⁵ American costs shot up after Three Mile Island, while over the same years France and Korea held their costs very nearly flat by building standardized designs repeatedly. Under the Messmer plan, France's national nuclear programme, France connected 58 units to the grid between 1977 and 1999, 52 of them concentrated in the first fifteen years. The same technology escalated in one country and stayed stable in another, on the difference in regulation and politics.

A scarcity chosen

What happened in the 1970s was a choice. Led by the United States, much of the West stopped the expansion of energy. Partly as a legitimate response to accidents and to the environment, partly as an adaptation to a price cartel, and partly as the West's own doubt about whether to expand further. Whatever the reason, the consequence is the same. And the problem is not the stepping down. It is that nothing was prepared to fill the space stepped down from. For the half-century since, the advanced countries have rebuilt their economies, their institutions and their values on the premise that energy does not grow. The scarcity this chapter is about is that rebuilding. It was a scarcity chosen, not a scarcity given.

Seen from the theory of resources this should not be surprising. Resources are not, they become, as Zimmermann put it.⁶ Coal, uranium and wind were not resources

² The deceleration of productivity. Robert J. Gordon, *The Rise and Fall of American Growth*, Princeton University Press, 2016, Figure 1-2. Total factor productivity grew at about 0.50 percent a year from 1890 to 1920, 1.89 percent from 1920 to 1970, and about 0.65 percent from 1970 to 2014. "About a third" is the ratio of 1.89 to 0.65, and the end point of the comparison period is 2014.

³ The residual in the useful-work model. Robert U. Ayres & Benjamin Warr, *The Economic Growth Engine*, Edward Elgar, 2009, and the same authors' "Accounting for growth: the role of physical work," *Structural Change and Economic Dynamics* 16(2), 2005, 181-209.

⁴ The freezing of American nuclear power. IAEA, *50 Years of Nuclear Energy*, GC(48)/INF/4, Attachment 3, 2004. https://www.iaea.org/sites/default/files/gc/gc48inf-4-att3_en.pdf

⁵ Construction costs are settled by institutions. Jessica R. Lovering, Arthur Yip & Ted Nordhaus, "Historical construction costs of global nuclear power reactors," *Energy Policy* 91, 2016, 371-382. The French figures are counted from years of grid connection in the IAEA power reactor data: 58 units from Fessenheim 1 in April 1977 to Civaux 2 in December 1999, of which 52 were connected by 1991. Nuclear's share of generation reached its maximum at 78.4 percent in 2005 on Eurostat's total-generation basis.

until human beings held the knowledge and the skill to draw on them for support. Scarcity, then, is not a property of matter but a function of the knowledge and skill of whoever draws on the potential resource. The same point arrives from the side of efficiency. Raise a truck fleet from 4.3 to 6.2 miles per gallon and the stockpile of diesel on hand has effectively grown by more than a third. Physics sets a ceiling; it does not set the scarcity.

3.2 A supply that swings against us

The structural difficulty with renewables, in one line, is that they swing in the opposite direction to human need. In the second half of that transitional age, this was the main exit the advanced countries chose.

There are four difficulties. First, the output cannot be commanded. Solar generates only in daylight, and wind rises and falls with the weather. A system that must match supply and demand instantaneously and continuously takes in a large volume of generation nobody can give an instruction to. That is a structural contradiction. Second, it requires backup at all times. The higher the share, the more fast-responding plant has to be held on standby, ready to be brought up at any moment. That means owning the fleet twice over, which makes for a high-cost system running at low utilization. Third, surplus and shortage occur at the same time. In clear, windy conditions generators are ordered to curtail output, and prices go negative, meaning the generator pays to have its electricity taken; under cloud, in still air, at night, there is a large deficit. Past a certain share it destabilizes the grid as a whole. Fourth, its energy density is low. Output per unit of area is small, so it demands vast land and long transmission networks. It scales badly against exponential demand, in land and in grid alike.

None of this is anti-renewables rhetoric. It is established knowledge in energy economics. Lion Hirth's classic study showed that the market value of wind, meaning its ratio to the average power price, falls from about 110 percent near zero penetration to between 50 and 80 percent at a system share of 30 percent.⁷ The more it generates, the more everybody else is generating in the same hours, so it destroys the value of the very electricity it produces. The systematic estimates by the OECD Nuclear Energy Agency quantified how, as the share of variable renewables rises to 50 and then 75 percent, the system-level costs lying outside generation cost, meaning backup, grid reinforcement and balancing, pile up along a steepening gradient.⁸ That the IEA extended its generation-cost metric from LCOE, the levelized cost of electricity, to VALCOE, the value-adjusted version of it, is a confession from the institutional side that comparisons ignoring this decline in value had been leading policy astray.

Germany is the implemented case. The price of solar panels did indeed collapse. Electricity as a system did not get cheaper. The renewables surcharge under the EEG peaked at 6.88 cents per kilowatt-hour in 2017 and pushed household electricity prices to the highest level in the world.⁹ The number of hours in which German wholesale power prices were negative was 457 in 2024 and 573 in 2025, a record for the second year running. The time during which generators pay money to have their electricity taken off their hands passed three weeks a year. The gap in

⁶ Resources are not, they become. Erich W. Zimmermann, *World Resources and Industries*, Harper & Row, 1951, p.15. The quotation and the fuel-economy example follow Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 6, which sets out resource scarcity as a function of the knowledge and skill of the actor drawing on the potential resource.

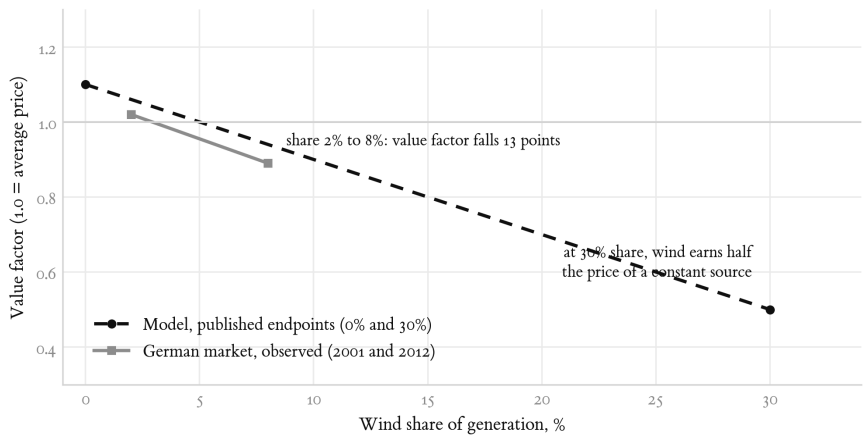
⁷ The market value of wind, and Figure 3-2. Lion Hirth, "The market value of variable renewables: The effect of solar wind power variability on their relative price," *Energy Economics* 38, 2013, 218–236. The statement that the value of wind falls from 110 percent of the average power price to between 50 and 80 percent as penetration rises from zero to 30 percent is from the paper's abstract; the observed German values (1.02 at 2.0 percent penetration, 0.89 at 8.0 percent) are from Table 3. The first is a model calculation and the second is realized day-ahead market outcome. They are different kinds of evidence.

⁸ System-level costs. OECD Nuclear Energy Agency, *The Costs of Decarbonisation: System Costs with High Shares of Nuclear and Renewables*, OECD Publishing, 2019.

⁹ The EEG surcharge. The EEG surcharge series published by the four transmission system operators (50Hertz, Amprion,

large industrial power prices, roughly 0.113 euros per kilowatt-hour in the EU against roughly 0.076 in the United States, is the bottom line of this structure.¹⁰

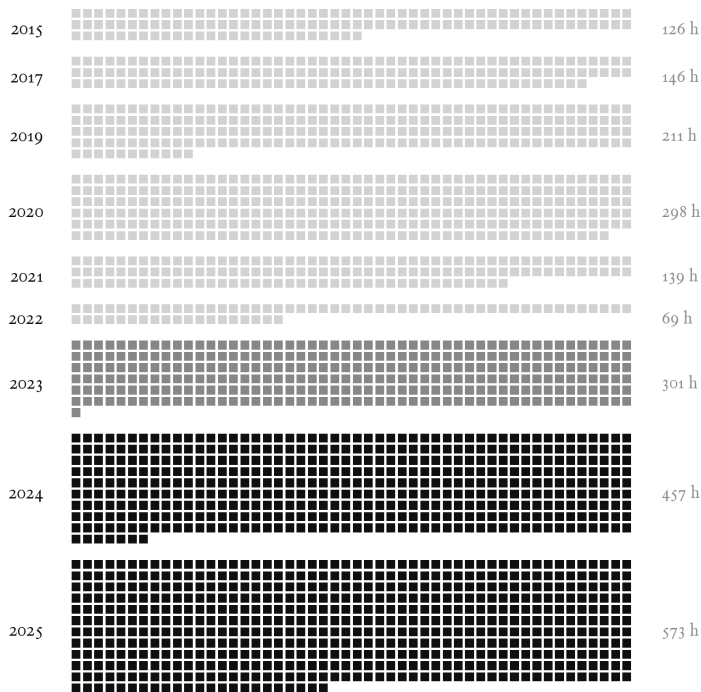
Figure 3-2 The more it generates, the less its output is worth



Only the values stated in the source are plotted. System costs at high VRE shares are treated in the text (OECD-NEA 2019). Source: Hirth (2013), Energy Economics 38, Table 3 and the mid-term EMMA results.

Figure 3-3 When electricity costs less than nothing

one square = one hour in which German wholesale power sold below zero;
each row is 50 hours



Congestion management cost EUR 3.3bn in 2023, passed through to consumers as network charges. Source: Bundesnetzagentur / SMARD; Bloomberg (2026); Montel.

The conclusion is not that renewables should be excluded. It is that renewables should be laid on top of firm generation, nuclear and fossil, as an auxiliary, and that the sound way to place them is as the optimal complementary source for a given geography. A society that handed the baseload of its civilization to a source that

TenneT, TransnetBW). The maximum was 6.88 cents per kilowatt-hour in 2017, followed by 6.756 in 2020 and 6.50 in 2021, and it was abolished on 1 July 2022. Energy-intensive industry received a reduction under an exemption, so not every consumer was paying this level.

¹⁰ The international gap in industrial power prices. On Eurostat `nrg_pc_205` (excluding taxes that are refundable, such as value added tax), the EU27 band above 150,000 megawatt-hours a year was 0.1113 euros per kilowatt-hour in 2025; the United States figure is the industrial-sector average of 8.62 cents per kilowatt-hour in EIA's *Electric Power Annual* (2025), converted at the annual average rate to roughly 0.076 euros per kilowatt-hour. For the mid-sized band of 500 to 2,000 megawatt-hours the figure is 0.187 euros and the ratio is about 2.5 times, but since the American industrial-sector average is a weighted average dominated by large consumers, matching bands means comparing large consumers with large consumers.

¹¹ Negative-price hours and congestion-management costs, and Figure 3-3. The counts of negative-price hours are the Federal Network Agency's published SMARD values (457 hours in 2024, 573 in

swings against human need distributed the cost of doing so, in the form of the electricity bill, to households in the end.

3.3 Tying one's own hands

There is a strange asymmetry of information here. Citizens, in asserting sustainability, impose an energy constraint on themselves, and do not notice that they pay for it in the electricity bill and the living costs it pushes up.

To be exact. The goal of sustainability is legitimate in itself. The problem is that the accounting between the goal and the means is not disclosed to the citizen. To choose a generation mix is to choose an electricity price curve. To choose an electricity price curve is to choose, as one bundle, the prices of housing, food and services. And to choose those prices is, in the end, to choose the number of children a household can have. If the whole of that chain were printed on the ballot paper, it would be a democratic choice. If it is not printed there, it is a society tying its own hands under an asymmetry of information. The society chose it itself, and the price system distributed it to every citizen. All that was missing was the awareness of having chosen.

3.4 The swelling of the virtual, the stalling of the physical

Alongside energy there is a second constraint: a skew in the allocation of capital and of talent. The development of information technology over the past thirty years can be called the second stage of the externalization of the brain described in Chapter 2. The internet, the smartphone, cloud computing, social platforms as an industry. These built an enormous economy in virtual space. But virtual space is a layer of computation constructed on top of physical infrastructure and energy. Servers, communication networks, devices, semiconductor fabs, logistics hubs. Every one of them demands vast investment in plant and vast quantities of electricity.

The virtual world alone swelled exponentially, while the physical world, meaning cities, transport, energy infrastructure and housing, stalled at the level of decades ago. This asymmetry was the inevitable consequence of the law of competition set out in Chapter 2. Under an energy constraint, capital flows in whichever direction eats less energy. But if the asymmetry persists, the expansion of the virtual world will itself in time be capped by the lag in physical infrastructure and energy supply. The electricity problem of AI in the 2020s is in fact the first surfacing of that cap. A virtual world that swelled for thirty years while leaving the physical world behind has begun, for the sake of its own next step, to demand the reconstruction of the side it left behind.

3.5 Why the science of the 1910s is still not built

Scientific discoveries at the level of the 1910s have still not been put to practical use. Fusion is the leading case. In 1920 Arthur Eddington proposed that the energy source of the stars might be the fusion of hydrogen.¹² Physics knew the principle a century ago. Research on controlled fusion began in the 1950s, and ITER has

2025). Congestion-management costs, in the same agency's annual publication, were 3.335 billion euros in 2023, 2.954 billion in 2024 (final), and 3.071 billion in 2025 (provisional). The 2.776 billion euros originally published for 2024 was a provisional figure, and the final figure was revised upward to about 3 billion. 2025 is higher than the year before: the cost is not coming down. These are recovered as transmission network charges, but in some years federal subsidy has partly offset those charges, so it cannot be said outright that the whole amount was passed through to consumers.

¹² Eddington. A. S. Eddington, "The Internal Constitution of the Stars," *Nature* 106(2653), 1920, 14–20 (the presidential address to Section A of the British

become an international undertaking running for more than half a century. What has obstructed implementation is not knowledge.

There are two ways to read this stagnation. The first reading is that ideas themselves are running dry. On the well-known empirical work of Bloom and colleagues, research productivity in the United States halves roughly every thirteen years, and the research input required to sustain Moore's law has swollen to eighteen times its 1971 level.¹³ The same result takes ever more minds.

The second reading is this thesis's own. Ideas did not run dry; implementation was put on rations, of energy and of capital. The science of the 1910s, fusion and the full exploitation of fission alongside it, demands enormous energy density and enormous up-front investment. Across the half-century in which hesitation about expanding energy coincided with the thinning of capital flowing into the physical world, as described in the previous section, the sciences that were most capital-intensive and most energy-centered were the first to stop being funded for implementation. The decline in research productivity that Bloom and colleagues document is not inconsistent with this reading either. In an era when the fruit that was cheap to pick hung disproportionately on the virtual side, the fruit on the physical side was not hard to pick but unpicked. Which of the two readings is correct is being put to the experiment right now. If the speed of implementation recovers in the 2020s, the decade in which computing demand began to force energy investment, then what had run dry was not ideas but capital.

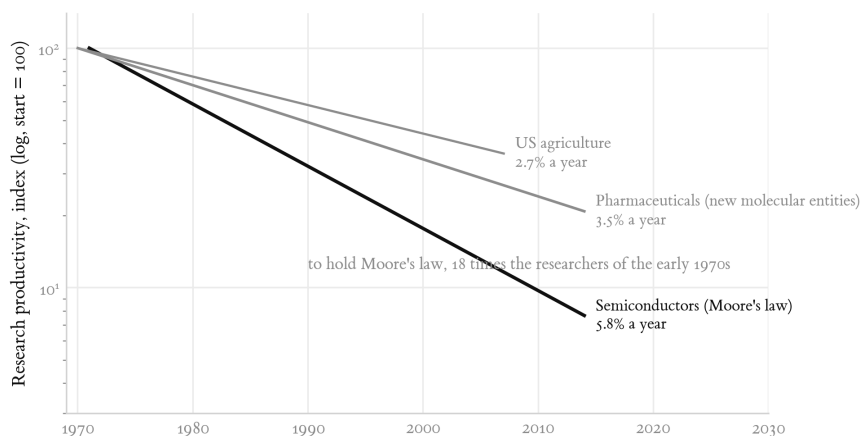
There is groundwork in management thought for separating invention from implementation. Lusch and Vargo write that invention is not what carries a society to prosperity. An invention that never connects to a body of people who use it stays a mere object, unrealized and often a waste of the resources that went into it. Gutenberg, Newcomen, Edison and Ford were, more than inventors of devices, the people who pushed their own solution up into an institution.¹⁴ Invention is a problem of engineering; implementation is a problem of institutions and design. The physics of a hundred years ago had only the first.

Association for the Advancement of Science, Cardiff, 24 August 1920). Eddington himself entered the reservation that it did not much matter whether the proposal touched on the true source of the heat, so this is a proposal rather than a proof. He rests it on the results of Aston's mass spectrometry.

¹³ Are ideas getting harder to find, and Figure 3-4. Nicholas Bloom, Charles I. Jones, John Van Reenen & Michael Webb, "Are Ideas Getting Harder to Find?" *American Economic Review* 110(4), 2020, 1104–1144. Research productivity declines at 5.3 percent a year, for a half-life of about thirteen years. On Moore's law, the "eighteen times" is not a head count but effective research effort, deflated by high-skill wages.

¹⁴ Invention and implementation. Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 10, "Innovating versus inventing."

Figure 3-4 Ideas are getting harder to find



Each line is drawn at the average annual rate of decline the paper reports for that sector, not at an assumed rate. Source: Bloom, Jones, Van Reenen & Webb (2020), *American Economic Review* 110(4), and its data appendix.

3.6 Three observational indicators, and eight billion robots

If the stagnation of the advanced countries is to be decomposed, the three indicators to watch are automation, energy cost, and the speed at which research and development gets implemented. Countries in which all three improve together rise; countries in which all three deteriorate together sink.

The total quantity of energy is going to fall short. Treat the eight billion humans alive now as robots with computing capacity. On that count alone, the total of drive energy, meaning food, and computing energy, meaning the brain, is already enormous. Now suppose we newly create advanced robots equivalent to another eight billion units: automated plant, humanoids, data centers. Very nearly the same quantity of energy again, or more, will be required on top. In a world where energy generation grows only by something like a factor of 1.2 while the requirement doubles, what happens? Energy prices rise, the whole price system is pushed up, and household disposable income falls. And so our population fell. Demand, moreover, does not stop at eight billion units. There is no limit to it. We will automate more, move more, compute more.

A world in which supply does not expand while a new class of energy consumer, the machine, begins to take its seat beside the human. What happens to the human household there?

Chapter 4. On Population

4.1 The classical Malthusian structure

In the debate over falling fertility in the advanced economies, a change in values and a failure of policy are the standard explanations. People came to weight career and free time, and to see children as a cost. Or the government's family policy is weak, and childcare, housing and education are inadequate. Both are partly right. What this chapter asks is whether they are causes.

In the world Malthus analyzed the constraint variable was unambiguous. It was food production. Where land and agricultural technique have a ceiling, a rising population means falling grain per head, and famine, epidemic, war, later marriage and contraception push the population back down again. "Population is permanently tethered to the neighborhood of the food constraint." That is the classical Malthusian trap. Nitrogen fixation by the Haber-Bosch process in the early twentieth century changed that structure dramatically. The spread of artificial fertilizer lifted grain yields enormously, and in the advanced countries a shortage of total calories ceased, at the macro level at least, to be the binding constraint.

Malthus's schema was not itself wrong. What was wrong was the premise that agriculture would go on being the bottleneck forever. The structure, that population is held down by the dominant physical constraint, holds. The bottleneck simply moved out of agriculture into another domain.

4.2 The household Malthus

Nobody in an advanced country today gives "there is nothing to eat" as the reason for not having children. And yet fertility sits pinned at historic lows.

The modern "food" is, in substance, the household's economic capacity. But the household's economic capacity and its constraint are themselves second-order variables, settled by the energy structure and the level of technology.

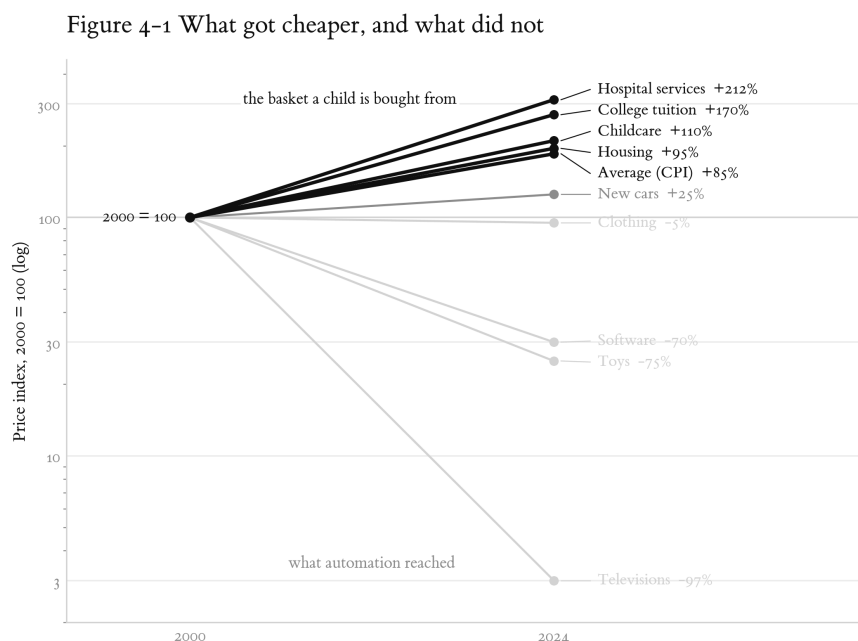
Households read off wages and the price system whether a life is affordable at all. And the price system is itself a function of the energy constraint and of technological structure. The price of a good or a service is set, roughly, by energy cost, manufacturing cost (including the extraction, processing and transport of materials), transport cost, computing cost, and the labor cost that has not been automated away. Energy is the base cost common to almost all production, transport and computation, and labor cost is heavily affected by energy and by the level of technology, which is to say the degree of automation. Hence,

Price vector of goods and services $p = f(\text{energy price } P_E, \text{ degree of automation } A)$

The "labor cost that has not been automated away" has a proper name in economics. It is Baumol's cost disease.¹ When wages rise in a sector where

productivity grows, such as manufacturing, the sectors hiring from the same labor market have to raise wages too or they cannot hire at all: education, healthcare, childcare, elder care. Their productivity has not risen and their wages have, so their prices keep rising. Helland and Tabarrok's work showed that the main cause of the surge in American education and healthcare prices is this Baumol mechanism.¹

The cost items that make up raising one child are almost all Baumol sectors plus housing. Childcare, education, healthcare, and a house. In the famous decomposition of American consumer prices, since 2000 hospital services have risen more than 200 percent, college tuition around 170 percent, and childcare more than 100 percent, while the prices of televisions, toys and software have fallen by anything from tens of percent to more than 90 percent.² The goods automation reached collapsed in price; the person-to-person services it did not reach soared. Raising a child is the act of buying, twenty years at a time, precisely the goods that automation has reached least. On the Brookings Institution's update of a US Department of Agriculture estimate, raising one child born in 2015 to the age of seventeen costs about 310,000 dollars.³ In Korea, spending on private tutoring alone hit a record 29.19 trillion won in 2024. Measured by how many years of income it takes to buy a house, Seoul is among the worst in the world.⁴



Log scale, so equal slopes are equal rates of change. Baumol sectors rise; automated goods collapse. Label positions on the right are spread for legibility; the line ends are exact.
Source: US Bureau of Labor Statistics, consumer price data, 2000–2024.

What the household faces is the budget constraint composed of that price vector and its wages. As Chapters 1 to 3 showed, in the advanced importing countries after the 1970s households carried the cost of the energy transition over a long period, living costs rose faster than real wages, disposable income was pared back, and births were suppressed to that extent. The modern population-suppressing structure that deserves the name “the household Malthus” is not a matter of subjective household judgment or of values but a secondary constraint derived mechanically from energy, technology and the price system.

¹ Baumol's cost disease. William J. Baumol, “Macroeconomics of Unbalanced Growth: The Anatomy of Urban Crisis,” *American Economic Review* 57(3), 1967, 415–426. For the application to American education and healthcare, Eric Helland & Alex Tabarrok, *Why Are the Prices So Damn High?*, Mercatus Center, 2019.

² The chart of the century, and Figure 4-1. Calculated by the author from US Bureau of Labor Statistics consumer price index series retrieved directly through the API. From January 2000 to January 2024: hospital services (CUUR0000SEMD) +248.9 percent, college tuition (CUUR0000SEEB01) +184.6 percent, childcare and nursery school (CUUR0000SEEB03) +134.3 percent. The body's “more than 200 percent”, “around 170 percent” and “more than 100 percent” are all more conservative than the measured values.

³ The cost of one child. Isabel V. Sawhill, Morgan Welch & Chris Miller, “It's getting more expensive to raise children. And government isn't doing much to help,” Brookings Institution, Aug 30, 2022. The estimate is 310,605 dollars, and it applies to a middle-income two-child family with a child born in 2015, not to the population average. It updates the US Department of Agriculture's 2017 estimate of 233,610 dollars for future inflation at 4 percent.

⁴ Korean private tutoring costs and housing. Statistics Korea and the Ministry of Education, “Results of the 2024 Survey of Private Education Expenditure on Elementary, Middle and High School Students,” 13 March 2025. Total 29.1919 trillion won, up 7.7 percent year on year, participation rate 80.0 percent. The number of pupils fell 1.5 percent, so the rise in the total comes from a rise in spending per head. Seoul's ratio of house prices to income is 24.5 to 27.2 in Numbeo's international comparison (2025–2026), but roughly 10.5 in Korean official statistics (KB Real Estate). Numbeo is user-submitted, so “among the worst in the world” swings widely with the source.

Expected number of children $n^* \approx g(\text{household disposable income } Y_d \div \text{expected cost per child } C_{\text{child}})$ Here C_{child} is a composite of the effective prices of housing, education, childcare and healthcare, and each of those terms, being a Baumol sector, rises faster than anything else inside p .

4.3 Economics is arriving at the same place

This hypothesis is consistent with where the economics of fertility has got to. Gary Becker formalized childbearing as an economic choice of the household, and Becker and Lewis showed the quantity-quality trade-off, the mechanism by which rising income raises investment per child, the quality, rather than the number. And the modern summing-up, the survey by Doepke and colleagues, reports a decisive fact.⁵ The negative income gradient that once held, richer countries and richer households having fewer children, has flattened inside the advanced world and in places has begun to invert. What now separates fertility rates is not income itself but the cost of combining career and family: what has to be given up to have a child, and what actually has to be paid. The economics of fertility is moving from cultural determinism to a theory of cost structure.

On housing, an American panel analysis shows that a rise in house prices slightly increases births among owner-occupiers while clearly reducing them among renters.⁶ Those who own a house are richer for it; those who do not are left with the higher price and nothing else. For the young, house prices are the entrance fee for having a child.

On employment, Adserà's international comparison shows that high unemployment and unstable employment push fertility down systematically.⁷ Japan is the decisive instance. The employment ice age, with youth unemployment spiking and the share of non-regular employment doubling, ran from the late 1990s through the 2000s, and marriage and births collapsed in its wake.⁷ Japanese fertility fell not because values changed first but because the income outlook of the young collapsed first.

In Korea, cumulative spending on low-fertility measures since 2006 has reached roughly 380 trillion won, and the total fertility rate (TFR) arrived at after those twenty years is 0.72 (2023), and 0.55 in Seoul.⁸ Twenty years of handing out a sum equivalent to several percent of GDP, and the price structure itself did not move.

Widen the view to the world and this is not a local phenomenon. On the United Nations' World Population Prospects 2024, world TFR has fallen to 2.25 and two thirds of the world's population live in countries below replacement, the level at which a population merely holds steady. The projection from the Global Burden of Disease study published in The Lancet has world TFR heading for 1.83 in 2050 and 1.59 in 2100. Thailand's TFR is 0.86, already below Japan's.⁹

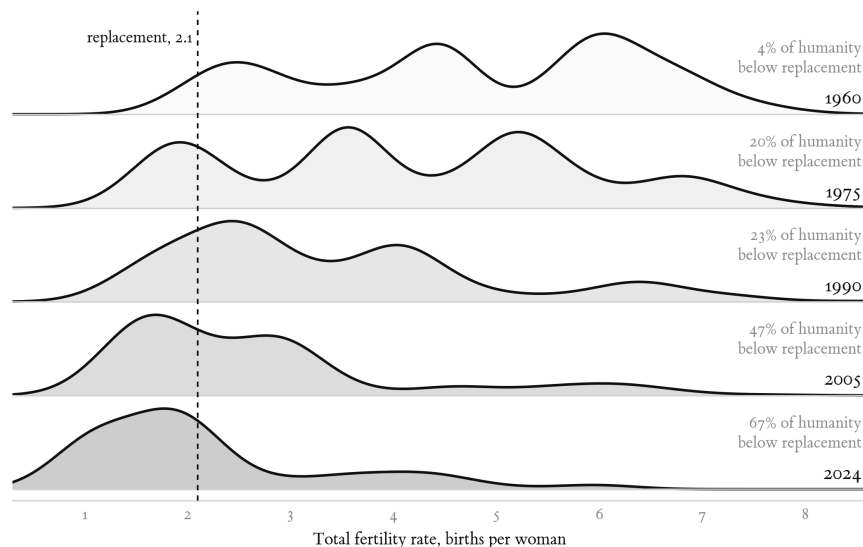
⁵ The economics of fertility. Gary S. Becker, "An Economic Analysis of Fertility," 1960. Gary S. Becker & H. Gregg Lewis, "On the Interaction between the Quantity and Quality of Children," *Journal of Political Economy* 81(2), 1973, S279–S288. Matthias Doepke, Anne Hannusch, Fabian Kindermann & Michèle Tertilt, "The Economics of Fertility: A New Era," NBER WP 29948 / *Handbook of the Economics of the Family*, 2022.

⁶ Housing and births. Lisa J. Dettling & Melissa S. Kearney, "House prices and birth rates," *Journal of Public Economics* 110, 2014, 82–100.

⁷ Employment and births. Alicia Adserà, "Changing fertility rates in developed countries: The impact of labor market institutions," *Journal of Population Economics* 17, 2004, 17–43. On Japan, the series of studies by Yuji Genda on the employment ice age and youth employment.

⁸ Korean low-fertility spending, and Figure 4–3. The cumulative total is a press-reported figure based on the National Assembly Budget Office's tabulation, roughly 379.8 trillion won from 2006 to 2023. This number is not published by a statistical agency as a single series; it is a press aggregation built up plan period by plan period. Its scope differs from the 280 trillion won (2006–2021) of the Presidential Committee on Low Fertility and Aged Society, which combines low-fertility and aging measures. The TFR is from Statistics Korea, "2023 Birth Statistics" (0.72 nationally, 0.55 in Seoul). It has since recovered for two consecutive years, to 0.75 in 2024 and 0.80 in 2025 (0.63 in Seoul). This thesis does not use the recovery side as evidence because interpretations of that recovery diverge.

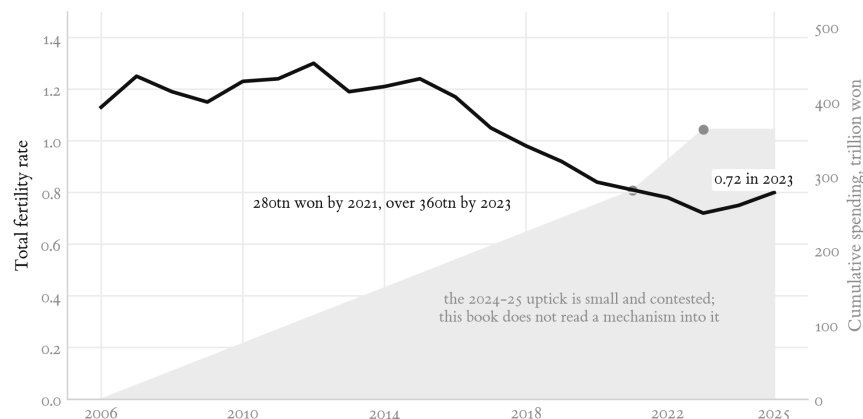
Figure 4-2 The whole distribution moved: world fertility, 1960–2024



Each ridge is the population-weighted distribution of national fertility in that year. The share of humanity living in a below-replacement country went from 3.9% to 67.4%.
Source: World Bank Open Data (SP.DYN.TFRT.IN and SP.POP.TOTL), 217 countries, retrieved 2026.

⁹ World fertility rates, and Figure 4-2. United Nations, DESA, *World Population Prospects 2024: Summary of Results* (world TFR 2.25; two thirds of the world's population living in countries below 2.1). GBD 2021 Fertility Collaborators, *The Lancet* 403, 2024 (1.83 in 2050, 1.59 in 2100). Thailand is from the Institute for Population and Social Research, Mahidol University, *Mahidol Population Gazette* Vol.35, January 2026 (registration basis, TFR 0.86). The United Nations' model estimate of around 1.2 is higher than this, because it places a larger resident population rather than the registered population in the denominator. The widely circulated value of 0.87 can be traced no further back than the CIA World Factbook and reproductions of it, and is not used. Figure 4-2 uses World Bank Open Data SP.DYN.TFRT.IN and SP.POP.TOTL, 217 countries, retrieved 2026.

Figure 4-3 South Korea: money spent, and what it bought



Only the published cumulative totals are marked; the shaded area interpolates between them.
Source: Statistics Korea (TFR); cumulative low-fertility spending as published (280tn won to 2021; over 360tn to 2023).

4.4 China, the largest case

The site where this mechanism is running at the largest scale in the world is China. The merits of the political system are not the subject. Only the arithmetic of population and economic structure is.

In 2025 China's annual births fell to 7.92 million. That is a drop of 17 percent in a single year from the previous year's 9.54 million, less than half the 2016 figure, and the lowest since the founding of the republic. The population fell by 3.39 million to 1.405 billion, a fourth consecutive year of decline, and the rate of decline more than doubled against the previous year. Marriage is the leading indicator, and marriage registrations in 2024 numbered 6.10 million, down 20 percent on the year, the largest fall since the statistics began. The 2013 peak was 13.47 million.¹⁰

¹⁰ Chinese population statistics. National Bureau of Statistics, "Statistical Communiqué of the People's Republic of

The population base is already fixed

That it will not reverse is already settled once you look at the cohorts, meaning the groups born in the same year. The female population aged 20 to 34, which carries 85 percent of births, goes from 105 million in 2025 to 58 million in 2050, close to halving, and that is fixed by the number of people who have already been born. This thesis's own estimate lays over that cohort structure the observed trends in marriage rates and in marital fertility, meaning how many children married couples have, and stacks up annual births from them. The implication is simple. Births, which ran at around 20 million a year in the 1990s (20.67 million in 1996), fall below 6 million around 2032 and flatten out at 5 to 6.5 million a year from the late 2030s into the 2040s.

Births in a given year are the female population of reproductive age in that year multiplied by age-specific fertility rates and summed. China's mean age at childbearing is 29.2 to 29.3 on the United Nations' estimates;¹¹ the distribution of births spreads from roughly 18 to 40, centered on that mean with a slight tail to the right. Every woman who will give birth in 2050 has been born by 2032 at the latest. The population base is already fixed. The only degree of freedom left is the level of age-specific fertility, which is to say the TFR.

The median birth year of the mothers of the 7.92 million children born in 2025, backed out from the mean age at childbearing, falls around 1996. Births in 1996 were 20.67 million. By 2003 they had fallen to about 16 million.¹⁰ Mechanically from there, births in 2032 shrink to around 6 million. Actually applying the age-specific fertility distribution and summing gives about 6.55 million if TFR is held at its 2025 level of 0.93, and about 5.97 million if TFR falls to 0.75.

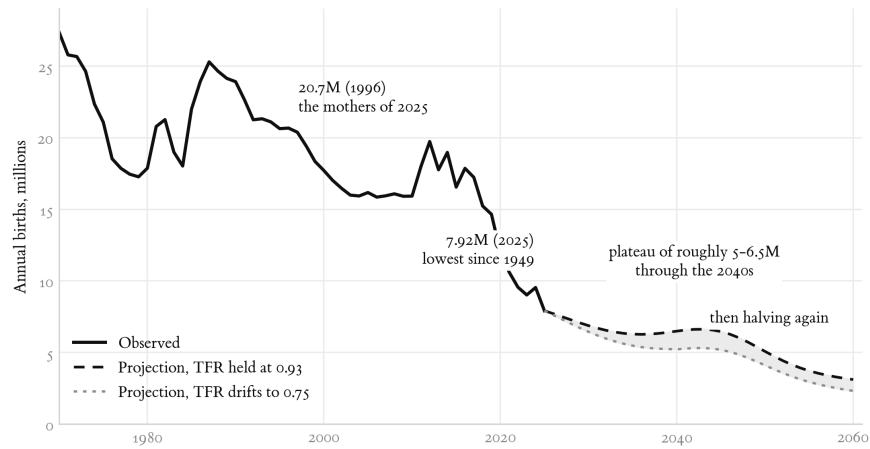
The number of mothers born between 2003 and 2010 is flat, and between 2011 and 2017 births jumped temporarily. That structure gives the estimated line three phases. Births from about 2033 to 2040 stabilize at 5 to 6.5 million (they enter the labor market between 2055 and 2062). The years from about 2041 to 2047 are the last opportunity to raise the number of births (labor market entry between 2063 and 2069). And from 2047 the line enters a phase of halving as the years pass (2070 onward). The estimate for 2060 is 3.12 million even with TFR held, and 2.32 million at a TFR of 0.75.

This reading is still an optimistic one. The held-constant scenario does not price in any further fall from the current TFR of 0.93. Young people worn down by 996, meaning nine in the morning to nine at night six days a week, and by involution, the grind that leads nowhere, are leaving Shanghai, Hangzhou and Shenzhen for cheaper cities such as Chengdu and choosing to lie flat and stop trying. Look at that and there is nothing unreasonable about assuming a TFR that falls to somewhere between 0.7 and 0.8. In that case births in the late 2030s come to 5 million a year, which is about the American level. A country of 1.4 billion people and a country of 340 million would be producing the same number of children.

China on the 2025 National Economic and Social Development," 28 February 2026. Births of 7.92 million in 2025 (birth rate 5.63 per thousand), 9.54 million in 2024, year-end population of 1,404.89 million, down 3.39 million on the previous year end, a fourth consecutive year of decline. Marriages are from Ministry of Civil Affairs statistics: 6.106 million in 2024 against 7.682 million the year before, down 20.5 percent; the 2013 peak was 13.469 million. The 15.99 million births in 2003 are from that year's communiqué. The 20.67 million for 1996 is a yearbook series value; that year's communiqué gives only a birth rate of 16.98 per thousand and a year-end population of 1,223.89 million.

¹¹ China's age at childbearing and TFR. Mean age at first birth was 28.63 in 2024 in the *China Population and Employment Statistics Yearbook 2025* (27.22 in 2020, 27.84 in 2021). The 2025 value is unpublished. Mean age at childbearing is 29.15 in 2024 and 29.28 in 2025 on the estimates in the United Nations' *World Population Prospects 2024*. The National Bureau of Statistics does not publish a total fertility rate, and the 0.93 this thesis uses is a private estimate. The *China Population and Employment Statistics Yearbook* gives 1.14 for 2024 and the United Nations estimate is about 1.01; the sources diverge widely.

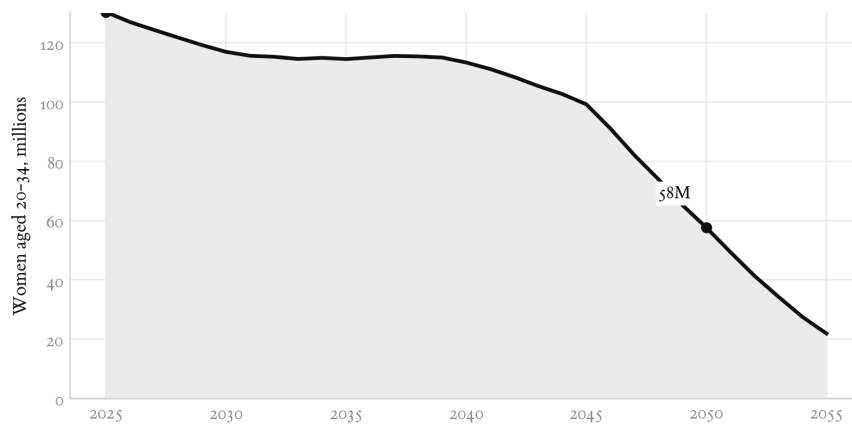
Figure 4-4 China: annual births, 1970–2025 and projected to 2060



Projection applies an age-specific fertility schedule (mean age at childbearing 29.7, drifting to 30.5) to female cohorts already born, calibrated to the 2025 outturn.

Source: Observed: National Bureau of Statistics of China. Projection: author, cohort-component model.

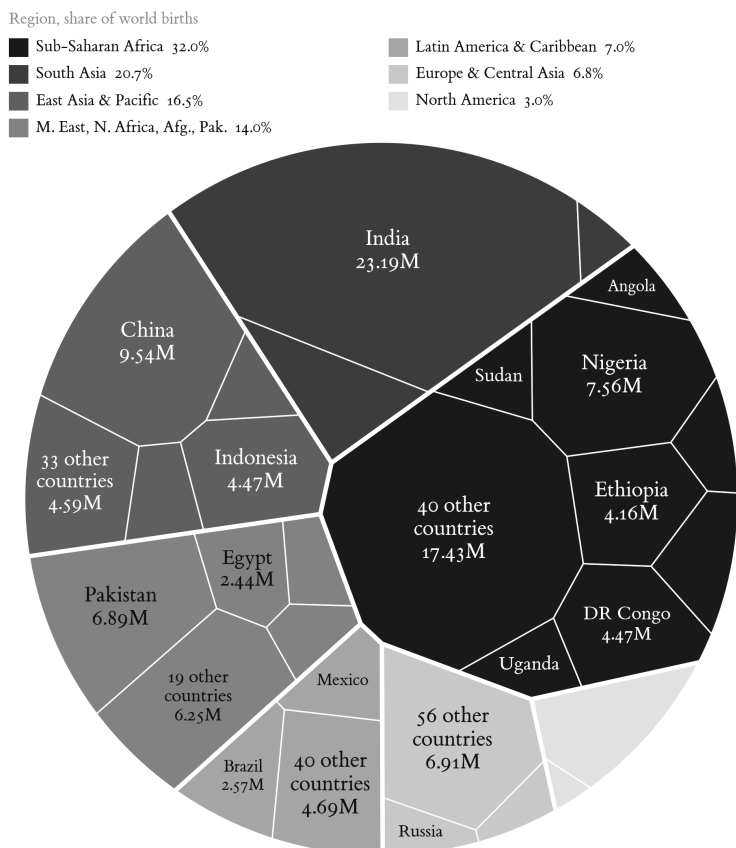
Figure 4-5 China: women of peak childbearing age, already determined



Every woman in this chart has already been born. The decline is arithmetic, not forecast.

Source: Derived from NBS birth series and census sex ratios.

Figure 4-6 Where the world's children are born, 2024



Voronoi treemap: every cell's area is proportional to that country's births, and every region's area to its share. Smaller countries are pooled inside their region. Sub-Saharan Africa and South Asia together hold more than half. China's own registration figure for 2025 is lower still, at 7.92M.

Source: World Bank Open Data, crude birth rate x population, 2024; 132.3M births across 217 countries.

The same mechanism as Korea

Persistently high youth unemployment reproduces, unchanged and on a far larger scale, the suppression of births through a collapsed income outlook seen in section 4.3. The behavior of not moving when money is handed out, and moving only in the years when the income outlook moved, is the same as Korea's.

Population is the variable that is fixed before any other, and the one that moves last.

4.5 Values are a tertiary variable

Values are not a primary cause. They appear two or three steps down from a structure that sits below them. Wanting to put a career first, not wanting to give up free time, seeing children as a burden. In most cases these values appear at the final stage of a chain that runs like this.

Energy and technology structure → the price system and the wage structure → the household's budget constraint and its anxiety about the future → weighting toward career and toward fewer children as a "rational" principle of action → that principle later justifying itself as a "value"

¹² The Chinese estimate, and Figures 4-4, 4-5 and 4-6. The estimates in Figures 4-4 and 4-5 use this thesis's own cohort component method. An age-specific fertility distribution is applied to the female cohorts already born, and calibrated on the 2025 actual. Figure 4-5 is derived from the National Bureau of Statistics birth series and the census sex ratio at birth. Figure 4-6 uses World Bank Open Data (SP.DYN.CBRT.IN × SP.POP.TOTL, 2024, 217 countries), with the full table at [References/worldbank_fertility_latest.csv](#). The World Bank does not publish absolute numbers of births, so these are values derived as the product of the crude birth rate and total population, and they do not agree with each country's confirmed registration-based figures.

People are not choosing values freely in a vacuum. They run a calculation of gains and losses inside the macro structure they are given, and the principle of action they adopt as a result is simply being put into words as a “value”.

The strongest counter-evidence against this claim is the work of Kearney and colleagues. Regress the fall in American fertility since 2007 on state-level economic variables, unemployment, house prices, childcare costs and student debt, and no economic variable explains more than a small part of the fall.¹³ Their conclusion is a shift in priorities across cohorts, which is to say a change in how the younger generation designs a life.

There are two answers. First, a shift in priorities is not the end point of an explanation. It is the thing to be explained. When the priorities of a generation move all at once, that is not a change in taste occurring in a vacuum; it is a cohort-level adaptation to the price system that generation watched while growing up and to the struggle it watched its parents have. To detect a change in a tertiary variable and call it a primary cause is close to calling the movement of a thermometer needle the cause of the heat.

Second, the two are measuring different things. What Kearney and colleagues measure is how much moves inside the present regime when a variable is nudged: how much of the variation in births is explained by variation in unemployment and house prices from state to state. What this thesis is talking about is the regime itself: what happens to births when the effective price of energy falls by an order of magnitude, when a productivity revolution reaches the Baumol sector, and when the wealth produced is distributed to households. These are not the same question.

Even so, the credit for burying the naive version of the price account belongs to that study. Checks do not move it; only structure does.

4.6 Population is the anchor

The speed at which population returns sets the speed of everything else.

That is not an argument for increasing population. The collapse of population is neither an objective nor an enemy but one of the symptoms counted in section 1.4. Take the constraint off and every one of the symptoms moves toward resolution, for the simple reason that they were only ever what that larger constraint made them.

The degrowth cast of mind has to be broken all the same: the idea of accepting a falling population in order to hold down the consumption of resources. There is no ground for suppressing population. Population is a result, and pressure is being applied to it now only because there is a constraint on the supply of energy. If energy were inexhaustibly abundant and supplied cheaply and in volume, the rate of economic growth would by rights explode, and with redistribution working, our population would be free to rise again. As we saw in Chapter 1, societies of abundance have in fact reproduced their populations. In Britain during the industrial revolution, in the postwar United States, in the Gulf during the oil boom.

Elon Musk’s dream of becoming a species living on more than one planet is not a mission but a result that follows once the constraint comes off. Abolish the energy

¹³ Counter-evidence to the price account. Melissa S. Kearney, Phillip B. Levine & Luke Pardue, “The Puzzle of Falling US Birth Rates since the Great Recession,” *Journal of Economic Perspectives* 36(1), 2022, 151–176.

constraint on Spaceship Earth and our civilization turns necessarily toward expansion, with expansion in the direction of space following as part of it.

Falling fertility is not a cultural phenomenon but a price phenomenon. Only a collapse in the cost of energy and robotics, therefore, can reverse the demographics. That is the end of the diagnosis. What comes next is what happens when that constraint comes off.

Chapter 5. On Abundance

5.1 The traveler with 2GB

On a week away, with only 2GB of roaming data, we hold back as far as we can. Whether a thing is finite or infinite works on the mind out of all proportion. Give us genuinely unlimited data and we keep the phone running and do anything at all with it: quite possibly out among the sights from morning to night with a voice agent open the entire time. We decide which businesses to put capital into from inside a constraint. Latent business models branch like a tree, and to stop short because a constraint is in the way may be to cut away, wholesale, every branch that would have flowered further on. And from inside the constraint we cannot, in principle, imagine what we would do in a world without it.

On energy, our civilization is that traveler with 2GB.

So what happens when the cost of energy structurally collapses is not a discount on existing demand but an explosion of demand. One fall in price unlocks the next use, and that unlocks the one after it. This thesis calls that chain the cascade. Three centuries of evidence stand behind the claim.

5.2 Three centuries of light

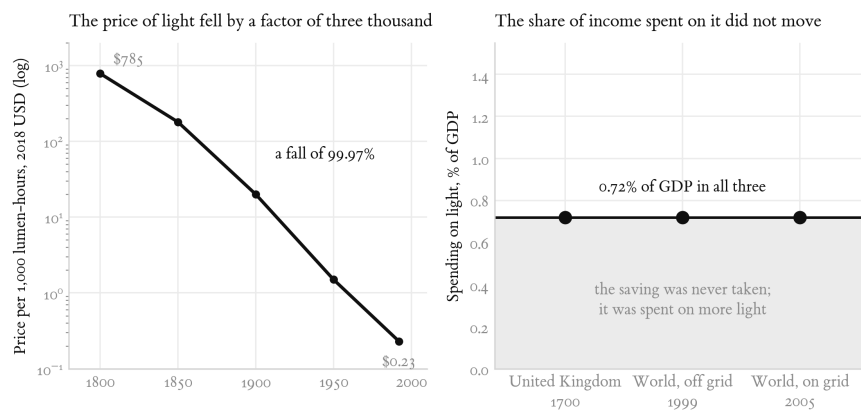
One of the goods whose cost humanity has driven down for longest is artificial light. William Nordhaus, in a classic study, measured the true price of light from open fire through tallow candles, whale oil, gas lamps, incandescent bulbs and fluorescent tubes, and showed that the official price indices had overstated the rise in the price of light by between 900 and 1,600 times. In real terms, the price of a thousand lumen-hours fell by 99.97 percent between 1800 and 1992.¹ Setting computation aside, no good in human history has become as cheap as light.

So when light became three thousand times cheaper, did humanity spend less on light? On the work of Tsao and colleagues, humanity has for three centuries, across six continents and five distinct lighting technologies, gone on spending consistently about 0.72 percent of GDP on light.² Exactly to the extent that efficiency rose, and by exactly that much, humanity consumed more light. It invented the city at night, the twenty-four-hour factory, the screen that never sleeps. Economics calls it the rebound effect when the saving efficiency was supposed to deliver is absorbed straight back into higher use. For light it is close to 100 percent.

¹ The true price of light. William D. Nordhaus, "Do Real-Output and Real-Wage Measures Capture Reality? The History of Lighting Suggests Not," in *The Economics of New Goods*, University of Chicago Press, 1996, ch.1. The ratio of the traditional price index to the true price stands at 1992 at 910 times (Light II) to 1,632 times (Light I). The real price of a thousand lumen-hours fell from 429.628 cents in 1800 to 0.124 cents in 1992, a fall of 99.971 percent. This is the series for the best available technology, not the series for an average household.

² 0.72 percent of GDP. J. Y. Tsao, H. D. Saunders, J. R. Creighton, M. E. Coltrin & J. A. Simmons, "Solid-state lighting: an energy-economics perspective," *Journal of Physics D* 43(35), 2010, 354001. Across three centuries the world has gone on spending roughly 0.72 percent of GDP on light. The underlying data are Tsao & Waide, *LEUKOS* 6(4), 2010 (three centuries, six continents). The formulation in terms of five technologies is H. D. Saunders & J. Y. Tsao,

Figure 5-1 Three centuries of light: the purest case of the cascade



The three points on the right are the independent empirical settings the paper reports, three centuries apart, from tallow candles to grid electricity.
Source: Left: Nordhaus (1996), in *The Economics of New Goods*. Right: Tsao et al. (2010), *J. Phys. D* 43(35), 354001.

“Rebound effects for lighting,” *Energy Policy* 49, 2012, 477–478. That paper is about the rebound in energy consumption rather than about the GDP share.

This is the purest modern version of the paradox Jevons saw in coal. A collapse in cost is absorbed not as a contraction of spending but as an explosion of consumption. There is no reason why what happened to light should not happen to energy as a whole. Light, after all, is only one form of energy.

5.3 Anatomy of the demand explosion

The more the efficiency of energy conversion in the physical world improves and the further cost falls, the more the business models that have been held down come into flower, the more energy demand expands without limit, the more uses multiply, and the more energy production rises again in a virtuous circle.

$$E \uparrow \rightarrow P _ E \downarrow \rightarrow \text{application demand } D \uparrow \rightarrow \text{investment } \uparrow \rightarrow E \uparrow$$

What gets unlocked can be drawn concretely from the cost structure.

Water. Reverse osmosis desalination has come down to roughly 0.5 dollars per cubic meter, and between 30 and 50 percent of that cost is energy.³ Cut the price of energy to a fraction of itself and fresh water stops being the binding variable of arid regions.

Food. Energy accounts for 40 to 70 percent of the operating cost of vertical farming.³ In a world of cheap electricity, agriculture is cut loose from land and climate and moves into the city.

Cities. Whether a city can be built from nothing is the product of institutions, energy and labor input, and fusion and automation set that product high in any location at all.

Space. Launch cost has fallen twentyfold, from roughly 54,500 dollars per kilogram on the Space Shuttle to roughly 2,720 dollars per kilogram on Falcon 9. Starship is spoken of as aiming two orders of magnitude below that.⁴ Complete the third order of magnitude and orbit becomes an industrial location.

³ Water and food. The desalination cost and its energy share are approximate. For indoor vertical farms there are measurements putting lighting at 46 to 75 percent of total energy consumption and climate control at 22 to 52 percent (University of Manitoba). The breakdown by end use, lighting at 60 to a little over 80 percent and climate control at 20 to a little over 30 percent, sits toward the middle of the range in the peer-reviewed literature, and depending on facility type and climate the ranking of lighting and climate control reverses.

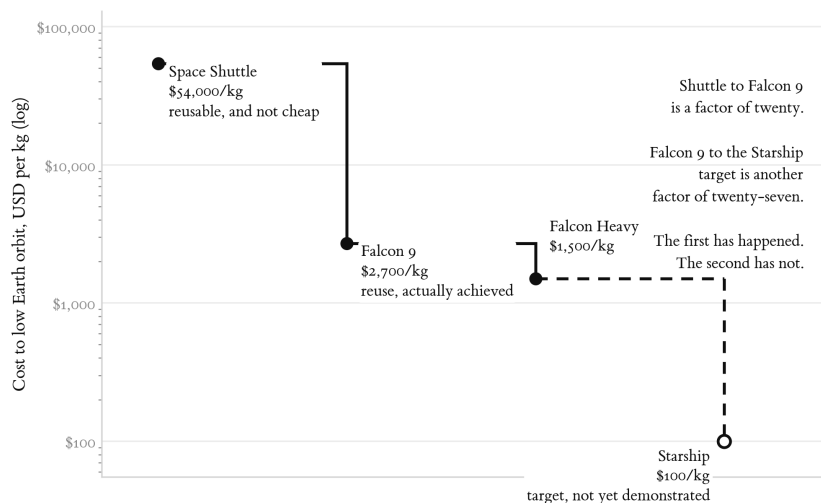
⁴ Launch cost and Figure 5-2. Harry W. Jones, “The Recent Large Reduction in Space Launch Cost,” *48th International Conference on Environmental Systems, ICES-*

Computation. As Chapter 2 showed, that the leading edge of the demand explosion is computation is no longer a forecast but an observation.

The constraint on a civilization is industrial energy demand, not household demand. The macro sketch is drawn by the IEA itself. Electricity as a share of final energy consumption rises from about 20 percent today to more than 50 percent by 2050 in the net-zero scenario, and world electricity demand rises by between 75 and 150 percent by 2050 depending on the scenario.⁵ And that estimate has not yet priced in a world in which consumption per head in the global south moves up toward OECD levels. Recall the arithmetic of eight billion robots at the end of Chapter 3. There is no upper bound on demand.

2018–81, 2018. The Space Shuttle carried 27,500 kilograms to low Earth orbit at roughly 1.5 billion dollars per flight, that is 54,500 dollars per kilogram. Falcon 9 carries 22,800 kilograms for 62 million dollars, 2,720 dollars per kilogram. SpaceX has published no official unit-cost target for Starship. The talk of a fall by orders of magnitude comes entirely from Musk’s own statements, and as of mid-2026 no commercial price exists. Compared on delivery to the ISS, moreover, the Shuttle’s 105,800 dollars per kilogram against roughly 25,000 dollars per kilogram for Falcon 9 with Dragon makes the difference a factor of four rather than twenty.

Figure 5–2 Orbit: how many orders of magnitude have actually been crossed



Solid steps have flown and been priced. The dashed step is a target, not an achievement. Complete the third order of magnitude and orbit becomes an industrial location rather than a mission.

Source: Jones (2018), ICES-2018–81; CSIS Aerospace Security; published vehicle targets.

⁵ The outlook for electrification. IEA, *World Energy Outlook 2022* (net zero emissions scenario; electricity above 50 percent of final energy consumption in 2050, world electricity demand up by 75 to 150 percent by 2050 depending on the scenario).

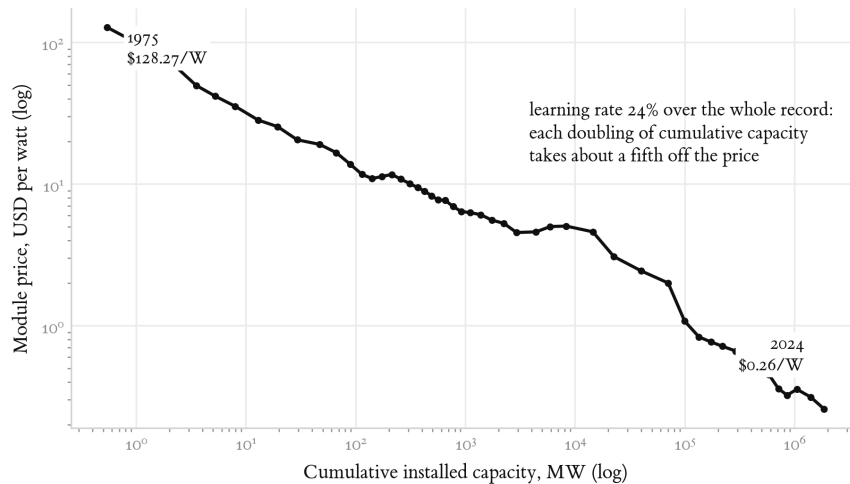
5.4 The order of supply, and Wright’s law

If demand grows without limit, the question moves to supply. What supplies it, in what order, on what cost curve? The order follows from the arguments of Chapters 3 and 4. At the top, fusion. The transitional period before it is everywhere is dominated by fission, both conventional large reactors and small modular reactors (SMRs) built in series in a factory. Fossil fuels as a bridge. And renewables riding on top of firm power as a supplement.

What sets the cost curve is Wright’s law: unit cost falls by a constant rate for every doubling of cumulative production. Solar panel prices fell 20 to 25 percent with every doubling of cumulative capacity, and module prices dropped roughly 90 percent in the 2010s alone. Way and colleagues at Oxford, having tested learning-curve forecasts against more than fifty technologies, concluded that a rapid energy transition comes out roughly 12 trillion dollars cheaper, net, than a slow one.⁶ Cost reduction is not a function of will but of volume.

⁶ Wright’s law and Figure 5–3. T. P. Wright, “Factors Affecting the Cost of Airplanes,” *Journal of the Aeronautical Sciences* 3(4), 1936, 122–128. The solar series is Our World in Data, “Solar photovoltaic module price vs. cumulative capacity” (CC BY 4.0, drawing on Nemet 2009 and IRENA). The

Figure 5-3 Wright's law in solar modules, 1975-2024



Observed series, not a fitted curve. The learning rate is estimated from these points.
Source: Our World in Data, solar PV prices vs. cumulative capacity (CC BY 4.0); IRENA.

conclusion that the faster transition is the cheaper one is Rupert Way, Matthew C. Ives, Penny Mealy & J. Doyne Farmer, “Empirically grounded technology forecasts and the energy transition,” *Joule* 6(9), 2022, 2057–2082.

So nuclear and SMRs turn on how many copies of one design get built. As Chapter 3 showed, the cost escalation of reactors is a product of institutions rather than of physical law, and France and Korea, repeating a standard design, held their costs stable. What SMRs aim at is the conversion of a reactor from a civil engineering project built on site into a product manufactured in series in a factory, putting nuclear on Wright’s law for the first time in its history.

The order book that mass production presupposes is already being built. Google contracted for up to 500 megawatts, Amazon for more than 5 gigawatts by 2039, Meta tendered for up to 4 gigawatts, and Microsoft locked in 835 megawatts on a twenty-year contract through the restart of Three Mile Island unit 1. The capacity the four announced or tendered across four months in the autumn of 2024 comes to something on the order of 10 gigawatts.⁷ OpenAI’s Stargate has reached roughly 7 gigawatts of planned capacity and more than 400 billion dollars of investment.⁸ In fusion, Microsoft signed the world’s first fusion power purchase agreement with Helion, and Google placed a 200-megawatt one on CFS’s ARC reactor.⁹ On the machines themselves, Helion’s Polaris became the first privately funded device to measure fusion on deuterium-tritium fuel, reaching 150 million degrees, and CFS’s SPARC has set its first-plasma target at 2027. These are targets, not achievements.¹⁰

But neither an order book nor an ignition is a price. SMRs and fusion alike carry large up-front construction costs while their fuel costs once running are low, which gives them the property data centers and semiconductor fabs have: once the initial investment is cleared, the marginal cost of expanding use is low. Between the ignition of the first machine and a tenfold reduction in the cost per unit lie several decades and several hundred reactors. First power reaches the grid at the end of the 2020s and through the 2030s, prices genuinely move in the 2040s, and even the 2040s may be too early. Diffusion itself is not the hard part, because fusion is compatible with the existing grid.

If energy settles the speed of AI, energy is also an instrument of governance over it. The electricity large-scale AI takes for training and inference cannot be hidden, cannot be smuggled, and can be counted.¹¹

⁷ Nuclear procurement by the technology majors. Google and Kairos Power, up to 500 megawatts (14 October 2024; Google’s own wording is the world’s first corporate agreement to purchase nuclear energy from multiple small modular reactors). Amazon and X-energy, more than 5 gigawatts by 2039 (16 October of the same year). That, however, is an equity investment and a joint deployment target rather than a power purchase agreement, and the capacity actually settled on the day was roughly 320 megawatts with Energy Northwest and more than 300 megawatts with Dominion. Meta’s tender for 1 to 4 gigawatts (3 December of the same year). That tender bore fruit on 9 January 2026 in agreements totalling up to 6.6 gigawatts, including up to 2.8 gigawatts with TerraPower, more than 2.1 gigawatts with Vistra and up to 1.2 gigawatts with Oklo. Microsoft and Constellation, roughly 835 megawatts at Three Mile Island unit 1 (renamed the Crane Clean Energy Center) on a twenty-year contract (20 September of the same year). This is the restart of an existing reactor

5.5 The effect on population

If fertility reverses, it will reverse when the price of a child falls. As Chapter 4 showed, the fall in fertility is a price phenomenon. And the condition for that arrives for the first time when the marginal cost of labor collapses in the wake of energy.

The sequence runs like this. If the mass production of humanoids and automated plant begins at the end of the 2020s, the price of energy P_E and the degree of automation A , in the framework of Chapter 4, move at the same time. The effective prices of the components of the expected cost per child C_{child} , meaning housing, education, childcare and healthcare, the Baumol sectors, can then fall structurally for the first time. There has only ever been one known cure for Baumol's cost disease. Raise productivity in the sick sector directly, with technology. Physical AI is another name for the productivity revolution in the Baumol sectors.

So the place for capital changes. Family-policy subsidies do not produce the reversal; Korea's roughly 380 trillion won is the largest instance of the attempt on record. There is little point in committing the bulk of scarce investment resources to pro-natal spending. The same capital belongs in R&D on energy and on automation. The price of a child lives in the Baumol sectors, and what reaches the Baumol sectors is productivity, not transfers.

The structure is the mirror image of the classical Malthusian trap. In Malthus's world population was bound by the food constraint, and it grew again when technology loosened that constraint. What stands where food stood is energy. Stagnation in growth and in population is, in the first instance, an energy problem. Let the price of energy fall dramatically and available energy per head expand, and new business models arrive one after another, the price of a child falls, and fertility returns.

5.6 How long it actually takes

The physical world runs on a different clock. Delivering physical AI and energy abundance in full within the 2030s is probably not achievable. Software has a zero marginal cost of copying; robots have to be built one at a time, and factories, parts, regulation, accidents and repairs all proceed on physical-world time. In the terms of Chapter 2, C_{pred} falls exponentially while C_{config} is bound by physics. Confusing the speeds of the two is the single most common error in this field.

The unit cost of humanoids is indeed beginning to ride Wright's law. On Morgan Stanley's estimate, the bill of materials for Tesla's Optimus Gen2 currently runs at 50,000 to 60,000 dollars, and Musk talks about a sale price at volume of 20,000 to 30,000 dollars. But on the January 2026 earnings call Musk himself said the machine is "not being used in any meaningful way in the factory".¹² Unit cost is falling, while quality is still in the middle of being demonstrated. Unit cost and quality advance separately.

This is a story about the advanced economies alone. A substantial share of the world's population has still not reached stable electricity, water and sanitation, or

rather than new build, and although the target was pulled forward from 2028 to 2027, as of mid-2026 it remains under environmental review at the Nuclear Regulatory Commission. Adding the four companies' figures gives something on the order of 10 gigawatts, but the new capacity settled at that point was between roughly 1 and 1.6 gigawatts.

⁸ Stargate. OpenAI, "OpenAI, Oracle, and SoftBank expand Stargate with five new AI data center sites," Sept 23, 2025 (planned capacity close to 7 gigawatts, more than 400 billion dollars over three years, and a path to the 10 gigawatts and 500 billion dollars announced in January). The Michigan announcement in October of the same year updated this to more than 8 gigawatts and more than 450 billion dollars. Confirmation that 10 gigawatts and 500 billion dollars had been secured by the end of 2025 has not been obtained.

⁹ Fusion power purchase agreements. Helion Energy, "Helion announces world's first fusion energy purchase agreement with Microsoft," May 10, 2023 (50 megawatts or more, target operation in 2028, with penalties for non-delivery). Commonwealth Fusion Systems, "Google and Commonwealth Fusion Systems Sign Strategic Partnership," June 30, 2025 (200 megawatts of the 400-megawatt output of the first ARC machine, early 2030s, Chesterfield County, Virginia).

¹⁰ Where fusion has got to. Helion Energy, "Helion Achieves New Industry-First Fusion Energy Milestones," Feb 13, 2026. Polaris became the first privately funded device to measure deuterium-tritium fusion, reaching 150 million degrees. One hundred million degrees was a record achieved in 2021 by a different device, Trenta. Polaris is not generating electricity and does not claim scientific breakeven. The power for Microsoft is to be supplied by a different machine, Orion. SPARC's first plasma in 2027 rests on reporting of statements by CFS, and no official announcement from the company stating that date has been confirmed. The original target was 2025.

¹¹ The chokepoint of computation. Girish Sastry, Lennart Heim, Haydn Belfield et al., "Computing Power and the Governance of AI," arXiv:2402.08797, 2024.

¹² Humanoid unit cost and deployment. Morgan Stanley (Adam Jonas) built the Optimus Gen2 bill of materials up from component supplier quotes in June 2024 at 50,000 to 60,000 dollars, excluding software.

basic education. That is a matter of capital cost, of grid and of procedure, not of capacity or of will. On the same planet on which we are discussing fusion and humanoids. And the timeline this chapter has counted does not apply to them.

We look at what is around us and assume that humanity as a whole is moving at the same speed. What is fast is only what is around us, in Silicon Valley or the UAE. The claim of this thesis is not about when it arrives but about what the constraint is. The identification of the constraint can be correct while the speed of its removal differs completely from place to place according to capital, institutions and geography.

The possibility that capital markets break first. The equation of Chapter 1 has a variable in it that has not yet been made explicit: the appetite of capital markets to invest. In Bank of America's fund manager survey of February 2026, the share of respondents saying that companies are overinvesting in plant reached the highest level in the history of the survey, and 25 percent named an AI bubble as the extreme downside they were most wary of.¹³ This is the most direct skepticism the financial markets offer about this thesis's supply-side scenario. Let AI investment break badly once and the expansion curves of energy and computing power themselves bend. At that point the diagnosis of what the constraint is stays correct, and only the date it comes off slips back by a decade or more.

Growth itself will not be exponential. The market of 2026 prices AI as though it will be. But what sets the speed is not how much power can be generated; it is whether a new load can be connected to the grid.

Generation itself has grown at that rate. As Figure 5-3 shows, cumulative installed solar capacity climbed six orders of magnitude, from 0.54 megawatts in 1975 to 1.85 million megawatts in 2024, and China accumulated 1,840 gigawatts of wind and solar alone by the end of 2025, 47 percent of its total installed capacity.¹⁴ The constraint sits between the generator and the load: the queue for new grid connections, manufacturing capacity for transformers and switchgear, land acquisition and permitting for transmission lines, substation construction. Every one of these compresses only in units of years, however many factories you add, and every one of them tightened worldwide at the same time from the early 2020s.

Two implications follow from this. Prices in the transition rise rather than fall, because demand explodes while supply is held. And the scarce asset is not the generating plant but the site already connected to the grid, and the position in the queue. What energy settles, then, is not how fast growth is, but whether growth is exponential at all.

5.7 The valley of demand

Supply grows at once; demand grows only at the speed of population.

As section 2.7 derived, in a world where supply is automated the scarce thing is not productive capacity but demand. And population, which is what demand is made of, is the slowest-moving variable in an economy. Several decades pass before demand catches up with supply.

The 10,000 to 20,000 dollar level is not a Morgan Stanley target but the sale price at volume that Musk talks about, 20,000 to 30,000 dollars. The range the firm publishes is far wider, from 10,000 to 300,000 dollars depending on application and configuration. On the fourth-quarter earnings call in January 2026 Musk said the machine is still in the research and development stage and not being used in any meaningful way in the factory. As of April 2026 mass production of Optimus had not begun; the plan is to repurpose the Fremont line once production of the Model S and Model X ends and to start in the second half of 2026. Statements of the kind that more than a thousand units are operating in the factory are in circulation, but there is no support for them.

¹³ Skepticism from the capital markets. Bank of America Global Research, *Global Fund Manager Survey*, February 2026 (conducted 6 to 12 February, published on the 17th). The global questions drew 162 respondents with 440 billion dollars of assets under management between them. The overinvestment response was 35 percent, the highest in a series going back to 2005. As the epicenter of a systemic credit event, private equity and private credit came first at 43 percent, and hyperscaler capex, at 30 percent, came second.

¹⁴ Solar and Chinese installed capacity. Cumulative installed capacity of 0.54 megawatts in 1975 and 1,852,359 megawatts in 2024 is the Our World in Data series (IRENA basis, grid-connected). IEA-PVPS puts the same end-2024 figure at 2.25 terawatts; the basis of aggregation differs. For China, the National Energy Administration's published figure for the end of 2025 is 1.84 billion kilowatts of wind and solar combined, 47.3 percent of total installed capacity (solar 1,200 gigawatts, wind 640 gigawatts).

Population grew exponentially after a constraint came off in the first and second industrial revolutions. As Chapter 1 showed, what drove the surge in English population after 1750 was not falling mortality but rising fertility, and what pushed it up was rising real income.

Even with distribution working and disposable income recovered, it takes time for fertility to move, and more time still for total population. Suppose a baby boom began in the 2030s: that generation enters the labor market in the 2050s. Through those years the largest supply capacity in history sits alongside a population that has not yet recovered. The trap China faces now is one the whole world will pass through once. China goes through it under the constraint, and the world goes through it with the constraint removed.

Of the four head-counts, automation can bypass one

For an economy to work, it needs at least four head-counts: labor supply, demand, risk-takers and taxpayers. Machines can replace only the first.

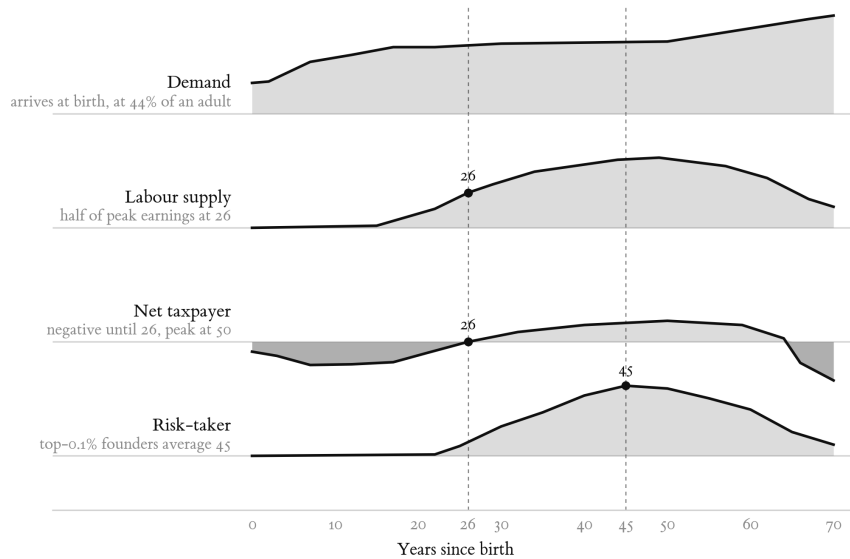
The other three are all head-counts. If nobody wants the thing, then however much you make, it is only inventory. How many people will put their own lives into a business nobody has tried is settled not by productivity but by the size of the population and by whether that population is allowed to fail. You can tax the wealth machines generate, but what carries that taxation, decides its distribution and maintains its institutions is a human community, and the scale of a community is its population. The meaning of a falling population is not a shortage of labor but that these three shrink at once.

The four do not come back at the same speed. The time it takes one human being to join each head-count can be read off the National Transfer Accounts, which tabulate by age what a person takes from the economy and what they pay into it.¹⁵ Demand is same-day: from the moment of birth a person is a consumer. Labor supply and taxpayers take twenty-six years, and for all twenty-six of those years the taxpayer is on the receiving side rather than the paying side. Risk-takers take forty-five. The mean age of founders of firms in the top 0.1 percent by growth is 45.0, and the probability of success goes on rising to nearly 60.¹⁶ The valley of demand is a valley because the four do not come back together. An argument for solving pension sustainability through fertility says nothing about the first quarter of a century, because it is not looking at this structure.

¹⁵ National Transfer Accounts. National Transfer Accounts Project, per capita profiles by age (this thesis uses the United States 2011 profile). <https://www.ntaccounts.org> Consumption per head at ages 0 to 4 is 44 percent of the average for ages 20 to 64, and 95 percent by ages 15 to 19. Labor income per head at ages 20 to 24 is 27 percent of peak; half of peak is reached at 26 and 80 percent at 34. The age at which net payments to the public sector turn positive is 26 in the United States, 22 in Japan, and 23 in Germany and Sweden; before that a person is a net recipient, at -0.14 of adult labor income at ages 0 to 4, -0.33 at 5 to 9 and -0.29 at 15 to 19, with net payment peaking at 45 to 59. This is a cross-section of a single year rather than a cohort followed through time, and the cross-section itself keeps sliding back: in the United States the age of turning net contributor moved from 24 in 1981 to 29 in 2011, and the peak age of labor income from 39 to 49, so for a child born today every delay in the body text is likely an underestimate.

¹⁶ The age of founders. Pierre Azoulay, Benjamin F. Jones, J. Daniel Kim & Javier

Figure 5-4 The four headcounts do not return at the same speed



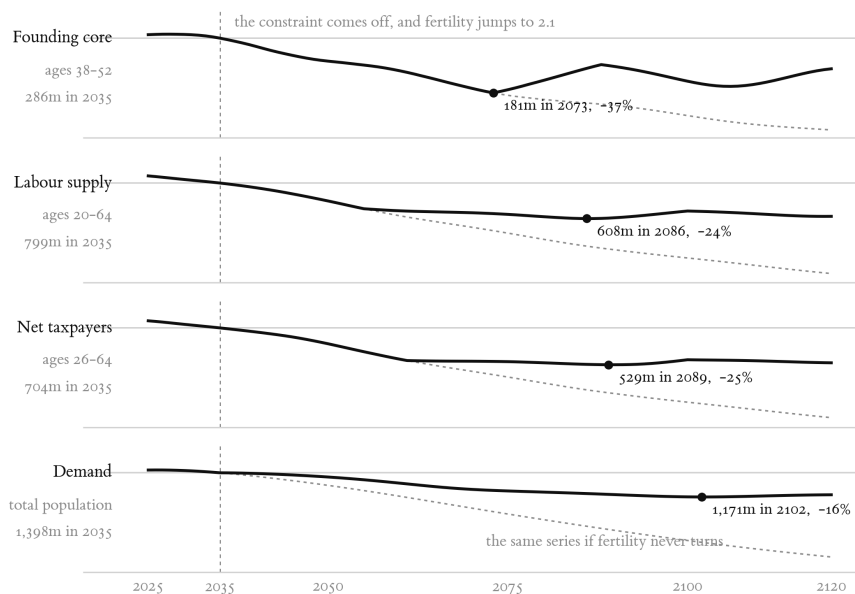
Each band shows the level reached by that age, relative to its own peak. Demand alone starts at the origin; the fiscal band is below the line until 26.
Source: National Transfer Accounts, United States 2011 per-capita age profiles; Azoulay, Jones, Kim & Miranda (2020).

Miranda, “Age and High-Growth Entrepreneurship,” *American Economic Review: Insights* 2(1), 2020, 65–82. Administrative data on 2.7 million founders of United States employer firms. The mean for all founders is 41.9, for firms in the top 0.1 percent by growth 45.0, and for successful exits 46.7. A founder of 50 reaches the top 0.1 percent at 1.8 times the probability of a founder of 30, and the rise continues to nearly 60. The mean for TechCrunch award winners is 31.

And even on the most optimistic assumptions available, the valley of demand lasts half a century. Suppose the constraint comes off completely in 2035, and that at that instant the total fertility rate of high-income countries jumps in a single step from 1.42 to 2.1 and stays there. This cannot happen in reality, but since no faster reversal is possible either, the valley this calculation produces is a lower bound.¹⁷ Even on that assumption, population follows the path in Figure 5-5.

¹⁷ The valley-of-demand calculation, and Figure 5-5. The assumptions of the calculation, and which way each assumption moves the result, are gathered in “Objections and Limits” at the end of this chapter.

Figure 5-5 The valley of demand, on the most optimistic assumption available



High-income countries, each row indexed to its own 2035 level. The total fertility rate is held at 1.42 and made to jump to 2.1 in 2035, with zero migration; no faster reversal is possible, so the valley drawn here is a lower bound. The dotted line is the same series without the reversal. The four floors do not fall in the same year.
Source: Author's cohort component projection; base population from World Bank data by age and sex (2025).

The investment implication comes in two stages

On the most optimistic scenario, the half-century from 2035 to roughly 2085 is the valley of demand. What can be relied on is returns that do not depend on head-count: stakes in automation capital itself, and the infrastructure of energy and computation. Businesses whose returns are proportional to the volume of demand sit in a structural headwind for that half-century. The founding cohort reaches its floor first, in 2073, the labor supply in 2086 and the fiscal base in 2089. Total population, which is what demand is made of, is still falling in the 2100s. Describing the world after the removal of the constraint is complete, and designing a way to survive the half-century that leads there, are two different jobs.

In a world short of demand, the strong position belongs not to the side that can make but to the side that can buy. And through the half-century of the valley of demand, the only places where population is still rising are East, West, Central and North Africa, the Levant, Pakistan, and parts of Central Asia and Central America, which are the lowest-income places in the world today. But demand is population multiplied by purchasing power per head, and what that part of the world holds now is future demand, not present demand. So for the side holding plant it cannot sell, building up the other side's ability to buy is not charity but the work of turning its own plant from a liability back into an asset.

If energy and labor genuinely become abundant and wealth is properly distributed, the household constraint loosens, fertility reverses and population returns. Distribution is a question of demand, of risk-taking and of the fiscal base before it is a question of fairness, and without designing it the valley of demand cannot be crossed.

Where does the wealth the cascade releases fall, if it is left to itself?

Chapter 6. On Capitalism

If what you have read so far has looked like a work of optimism, this chapter should correct that. This thesis is optimistic about supply and pessimistic about distribution. To put it precisely, the prediction here is that the redistribution of wealth will fail once, somewhere in the 2030s. And the first half of that prediction, the leftward turn in politics, has already begun.

6.1 The invention of the task

Our ancestors invented the task as the instrument by which wealth is distributed. Work was broken into discrete tasks, and income has been allocated through the performance of those tasks. The system worked because differentiated skill determined who could perform which task, and the value of the task a person could perform determined their income. Do X and you are paid Y. Reduced to its essentials, distribution in the modern economy runs on that one line.

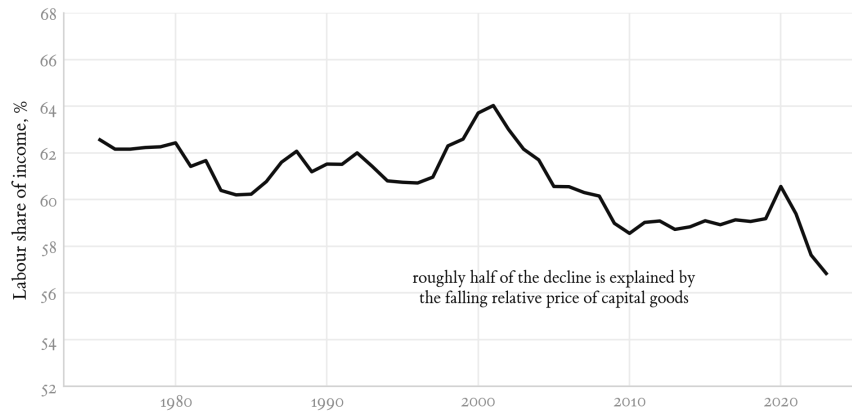
That line began to fail before AI. On Karabarbounis and Neiman's well-known measurement,¹ the labor share, meaning the portion of created value that goes to labor, has fallen significantly since the early 1980s across the majority of countries and industries. About half of that fall is explained by machines and computers getting relatively cheaper and firms replacing people with them. On the theoretical side, Acemoglu and Restrepo's task-based model identified the mechanism.² The displacement effect, by which automation hands a human task over to a machine, always lowers the labor share. The only thing that can offset it is the reinstatement effect, in which new tasks are created for humans and labor comes back. In the United States of recent years, displacement accelerated and reinstatement weakened. On their estimate, 50 to 70 percent of the change in the American wage structure is explained by task displacement.²

The problem is not the displacement of tasks but distribution. The automation of tasks is not new; it has run continuously since the industrial revolution. In that revolution, cheap labor and cheap energy moved workers out of agriculture and into manufacturing. Human skill stayed adaptable and stayed meaningful. What is different in the age of AI is that automation can remove the need for human labor itself, not task by task but across whole categories of work: displacement spreads across every kind of job while there is no sign of new human tasks appearing. The new industrial revolution will be driven by cheap automated labor and cheap fusion energy, and by the time it arrives the machinery of distribution will have gone first.

¹ The labor share and Figure 6-1. Loukas Karabarbounis & Brent Neiman, "The Global Decline of the Labor Share," *Quarterly Journal of Economics* 129(1), 2014, 61–103. About half of the decline is explained by the substitution of capital for labor driven by the falling relative price of investment goods. The American series in Figure 6-1 is Penn World Table 10.01 (FRED series LABSHPUSA156NRUG).

² The task-based model. Daron Acemoglu & Pascual Restrepo, "Automation and New Tasks: How Technology Displaces and Reinstates Labor," *Journal of Economic Perspectives* 33(2), 2019, 3–30. The same authors, "Tasks, Automation, and the Rise in U.S. Wage Inequality," *Econometrica* 90(5), 2022, 1973–2016.

Figure 6-1 The global decline of the labour share



US series shown. Karabarbounis & Neiman document the same downward trend across most countries and industries.
Source: Penn World Table 10.01 (FRED series LABSHPUSA156NRUG); Karabarbounis & Neiman (2014), QJE 129(1).

6.2 The link between task and income is already breaking

This is not a prophecy about the 2030s but an observation of the 2020s.

Brynjolfsson and colleagues at Stanford have dug a decisive fact out of American payroll data.³ In the occupations AI reaches most, employment of young workers aged 22 to 25 fell, relative to other age groups, while employment of experienced workers held steady. The fall came through a halt in hiring rather than through layoffs, and it is concentrated in occupations where AI substitutes for people rather than augmenting them. That gap stands at 19 percent as of June 2026. Anthropic's economic index points the same way.⁴ It is breaking at the entrance, at the young end. As Chapter 4 showed, youth unemployment and the collapse of births are already chained together in China; in the advanced economies, AI is starting the same chain at the young end.

There is evidence pointing the other way. Anthropic's economic index reports that in conversational use, uses that augment people (52 percent) slightly outnumber uses that replace them (45 percent).⁴ Use itself is skewed toward high-income countries and high-end knowledge work, and the speed gains are largest on tasks at the level of a university education. At the same time, where firms call the model from their own programs, coding that began as a collaboration is reported to be moving toward more fully automated procedures.

So the present answer to the question of whether AI destroys high-skill work first or amplifies it first is: both. Which is exactly why the question has to be posed differently. Not how many people AI has made redundant. Whether the entry-level jobs survive, who owns the output, and whether the fruit of productivity accrues to wages or to capital. This thesis takes the third of those, and what settles it is who holds the capital, not the employment statistics.

6.3 The democratic-socialist wave, and its failure

Politics always reacts. The reading has not changed for some time: through the 2030s, a democratic-socialist mood will rise in the major advanced economies. While this chapter was being written, the first half of the prediction began to come

³ The break between task and income. Erik Brynjolfsson, Bharat Chandar & Ruyu Chen, "Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence," Stanford Digital Economy Lab, 2025 (revised 2026). The authors themselves qualify this as correlation rather than causation, and the numbers move as the data is updated. The 13 percent of the August 2025 version is 16 percent on the November version and 19 percent on the June 2026 tracking.

⁴ Augmentation and replacement. Anthropic, *Anthropic Economic Index* (November 2025 data, published 2026). Use is skewed toward high-income countries and high-end knowledge work, and in enterprise API use a tendency is reported for coding that began as collaboration to move toward more fully automated workflows. In conversational use, 52 percent augmentation and 45 percent automation.

true. In November 2025, candidates standing as democratic socialists won the mayoralties of New York and Seattle. Gallup's survey that year put favorable views of capitalism down to 54 percent.⁵ In an era when the young are beginning to be shut out at the entrance, this current is not a coincidence. It is the first political sign of the crisis of distribution.

But the remedies this wave puts forward will fail. And the mechanism of failure is not the old point about high tax rates blunting the will to work but capital flight.

Tax financial income heavily and the agents that generate the wealth leave. Capital moves, and so do the people who hold it. Capital does not vanish at that moment; it moves to a country with better terms. This, exactly, is why institutions, the tax code, the business environment and the depth of the cluster are decisive. When you design distribution, what you are dealing with is not a citizenry but capital that can be redeployed across borders.

The graveyard of the wealth tax

Around 1990, twelve OECD countries levied a tax on net wealth. Three remain: Norway, Spain and Switzerland. France's wealth solidarity tax (ISF), at the end of a long outflow of capital, was cut back in 2018 to the property wealth tax (IFI), which reaches real estate only. Norway in 2022 created a new band for net wealth above 20 million kroner with a top rate of 1.1 percent, and in the two years that followed, departures of the wealthy jumped. Røkke, the country's best-known entrepreneur, left for Lugano.⁶ What remains is one fact. Where capital and people can move, taxing wealth country by country has not historically been sustained.

The strongest answer to this weakness is not a higher rate but international coordination. The report Gabriel Zucman submitted to the G20 in 2024 proposed a coordinated standard imposing on the world's billionaires a minimum effective tax rate equal to 2 percent of holdings, and in the Rio declaration of the same year the G20 stated its intention to cooperate in securing the effective taxation of the ultra-wealthy.⁷ If what defeated single-country taxation was mobility, then building a common floor that covers the destinations as well is a coherent response. But internationalizing the floor still leaves the questions unanswered. Who values the assets? Which country enforces? How long can it be held politically? The wealth tax has not risen from the graveyard but reappeared in another form, as an institutional experiment that crosses borders.

The limits of cash transfers

On universal basic income (UBI) we now have the best experimental data there is. The unconditional cash study run by OpenResearch, funded by Sam Altman, gave a thousand people a thousand dollars a month for three years. The results: hours worked down 1.3 a week, the employment rate down about 2 percent, no measurable improvement in health indicators. The Finnish experiment likewise raised subjective wellbeing while its employment effect was close to zero.⁸ Cash buys autonomy and immediate peace of mind. On its own it changes neither the trajectory of a life nor a person's health. A cheque does not move the structure. Distribution policy arrives at the same conclusion this thesis reached about family policy in Chapter 4.

⁵ The leftward turn in politics. On 4 November 2025, Zohran Mamdani won the New York mayoral election and Katie Wilson the Seattle mayoral election (final count of the New York City Board of Elections, certified 2 December 2025; Seattle was 50.2 percent to 49.5 percent, certified 25 November). On opinion, Jeffrey M. Jones, "Image of Capitalism Slips to 54% in U.S.," Gallup, 8 September 2025 (among all adults, 54 percent favorable to capitalism and 39 percent to socialism).

⁶ The wealth tax and Figure 6-2. OECD, *The Role and Design of Net Wealth Taxes in the OECD*, Tax Policy Studies No. 26, 2018. Sarah Perret, "Why were most wealth taxes abandoned and is this time different?" *Fiscal Studies* 42(3-4), 2021, 539-563 (twelve OECD countries in 1990, three in 2020). Norway's 1.1 percent is the top marginal rate applying to the band above 20 million kroner of net wealth created in 2022, not the rate of the wealth tax as a whole (the band below is 0.95 percent, and 1.0 percent from 2023). Departures among those with net wealth above 100 million kroner rose from an annual average of 8.5 over 2014 to 2021 to 55 in 2022 and 50 in 2023, roughly a sixfold increase. A doubling is closer to the rate of increase in the 10 to 50 million kroner band, at 75 percent. Røkke's departure is from his open letter of 12 September 2022. The net effect on revenue is contested, and wealth tax receipts themselves rose over the same period.

⁷ International coordination. Gabriel Zucman, *A Blueprint for a Coordinated Minimum Effective Taxation Standard for Ultra-High-Net-Worth Individuals*, G20 Report, 25 June 2024. It remains at the stage of a proposal and a diplomatic agreement and has not been realized as an institution.

⁸ Cash transfers. Eva Vivalt et al., "The Employment Effects of a Guaranteed

None of this is an argument against the safety net. A country where several percent of the population is strung out on fentanyl is not something to call acceptable. What I reject is the view that the remedy is socialist policy. It was I who wrote, in an earlier jointly written memo, that we should move from capitalist arrangements to a socialist UBI and eventually to holding the means of production in common. Six months later, what was wrong turns out to have been the method.

6.4 Make every citizen an investor

This thesis puts forward an update to capitalism: a design that takes the death of task-based allocation as given and moves the ground of distribution from tasks to the ownership of capital. As the labor share falls structurally, the only way for households to survive is to move over to the capital side. Make every citizen an investor.

The institutional seedlings are already in the ground. Under a law enacted in the United States in July 2025, the federal government pays 1,000 dollars into an account for every child born between 2025 and 2028, invested in a low-cost fund tracking a broad index of American equities, and the accounts went live on 4 July 2026. Whatever its ideological parentage, every American newborn is now an index investor from the moment of birth. Connecticut's Baby Bonds are a state version of the same design.¹⁰ And the natural experiments have reported. On the Alaska Permanent Fund Dividend, the oil-revenue dividend paid to every resident of the state, Jones and Marinescu found the negative effect on aggregate employment to be zero, with only a slight rise in part-time work of 1.8 points. The dividend did not make people idle. Norway's Government Pension Fund Global is collective capital of roughly 380,000 dollars per citizen, a full-scale precedent for a sovereign fund in which every citizen holds a stake.¹¹

Placed in its historical lineage, what this proposal actually is becomes clearer. Through the Homestead Act the United States moved 270 million acres, about 10 percent of the national territory, into private hands through 1.6 million patents, and made a middle class. Postwar Japan, through land reform, moved 1.74 million hectares to 4.75 million tenant farmers and cut the tenanted share of farmland from 45.9 percent to 9.9 percent, laying the foundation of its postwar middle class.¹² Every broad middle class in history was made by turning citizens into owners. The nineteenth century distributed land. The twentieth distributed home ownership and pensions. What the twenty-first has to distribute is a stake in automated capital. The Homestead Act took land as its object only because land was then the root of production. If the root of production moves to energy, robotics and computation, what has to be distributed moves with it.

As a lineage of institutions it is not a new species either. The institutions humanity has generated for exchange sort into three: reciprocity, meaning mutual giving, redistribution, and market exchange.¹³ A developed economy is not any one of them but a hybrid of all three. Herbert Simon put it that the United States is not a market economy but a market-organization economy; more precisely still, a market-organization-government economy. Pensions, health insurance, income support, everything the national accounts call a transfer payment is redistribution and not market exchange. Distributing a stake to citizens is therefore not an

Income: Experimental Evidence from Two U.S. States," NBER working paper (OpenResearch), 2024. On Finland, Olli Kangas, Signe Jauhiainen, Miska Simanainen & Minna Ylikännö (eds.), *Evaluation of the Finnish Basic Income Experiment*, Ministry of Social Affairs and Health report 2020:15. Recipients worked on average six more days over twelve months (78 days against 72), with no difference in the first year. Life satisfaction, mental strain, self-assessed cognitive functioning, and trust in others and in institutions all improved.

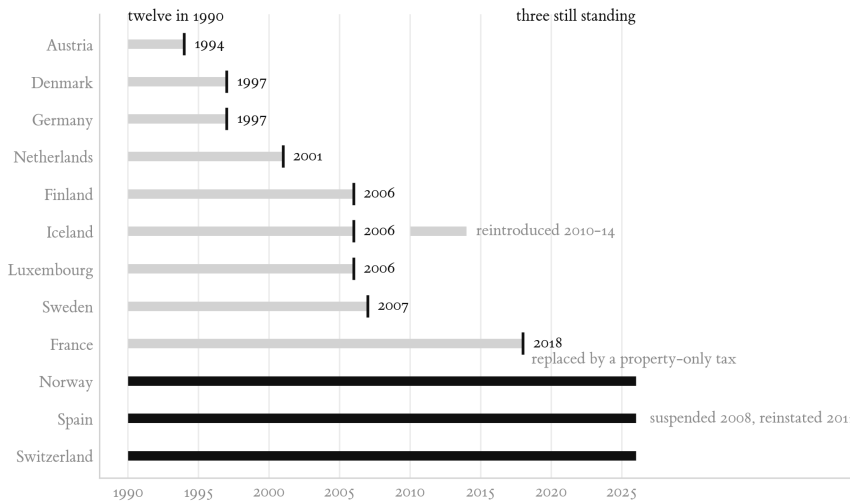
¹⁰ Accounts opened at birth. One Big Beautiful Bill Act (Public Law 119-21) §70204, Internal Revenue Code §530A and §6434. Enacted 4 July 2025. The Treasury contributes 1,000 dollars for each child who is a United States citizen born between 1 January 2025 and 31 December 2028, with additional contributions of up to 5,000 dollars a year permitted (employer contributions up to 2,500 dollars a year). Investment is restricted to funds tracking a broad index of American equities such as the S&P 500, and annual fees and expenses must not exceed 0.1 percent. Contributions to the accounts became possible a year after enactment, on 4 July 2026. Connecticut's Baby Bonds is a scheme under which the state contributes 3,200 dollars for each child born on Medicaid on or after 1 July 2023, usable between the ages of 18 and 30 for buying a home in the state, starting a business, higher education, or retirement saving.

¹¹ The dividend and the national stake. Damon Jones & Ioana Marinescu, "The Labor Market Impacts of Universal and Permanent Cash Transfers: Evidence from the Alaska Permanent Fund," *American Economic Journal: Economic Policy* 14(2), 2022, 315-340. NBIM, *Annual report 2025* (21.286 trillion Norwegian kroner at the end of 2025, roughly 2.1 trillion dollars at 10.1 kroner to the dollar, about 380,000 dollars per citizen).

exception to a market economy. It is an update to one of the three, at the level of what gets distributed.

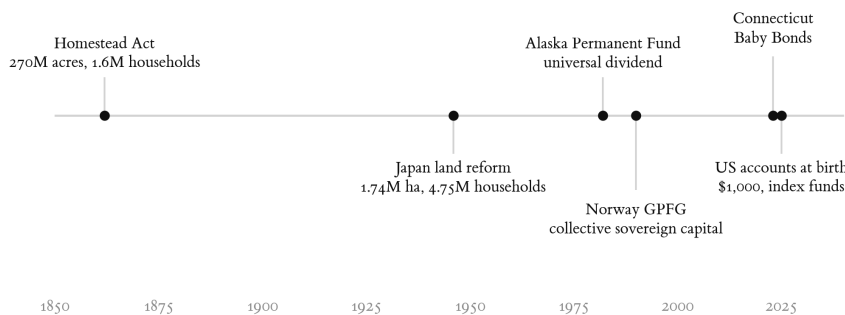
A sufficiently developed capitalism becomes a money tree that distributes wealth to every citizen. Productivity expands autonomously under AI, so the only question left is who holds the stake in the fruit.

Figure 6-2 The graveyard of the wealth tax



Each bar is one country's individual net wealth tax. Black bars are still in force.
Source: Perret (2021), Fiscal Studies; OECD (2018), The Role and Design of Net Wealth Taxes in the OECD.

Figure 6-3 Making citizens owners: a lineage



The nineteenth century distributed land. The twenty-first must distribute shares in automated capital.
Source: NARA/NPS; Dore (1959); Jones & Marinescu (2022); NBIM; US public law (2025).

¹² The lineage of citizen capital and Figure 6-3. On the Homestead Act, the United States National Archives and the National Park Service (more than 1.6 million applications processed by 1934, moving more than 270 million acres, 10 percent of the national territory, into private hands). The 1.6 million is a count of applications processed, not of the households that obtained land. Japan's land reform is from the Ministry of Agriculture, Forestry and Fisheries report to ICARRD (1.74 million hectares, 4.75 million tenant farmers). The fall in the tenanted share from 45.9 percent to 9.9 percent is given by Chiang, Fan & Hsu, IZA DP No. 18095, 2025, citing Kitamura (2022). On the economic assessment of the reform, Ronald P. Dore, *Land Reform in Japan*, Oxford University Press, 1959.

¹³ The three institutions of exchange. Karl Polanyi, "The economy as instituted process," in Polanyi, Arensberg & Pearson (eds.), *Trade and Market in the Early Empires*, Free Press, 1957, pp.243-269. Herbert Simon, *The Sciences of the Artificial*, MIT Press, 1996 (the market-organization economy). The three-way sorting, and the placement of transfer payments within it, follow Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 5.

6.5 The cost of becoming a capitalist is falling

The same technology that cuts the number of workers also lowers the minimum scale at which a business can be owned. Owning a business used to mean hiring people, paying them, and taking responsibility for their lives. That fixed cost set the minimum scale at which a firm is viable, its minimum efficient scale, and narrowed the number of people who could start one. AI and robotics lower the minimum efficient scale itself, because the volume of output one person can direct goes up. The number of business owners can therefore rise.

This does not happen automatically. Between the fall in minimum efficient scale and people actually owning businesses sits another condition: access to capital. Compute and robots both demand an initial investment. This is exactly where the universal stake of the preceding section does its work. Distributing a stake is not only a measure for supporting consumption. It is also the distribution, to every citizen, of the initial capital required to become an owner of a business.

What emerges is a new form: the capitalist combined with the machine. An individual produces, on their own judgment and on automated capital they alone direct, output at a scale that used to require a firm. If the influencer is the modern digital pedlar, what lies further along that lineage is a business owner far more capital-intensive and with far fewer employees. And between such people and the platform companies a strange relationship arises, neither employment nor trade. They rent compute, run their own capital on top of it, and return part of the proceeds to the lender.

From above, stakes are distributed; from below, the threshold for owning a business falls. Of the four head-counts counted in section 5.7 of Chapter 5, it is this movement from below that acts on the risk-taker term. Left alone, though, a new form of the distribution problem arises between those who lend compute and those who borrow it. If the threshold falls without any design for dispersing ownership, what multiplies is not capitalists but business owners without capital, which is to say tenant farmers.

6.6 The AI commonwealth

This economic form can be regarded as a kind of communist structure, because AI functions as a commonwealth, a shared asset, that goes on improving productivity almost automatically. If energy prices fall to the limit and labor productivity rises, inflation falls toward zero and output per head is maximized. At the point where the growth rate of GDP per capita asymptotes to what is effectively infinity, control over the supply of goods and services rests with whoever engineered the energy supply and the AI robotics. This is precisely why the universal stake of section 6.4 has to be institutionalized before that point.

In the final phase, when AI supplies goods and services according to preferences, does a human being need monetary income in order to experience their value? The answer is no. It is scarcity, the finiteness of resources, that creates the need to allocate resources efficiently, and price is the mechanism that rations a scarce resource among competing uses. If the supply of AI services is genuinely unlimited and energy can be assumed to be unconstrained, the notion of price itself disappears.

But the logic of the AI commonwealth is entirely unlike the historical communism of the Soviet Union. The functions of planning and improvement are embedded in the AI system rather than in a party or a bureaucracy. Once energy has stopped being a binding constraint, this system can be almost indifferent to the question of redistribution in the traditional sense. If the constraint is removed completely, the commonwealth can raise the level of value available to every human being, and the reason to fight over who gets more fades. While an energy constraint remains, the system keeps a partial interest in how the surplus is allocated, and has to set

priorities for energy and resources among people, places and tasks. Soviet communism was built on harsh material scarcity, party control and central planning by humans; the AI commonwealth model is built on abundant energy and automated improvement in productivity.

The question that remains is who, or what, generates the energy that runs the AI. The ultimate answer is a human-led expansion of fusion energy. There is a problem of order here. AI may in time improve itself and maintain the energy infrastructure on its own, but humans have first to expand capacity far enough for that autonomy to emerge at all. And so long as humans direct AI and go on drawing benefit from it, control of energy allocation remains the instrument by which AI is overseen.

6.7 The four head-counts come back

Distribution is the only means of restoring three of the four head-counts at once. Of those four, labor supply, demand, risk-takers and taxpayers, automation can route around only labor supply; the other three are population itself. It is this asymmetry that keeps the valley of demand a valley in a phase when supply alone grows exponentially.

The design of distribution argued in this chapter is a device for raising the floor of that valley of demand. If disposable income recovers, the equation of Chapter 4 runs backwards. The expected cost per child, C_{child} , falls from the denominator side through energy and automation, while income from the ownership of capital lifts the numerator. The mechanism of the household Malthus is symmetric: it loosens in the same order in which it tightened. If the birth rate rises, population comes back. And if population comes back, demand comes back, the pool of people who take risks comes back, and the tax base comes back.

Here is the reason this thesis is not pessimistic about the death of task-based distribution. Losing income from tasks and the economy shrinking are not the same thing. The first is a question of where income comes from; the second is a question of head-counts. If the source of income can be rerouted onto the ownership of capital, the head-counts recover. If the rerouting fails, the head-counts disappear first.

And there is a temporal asymmetry in that order. Institutions of distribution can be made by legislation, so once the decision is taken they move within a few years. Population does not. A child born now is a consumer from the first day, but does not reach risk-taking or the tax rolls for another two decades. The design of distribution therefore has to be started twenty years before its effect appears. To institutionalize it in the 2030s is to buy the demand, the challengers and the tax revenue of the 2050s now.

As Chapter 5 showed, the design of distribution is a necessary condition. It is not a sufficient one. Distribution restores three of the head-counts, but the restoration runs on human time. Institutions can solve the problem of income. They cannot solve the problem of time. Which is why there is a need to hurry.

Section 6.6 described a world in which value reaches people without monetary income. So what is value?

Chapter 7. On Value

7.1 Smith's fork

There was never a line between economic activity and everything else.

To create value is to apply one's own competence for the benefit of another.¹ Under that definition, serving coffee in a cafe and a mother loving her child are the same act. The only difference is that one of them carried a price. The side that carried one was called the economy, sorted into industries, counted in statistics and added up into GDP. The side that did not was called nothing at all.

So treat the coffee as a production unit and pay seven dollars for it: that reading is the one to drop. What is happening is that I receive a benefit from what the person behind the counter does, and that person, with the wage they are paid, receives a benefit in the same way from someone else's work in their own hours. Each is putting their competence at the other's disposal, and money is only the vehicle. The story in which a production unit gets replaced by AI comes from mistaking the vehicle for the substance.

What AI accelerates is the part of value creation a machine can take over: making the coffee, moving the freight, drawing the plans, writing the code. The range of human action, that part included, is far wider. So what happens is that the counted part shrinks against the whole and the weight of the part that was never counted rises. Human beings are not switching to something else. What changes is what gets counted.

Adam Smith drew the line. The distinction itself, though, was not his invention. Aristotle separated use-value from exchange-value in the fourth century BCE, and the medieval Schoolmen explicitly held that use-value came first, on the grounds that exchange is motivated by human need.² What Smith did was not to distinguish but to choose.

In Book I, Chapter IV of *The Wealth of Nations*, Smith notes the two senses of the word value.³ Use-value, which expresses the utility of some object, and exchange-value, which expresses the power of purchasing other goods that possession of it conveys. Then he wrote the famous paradox. Nothing is more useful than water, and almost nothing can be bought with it. A diamond has scarcely any use-value, and it will exchange for a vast quantity of other goods. Immediately after setting the paradox down, Smith makes a quiet and decisive choice. He confines everything that follows to exchange-value.

The reason was practical, not metaphysical. His concern was not economics in general but how England, at the start of the industrial revolution, could grow wealthy through international trade. For that purpose exchange-value was the easier of the two to understand and it gave a standardized measure. And it was use-value that he called real value. The real value of a thing is the toil and trouble of acquiring it, because the necessities, conveniences and amusements of life can only be drawn from the labor of others, and the only thing anyone has to offer in return is their own labor. Having defined real value that way, Smith moved to the measurable

¹ Service-dominant logic. Stephen L. Vargo & Robert F. Lusch, "Evolving to a New Dominant Logic for Marketing," *Journal of Marketing* 68(1), 2004, 1–17 (more than 14,000 citations). The axiomatization is the same authors' "Institutions and axioms: an extension and update of service-dominant logic," *Journal of the Academy of Marketing Science* 44(1), 2016, 5–23. The book is Robert F. Lusch & Stephen L. Vargo, *Service-Dominant Logic: Premises, Perspectives, Possibilities*, Cambridge University Press, 2014. At the time of the 2014 book there were four axioms and ten foundational premises; the fifth axiom, which covers institutions and institutional arrangements, was added in the 2016 paper.

² The lineage of use-value and exchange-value. Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 1. Aristotle distinguished the two in the fourth century BCE, and the medieval Schoolmen advocated the primacy of use-value.

³ The paradox of water and diamonds. Adam Smith, *The Wealth of Nations*, 1776, Book I, Ch. IV.

one, nominal value, the price paid in the market. He did so knowing that value is in fact settled inside the experience of the person who receives it.⁴

His distinction between productive and unproductive is misquoted almost as often. Smith did not say that services have no value. The criterion was whether the result could be carried forward and exchanged again. A barber shaves and cuts a person's hair. However valuable that act, the person who received it cannot put it into another exchange. Receive a tangible good and it can be sent on into the next exchange. So the view grew up that a good comes with transportable, transferable value embedded in it.⁴ Smith called productive those activities that contribute to producing and distributing exportable tangible goods, and for the purpose of explaining how to increase national wealth it is the right line. The line was not wrong. Later generations took a distinction drawn for that purpose and began using it as the measure of what people are getting. The paradox of water and diamonds was not solved but shelved, and then the shelving was forgotten.

⁴ Real value and nominal value, and the criterion for productive and unproductive. Smith, *The Wealth of Nations*, pp.30–31 (the definition of real value) and p.314 (the productive-unproductive passage). The reading follows Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 2, which notes that the received view that Smith held services to be worthless is a misquotation, and that the criterion was whether the result could be carried forward and exchanged again.

7.2 Undoing a 250-year detour

There is a body of theory that undoes that detour of two hundred and fifty years: Vargo and Lusch's service-dominant logic (SDL), whose axiomatization comes down to five statements.¹

Axiom 1: applying competence for the benefit of another is the fundamental basis of exchange. Axiom 2: value is always co-created by multiple actors, including the beneficiary. Axiom 3: all social and economic actors are resource integrators, combining what they have on hand to bring value into being. Axiom 4: value is always uniquely and phenomenologically determined by the beneficiary. Axiom 5: value co-creation is coordinated through actor-generated institutions and institutional arrangements.

Goods are a mechanism for delivering competence to the other party, a vehicle for value, and nothing more. Money and organizations stand between the buyer and the seller, and that is what hides the fact that the two are putting competence at each other's disposal. What a human being needs is not a good but value. We have never, from Day 1, been buying coffee. We feel, and pay for, the value that comes through the coffee, which is to say the use-value that arises in context. Every economy was an experience economy from the start.

So SDL has no division into producers and consumers either. The picture in which one party creates value and the other uses it up renders the economy as a cycle of making stuff, selling stuff and destroying stuff. But is education consumed? Are books? Is entertainment? An entrepreneur, a salesperson, a customer, the head of a household, a parent driving the carpool, a grocery shopper and a restaurant patron are not different kinds of actor. Quite often they are, literally, the same person.⁵

And the assessment of value is heterogeneous all the way down. One person finds value in flying from Dubai to New Haven for a dense intellectual conversation. Another finds it in going to a music festival. The moment you grant that value is settled inside the experience of the person receiving it (Axiom 4), both are equally rational. And then Axiom 5. What gathers those heterogeneous assessments and

⁵ Counting units of output; the absence of producers and consumers; the inability of an enterprise to create value on its own. Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 1, the sections "There are no services," "There are no producers and consumers" and "Enterprises cannot independently create value." The logic of goods was carried over to intangible things as the habit of counting units of output:

coordinates them is not the market price alone. It is language, custom, contract and standards, the arrangements actors make for themselves.

To take each person's own assessment as the input and assemble from it the experience that carries value for them, which is to say who they meet, where they go, what they read, when they rest: that is the role of AI in the world to come. AI becomes a delivery system for value.

7.3 The measurable side won

Smith's line was disputed from the moment it was drawn, long before SDL. It survived because only tangible goods, the ones that passed through a market and carried a price, could be counted.

Jean-Baptiste Say held that production is the creation of utility rather than a change in the form of matter. John Stuart Mill went further. No human being can produce one particle of matter, so what labor produces is not objects but utilities. Why then, Mill asked, should all labor that produces utility not be accounted productive ⁶? The one who went furthest was Frédéric Bastiat.

The great economic law is this: services are exchanged for services. It is trivial, very commonplace; it is nonetheless the beginning, the middle, and the end of economic science. Once this axiom is clearly understood, what becomes of such subtle distinctions as use-value and exchange-value, material products and immaterial products, productive classes and unproductive classes?⁶

That was written in 1848. Even so, Say and Mill both took issue with Smith's classification and then built their own arguments on top of it. Only what could be measured came to be called the economy. What is new is not the claim but that the means of measuring use-value directly is coming into reach for the first time.

The line is still drawn the same way. What decides whether an activity counts as goods production or as services is not what the activity consists of but who performs it, as the Economic Classification Policy Committee of the US Bureau of Economic Analysis says in its own words:

providing lodging becomes producing and selling bed-nights, providing transport becomes producing and selling passenger-miles. But what is actually done is everything that keeps a guest comfortable and safe while they are away from home, and that does not cut into units. The original makes the point through a distinction between the words: services is a noun (bed-nights), whereas service is a verb (helping and assisting guests).

⁶ The objections to Smith. Jean-Baptiste Say, *A Treatise on Political Economy*, Wells & Lilly, 1821. John Stuart Mill, *Principles of Political Economy*, J. P. Parker, 1848. Frédéric Bastiat, *Selected Essays on Political Economy*, trans. S. Cain, ed. G. B. de Huszar, D. Van Nordstrand, 1964 (orig. 1848), p.162 (the quoted passage). The placement of all three, and the source text for the quotations, follow Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 2.

One and the same activity, such as painting, may be classified as goods or service production depending purely on the organization of the overall process of production. If the painting is done by employees within the producer unit that makes the good, it will be treated as part of the goods production, whereas if it is done by an outside painting company, it will be classified as an intermediate input of services. Thus, when a service previously performed in a manufacturing establishment is contracted out to a specialized services firm, data will show an increase in services production in the economy even though the total activity of painting may be unchanged.⁷

The statistics produce, by the way they classify, the very facts they are supposed to be observing. The boundary between the acts the economy counts and the acts it does not is settled in the same place: not by what the act consists of, but by whether it can be measured.

7.4 The size of the blind spot

What GDP measures is not value but the monetized part of value. Of the value a person creates in a day, only a part of it carries a price. Full-time work is booked as economic activity in GDP. A conversation with a friend, worth fifty dollars to each of them, is not booked at all. Incorporate, though, and bill the very same conversation as consulting, and it enters GDP.

That reading is borne out by what has actually been measured. Three sets of figures, in order. First, household production. In the satellite accounts of the US Bureau of Economic Analysis, unpaid household production such as cooking, cleaning, childcare and elder care accounts for twenty to twenty-five percent of output once the accounts are extended to take it in. Restated as an addition to published GDP, around thirty percent.⁸ Value placed outside the national accounts for no reason other than that it does not pass through a market. Second, time. Averaging the OECD time use surveys over ages 25 to 64, paid work is only four and a half hours a day, against four hours of unpaid work and just under five hours of leisure.⁹ More than half of human waking time is outside price from the start. Third, free digital goods. In the large-scale choice experiments of Brynjolfsson and colleagues, the median user's compensation for giving up Facebook for a month was about forty-eight dollars.¹⁰ For a good with a price of zero and near-zero contribution to GDP. Their proposed GDP-B is an attempt to measure the benefit users get over and above what they pay, which is to say the consumer surplus that never gets counted. All three ways of measuring point the same direction. The value we create through the tasks we are employed to do is the minority of the value we create in a day.

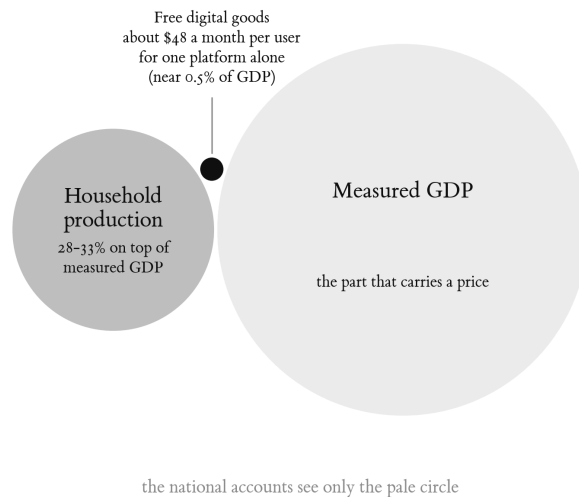
This is not a bug in GDP. It is the specification, Smith's proxy institutionalized over two hundred and fifty years. As the Stiglitz-Sen-Fitoussi Commission formally acknowledged,¹¹ we have measured what was easy to measure and called what we measured value.

⁷ Classification follows who performs the act, not what the act consists of. Bureau of Economic Analysis, Economic Classification Policy Committee, Issues Paper No. 6, "Services Classifications," US Department of Commerce, 1993 (the quoted passage). The same thing happens on the household side: people eat out more, children go to school, cleaning and yard work are hired in, evenings are spent at the cinema or the stadium, and the statistics record the growth of a service economy, but a hundred years ago all of this was done by family or extended family. The acts did not disappear; they crossed the line (same book, chapter 3).

⁸ Household production and Figure 7-1. Benjamin Bridgman, Andrew Craig & Danit Kanal, "Accounting for Household Production in the National Accounts: An Update, 1965–2020," *Survey of Current Business* 102(2), 2022. Non-market household production accounts for 22 percent (2019) to 25 percent (2020) of the satellite account's extended output. That is not a ratio to published GDP; restated as an addition to published GDP it corresponds to 28 to 33 percent. The numbers move between vintages of the accounts, so the vintage and the year both have to be stated. OECD estimates swing widely with the valuation method: for the G7, 14.7 to 25.6 percent on a replacement-cost basis and 43.7 to 68.6 percent on an opportunity-cost basis (van de Ven, Zwijnenburg & De Queljoe, OECD Statistics Working Papers 2018/04).

⁹ Time use and Figure 7-2. OECD Time Use Database, and Veerle Miranda, "Cooking, Caring and Volunteering: Unpaid Work Around the World," OECD Social, Employment and Migration Working Papers No. 116, 2011. On the

Figure 7-1 The economy the accounts do not see



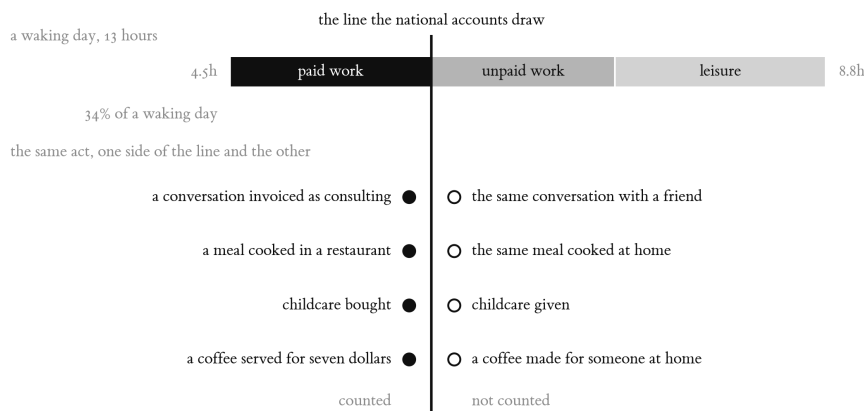
Areas are proportional to the published central estimates. Household production is drawn as an addition on top of measured GDP; the free-goods circle is the single-platform estimate extended to United States users.
Source: BEA household production satellite account; OECD; Brynjolfsson, Collis & Eggers (2019), PNAS 116(15).

OECD average for ages 25 to 64: paid work or study 4.5 hours, unpaid work 4.1 hours, leisure 4.7 hours, and personal care including sleep 10.8 hours. This is an average across all days including weekends, and an average across employed and non-employed people together. That is why paid work looks short. The values move a great deal by age band.

¹⁰ Free digital goods. Erik Brynjolfsson, Avinash Collis & Felix Eggers, "Using massive online choice experiments to measure changes in well-being," *PNAS* 116(15), 2019, 7250–7255.

¹¹ The critique of measurement. Joseph Stiglitz, Amartya Sen & Jean-Paul Fitoussi, *Report by the Commission on the Measurement of Economic Performance and Social Progress*, 2009.

Figure 7-2 The line drawn through a day



The paired acts are the same act in kind; only one of each pair carries a price.
Source: OECD Time Use Database, averages for ages 25 to 64.

7.5 Price's two jobs

Price does two jobs: it rations scarcity and it measures value. If the monetary institution is to retire, both have to go.

Money is not a natural object but an institution. Simmel drew it as a pure form of social relation, Graeber as a technology of credit and accounting that precedes coinage, Harari as a shared intersubjective fiction.¹² SDL's vocabulary is blunter: money is a claim on the future, a right to have some other actor apply their competence on your behalf one day.¹² A balance is not value itself but an inventory of those rights. An institution that loses its function is retired. With two jobs, retirement takes two conditions.

¹² The lineage that sees money as an institution. Georg Simmel, *Philosophie des Geldes*, Duncker & Humblot, 1900 (English translation *The Philosophy of Money*, Routledge & Kegan Paul, 1978). David Graeber, *Debt: The First 5,000 Years*, Melville House, 2011. Yuval Noah Harari, *Sapiens: A*

The first condition has already been observed in information goods. Where the cost of making one more unit, the marginal cost, is close to zero, competition pushes price down to it, which is to say to zero, and price stops working as a signal for allocating resources.¹³ As Chapter 5 showed, the collapse in the cost of energy and automation extends the same mechanics into the material world. When the marginal cost of goods and services goes asymptotically to zero, price as a rationing device loses the scarcity it was rationing.

The second condition is not settled by energy. As section 7.1 showed, Smith granted that use-value is the substance of value and then confined his inquiry to exchange-value. Price as an institution stands on one fact and no other, which is that at the time there was no way to observe use-value. So something observable was put in its place. SDL's fourth axiom states the same fact from the other side. Value is always uniquely and phenomenologically determined by the beneficiary. The original goes on: firms and government agencies may attempt to assess value, but they assess it from their own goals and their own context, and their assessment always diverges from the beneficiary's. Proxies such as economic efficiency and productivity can be measured, but actual value cannot be assessed.¹⁴

That premise, that it cannot be observed, is the one now breaking. Chapter 2 traced the history of rising fidelity in the reproduction of symbols. Language, writing, printing, the telegraph, the telephone, the internet. That sequence has a second effect that has never been counted. Each time fidelity rose, the range of economic activity that could be observed widened. The ledger made transactions observable, the telegraph made distant prices observable, and the smartphone made countless acts that had never carried a price into objects of measurement and billing. Riding in someone else's car, staying in a spare room, five minutes of advice: all of it used to sit outside monetization. What technology changed is not human desire but the resolution of observation.

And this rise has not yet reached its endpoint. As Ochiai has it, the twenty-first-century computer does not stay inside the screen: it rewrites reality itself and dissolves into the environment until it can no longer be seen.¹⁵ The place measurement is taken moves closer to the body as this proceeds. From the screen to the wristwatch, from the wristwatch to the smart contact lens, and then to the brain-machine interface. The point of observation moves from the transaction to the body, and from the body to the nervous system.

7.6 The retirement of money

If the very fact that a beneficiary felt a benefit becomes recordable, the reason for keeping a proxy disappears.

Suppose I talk with someone and get something deep out of it. Under the present institution that value vanishes between the two of us. But if heart rate, facial expression, and in time brain activity, record that I received a benefit, a claim can arise on the side that gave the value. Call it the right to three visits to a café. Without passing through money, the claim goes from the person who received the value to the person who gave it. The exchange SDL set out from the beginning, competence put at one another's disposal, gets implemented that way for the first

Brief History of Humankind, Harvill Secker, 2014. The term service rights, and the definition of money as a right to future service, are from Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 3.

¹³ The mechanics of zero marginal cost. Carl Shapiro & Hal R. Varian, *Information Rules*, Harvard Business School Press, 1998. Jeremy Rifkin, *The Zero Marginal Cost Society*, Palgrave Macmillan, 2014.

¹⁴ The fourth axiom and the impossibility of external assessment. Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 3. The original states that other actors might be able to measure surrogates such as economic efficiency or productivity but they cannot assess actual value.

¹⁵ The century of magic and digital nature. Ochiai called the twentieth century the century of the image and the twenty-first the century of magic: twentieth-century media confined reality inside a two-dimensional screen, while twenty-first-century computers rewrite reality itself. Yoichi Ochiai, *Mahō no seiki* (The Century of Magic), PLANETS, 2015. The same author's *Dejitaru neichā: seitaikei o nasu hanshinka shita keisanki ni yoru wabi to sabi* (Digital Nature: wabi and sabi by way of pantheized computers that constitute an ecosystem), PLANETS, 2018.

time. Take measurement as far as dissolving into the environment and the handover disappears from view as well. With AI assembling what is offered, the receiver's own body recording it, and the claim moving on the spot, there is nowhere left for either person to notice that anything was paid. Put a price on pleasure, though, and someone will manufacture pleasure. What that would actually look like, I cannot yet picture. The same measurement technology runs toward politics as well. In *Democracy for the 22nd Century*, Yusuke Narita set out aggregating preferences from unconscious data, from words and expressions down to the level of hormone secretion, in place of the vote.¹⁶ Democracy replacing the explicit vote with implicit data, and the economy replacing the explicit price with implicit value data, are the same substitution.

And the two costs set out in Chapter 2 both go to zero here. As the convergence cost of arranging matter goes to zero, price loses the thing it rations. As the convergence cost of predictive models goes to zero, price loses its reason to stand proxy. That is the sense in which the retirement of money is not a metaphor. The two do not arrive at the same speed, though. The rationing job goes for reasons of energy, on a scale of decades. The proxy job goes for reasons of measurement and of trust, and that is probably the harder of the two.

The place where money first gives way is not the side that measures value in data but the side of energy. The token that counts AI inference, and the carbon emission allowance. Both were designed as units of account, and both are converging on currency. For an AI company constrained by energy, the token is a unit of revenue and at the same time a unit of the scarce resource that has to be rationed. As Chapter 5 showed, the supply of compute is held down by connection to the power grid for a decade. Through that decade the scarcity is real. Anything issued, exchanged and stored as a unit of a scarce resource is performing the function of currency at that point. A world in which the quantity of energy allowed per head is the unit arrives before a world that measures value in data.

7.7 Machines work, people create

The further mechanization goes, the more conscious people become of their own value creation, and the more willing. Count a day by what was of use to someone rather than by what carried a price, and the size of what you make comes into view. It runs far past the frame of a full-time job, and far past the frame of GDP.

What has kept it out of view is the way employment counts. Hold a full-time post and sell your hours, and only the hours you sold are recorded as value you created. That shape is optimized for an institution that measures exchange-value alone. So is the firm. A firm is a boundary drawn where searching for people, contracting with them and monitoring them costs less inside than it does through the market. Searching, matching, contracting and monitoring are all problems of prediction. In the language of Chapter 2, the further the convergence cost of predictive models, C_{pred} , falls, the thinner the reason to draw the boundary. It thins, though, only where the resources to be integrated are information. The convergence cost of arranging matter, C_{config} , stays bound to physics, so reactors, fabs and grids keep the shape of a firm.

¹⁶ Unconscious data democracy. Yusuke Narita, *22-seiki no minshushugi: senkyo wa arugorizumu ni nari, seijika wa neko ni naru* (Democracy for the 22nd Century: elections become algorithms and politicians become cats), SB Shinsho, 2022.

What appears on the information side instead is co-creation raised for the occasion: a vision is set, the people holding the skills it needs gather, they build for a few months, and they disperse. Equity cannot form around a group that dissolves in a few months, because a share is a right to keep receiving whatever is left over for as long as the business runs. An enterprise cannot create value on its own in any case. What it can do is offer a value proposition; value arises when the beneficiary integrates it with resources of their own.⁵ The claim attaches not to the organization but to the use-value of the thing made. Investing becomes the purchase of a claim on use-value that has not yet arrived, rather than the purchase of a share in a company. So this form does not arrive until measurement is cheap.

What automation takes away is not the work but the compulsion. Grow up inside an arrangement where income is set by the quantity of tasks you get through, and automation can only be spoken of as having one's work taken away. If distribution is guaranteed by a stake in capital rather than by tasks, what automation hands over is not leisure but the state of being able to choose: what to make, for whom, and how far to take it. Step down from nine to five, and from being made to work for a wage, and what a human being does is not rest but the creation of value for someone.

And automation does not reduce what people do. There are two centuries of precedent for that. In 1930 Keynes predicted that within a hundred years a fifteen-hour week would be enough.¹⁷ The prophecy about wealth came true and the one about hours did not. Redesign distribution and it still does not come to fifteen hours. Lusch and Vargo set out how much of innovation has consisted of moving acts people used to perform for themselves onto appliances: knowing when to change gear moved to the automatic transmission, knowing how a word is spelled moved to the spelling checker, judging how much water a load needs moved to a sensor in the washing machine.¹⁸ Each time something moved, what people do did not shrink. The further AI and physical AI run construction on their own, the more work accumulates around them deciding what to build, who to build it for, and whether what got built was of any use. What changes is not the quantity of activity but whether the activity is compelled, and whether it appears in the ledger.

People then move on mission and on curiosity. Each holds a question of their own, as a founder does, and pushes it to the scale of an undertaking. SDL's authors propose the same inversion in their final chapter: entrepreneurial activity is the basic form of value creation, and management inside an established market is the special case.¹⁸

The ratio of section 7.4, monetized value against unmonetized, turns over here. The side that was counted shrinks against the whole, and the side that was never counted gets counted for the first time, as something the person made. What arrives is not idleness but diligence, standing for the first time outside compulsion.

¹⁷ Fifteen hours a week. John Maynard Keynes, "Economic Possibilities for our Grandchildren," 1930 (collected in *Essays in Persuasion*, 1931).

¹⁸ Innovation as the transfer of self-service acts, and the order of entrepreneurial activity and management. Lusch & Vargo, *Service-Dominant Logic*, 2014, chapter 6 (Exhibit 6.2 lists the automatic transmission, presence-sensing lighting, rain-sensing wipers, spelling correction and water-level sensors among others) and chapter 10, "Entrepreneurship versus management."

Chapter 8. On Clusters

8.1 The cluster, not the nation

The body of the world economy is not the nation but the cluster. A cluster is a bundle of metropolitan areas made continuous by economic activity, by commuting and trade and the movement of people, and it does not follow administrative boundaries.

Cut the world's land by the continuity of its lights at night and the lumps of the economy come out on their own. The 40 megaregions Florida and his colleagues extracted this way hold 17.7 percent of the world's population, and out of them come 66 percent of world output and 85.6 percent of its patents.¹

Agglomeration has not come apart, even as the cost of communication approaches zero and economic life moves into virtual space. What draws people is density.² Working from home in the 2020s was a natural experiment on this structure. Of those who left the downtowns of the twelve largest American metropolitan areas, 58 percent moved to suburbs of the same metropolitan area and 4 percent moved to rural areas.³ Embodied discovery, and the cocreation that happens between adjacent cities, have not yet been reproduced in virtual space.

The clusters this thesis takes up are nine. The San Francisco Bay Area, Texas, BosWash, the UAE, Northeast Asia (Japan-Korea-Taiwan), Loxbridge, the Yangtze Delta, the Pearl River Delta (GBA), and Bengaluru-Hyderabad. Drawn at their most generous, these nine come to about a quarter of world GDP.⁴

And seen in this unit, the argument about the social contract changes shape too. Taxation, elections and distribution all look as though they close at the border. But the places where value is made are far finer than the border, and the range over which talent and capital move is far coarser than it.

8.2 The center of gravity has gone to sea

As of the twenty-first century, it is not that clusters are arranged along the rim of the Indo-Pacific but that the cities along that rim are the ones that became clusters.

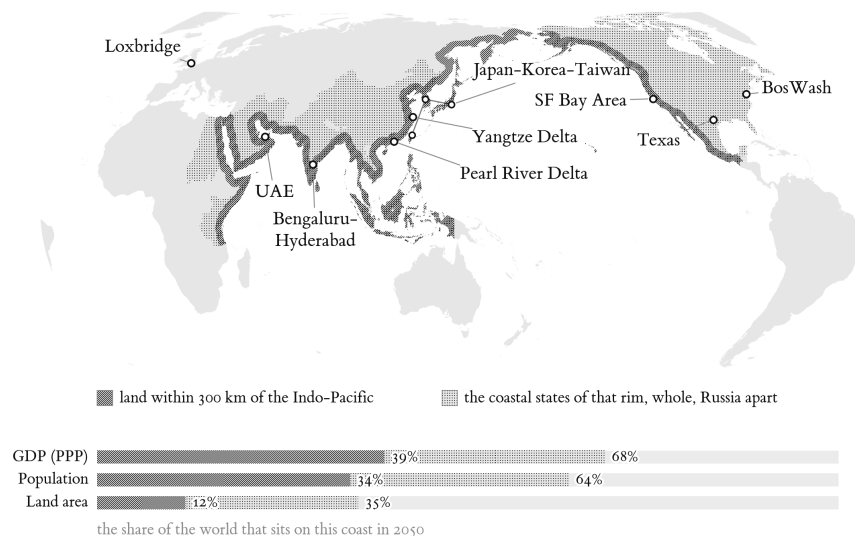
More than eighty percent of world merchandise trade by volume, and more than seventy percent by value, moves by sea.⁵ Access to the world market, to trade, is what the ocean supplies. And the physical substrate this thesis treats, the transformers and turbines and reactors and building materials, is all on the heavy side. Distance from a port is, directly, distance to construction. Against that, the band within 100 kilometers of the coast and below 100 meters of elevation is 9 percent of the world's land. On it live 27 percent of the world's people, and out of it comes 42 percent of world GDP.⁶

¹ Megaregions. Richard Florida, Tim Gulden & Charlotta Mellander, "The rise of the mega-region," *Cambridge Journal of Regions, Economy and Society* 1(3), 2008, 459–476. The figures are from the authors' working-paper version of October 2007. Output is estimated from light intensity and then renormalized to national GDP, so this is a spatial redistribution of world GDP rather than an independent measurement of output. The data are from around 2000 and contain almost none of the rise of China and India.

² Density and learning. Jorge De la Roca & Diego Puga, "Learning by Working in Big Cities," *Review of Economic Studies* 84(1), 2017, 106–142: a year worked in one of the top five metropolitan areas raises wages 3.1 percent more than a year worked elsewhere, and the difference stays with the worker wherever they go next. On the contribution of sorting to the urban wage premium: Pierre-Philippe Combes, Gilles Duranton & Laurent Gobillon, "Spatial wage disparities: Sorting matters!," *Journal of Urban Economics* 63(2), 2008, 723–742 (workers' skills explain 40 to 50 percent of spatial wage disparities). The productivity elasticity of density itself is around 0.04, that is about 2.8 percent for a doubling (Gilles Duranton & Diego Puga, "The Economics of Urban Density," *Journal of Economic Perspectives* 34(3), 2020, 3–26).

³ Working from home, and concentration. The concentration of tech employment in the top eight metropolitan areas rose from 35.2 percent in 2010 to 38.4 percent in 2020. Nicholas Bloom, Arjun Ramani & Joel Alcedo, "How working from home reshapes cities," *PNAS*, 2024 (the remaining destinations are another of the top twelve at 9 percent and smaller cities at 29 percent). The tech employment share is from Mark Muro & Yang You, "Superstars, rising stars, and the rest," Brookings Institution, March 2022. The hollowing of downtowns is real: the top twelve metropolitan areas lost

Figure 8-1 The Indo-Pacific coast, where most of the nine clusters sit



Two readings of the same rim. The black diagonals are land within 300 km of the Indo-Pacific, the Pacific and Indian oceans with the Gulf and the Red Sea, from the Bering Strait down to Mexico's southern border on one side and Tanzania's on the other. Islands go with the state that holds them; Australia, New Zealand, Papua New Guinea and Madagascar are out. The mesh is the coastal states of that rim taken whole, Russia apart because its body lies inland. The three clusters outside the band are coastal too, facing the Atlantic. Each bar shows the coastal states' total, and inside it the band. Equal Earth projection: areas on the page are proportional to areas on the globe, and Antarctica is off the frame but inside the land figure. Coastline SRTM15+ v2.7 (GMT earth_relief 02m), borders Natural Earth 10m, people and output on SSP2 from Murakami and Yamagata (2019), split inside each country on the 1990–2015 grids of Kumm, Taka and Guillaume (2018).

And over the past fifty years the center of gravity of that band has moved to the Indo-Pacific. Land within 300 kilometers of the Indo-Pacific alone, from the Bering Strait round to East Africa, is 12 percent of the world's land. By 2050, 34 percent of the world's people live on it, and 39 percent of world output comes off it.⁶ Take the littoral states of the Indo-Pacific together and they hold 69 percent of the world's people and produce 66 percent of world GDP at purchasing power parity. In 1980 the figure was 53 percent.⁷ On the other side of the same structure is a disadvantage. The 32 landlocked developing countries hold 7 percent of the world's population and 1.2 percent of its exports.⁸

Diamond argued that crops and domesticated animals spread only along latitudes, so that on a continent laid out east to west technique travelled fast.⁹ The shape of the earth dealt the initial conditions, with no reference to the effort of the people living on it. The present-day equivalent is the fact that heavy things go by sea. That the merchant cities strung along the rim of the largest ocean on earth grew giant is not a difference of effort; they were dealt that condition first.

The relative retreat of Europe, Russia and the Eurasian interior, the running out of breath in Latin America, and the slowness of the interior of Africa to take off, are one structure appearing in different places. In the language of Chapter 1, what the two shores of the Atlantic hold is inventory; where building is happening is the rim of the Indo-Pacific.

8.3 Migration is substantially urbanization across a border

People gather into cities in search of a better job opportunity, better wages, a better standard of living, and the chance of a useful accident.¹⁰ When that is done across a border we call it migration, but in substance it is urbanization.¹¹ Inside this

roughly 8 percent of their downtown residents.

⁴ The nine clusters. The total is this thesis's own aggregation; drawn at their most generous, taking whole states and regions, they come to 28.9 trillion dollars in 2024, or 25.9 percent of world GDP, and 22.0 percent redrawn at metropolitan grain. Sources: US Bureau of Economic Analysis GDP by county, IMF *World Economic Outlook*, local statistics for the Yangtze and Pearl River Deltas, PRS Legislative Research state GSDP, and ONS regional GDP for the United Kingdom. The total moves by ten to twenty percent with the drawing of the boundaries.

⁵ Heavy things go by sea. UNCTAD, *Review of Maritime Transport 2017*, UNCTAD/RMT/2017, p. x. The 2024 and 2025 editions state no such share, so the 2017 edition is cited. For air: IATA, *Air Cargo, Trade and Economic Growth in 2025*. The sea share by volume and the air share by weight belong to different accounting frameworks and do not partition world trade between them; land transport and pipelines are in neither.

⁶ Coastal population and GDP, and Figure 8-1. More than half of the cities above ten million people sit inside that band. M. Kumm et al., "Over the hills and further away from coast," *Environmental Research Letters* 11(3), 2016, 034010 (PPP, 2010). On tighter bands: A. G. Cosby et al., "Accelerating growth of human coastal populations at the global and continent levels: 2000–2018," *Scientific Reports* 14, 22489, 2024 (14.6 percent within 10 kilometers). The level is not disputed; the direction of travel is, and that argument is set out at the end of this chapter. The Indo-Pacific figures and Figure 8-1 are this thesis's own computation on a 2 arc-minute grid: land within 300 kilometers of the Indo-Pacific, with no elevation cut, each state's islands counted with it and Australia, New Zealand, Papua New Guinea and Madagascar left out, 17.79 million square kilometers, against all land including Antarctica, 149.9 million square kilometers. The 2050 shares take the part of each country's people and output that falls inside the band and carry it forward on the SSP2 country projections, letting that part move on its own 1990 to 2015 trend; frozen at its 2015 value it gives 34 and 38 percent instead. On the same basis the band held 36 percent of the world's people and produced 35 percent of world output in 2020. The second reading in the figure adds the coastal states of that rim whole, Russia apart, which carries

structure, the motive of a young Cameroonian heading for Paris and the motive of a young Russian from the provinces heading for Moscow are the same motive.

The pull points toward the city partly because the city takes less physical quantity per head. Double a city's population and output goes up 2.2 times while the electrical cable needed goes up only 1.8 times.¹² The physical quantity needed to keep one person falls as density rises.

For several hundred years the world's population has been structurally headed toward the city, and agglomeration into clusters is what is to be expected from here. Border control can slow the speed of it; it cannot structurally dam the flow.¹³

The pull has a clock on the supply side. Fertility in the countryside runs up to twice the urban rate,¹⁴ and that difference is what sends people to the city. But a countryside that keeps sending loses its women aged 20 to 39. What is moving is the reproductive base itself, so the countryside ages, births fall, and it empties out.¹⁵

What settles the population of a cluster is not the fertility rate of its country but the inflow from the countryside and from abroad. That a country's population is falling and that the cluster inside it is shrinking are two different propositions.

After several centuries of this, that the world's people might sit inside a few thousand megaregions would not be surprising.

8.4 Not countries but the terms of the equation

What settles a cluster's standing is not how many terms it has assembled for itself but whether it can reach the ones it lacks.

Civilizational competitiveness $C \approx$ Institutions $I \times$ Effective energy abundance $E \times$ Labor input L

$L \approx$ Deep tech $T \times$ Large capital K

I, institutions. On the side that calls in capital: ownership, investor protection, the tax regime and the speed of permitting, which comes down to whether the return on capital placed there can be believed to be preserved for the life of the asset. On the side that calls in people: that failure is allowed, social inclusiveness, and whether residency is detached from the employer.

E, effective energy abundance. How far a place has come toward power that can be drawn without watching the meter. As Chapter 5 showed, what binds a decision is not the level of the price but the sense that the supply is finite.

T₁, deep tech in bits. Foundation models and deep learning research, the proprietary data they are trained on, and the technology that settles physical-side production while holding no physical form of its own: the design tools and the design IP for chips. This is the chokepoint of direction 2 in Chapter 2.

T₂, deep tech in the physical. Materials, semiconductor production equipment, lithography, leading-edge logic, high-bandwidth memory, industrial robots, precision motors. This is the chokepoint of direction 1.

the three figures to 35, 64 and 68 percent in 2050. Coastline from SRTM15+ v2.7 by way of the GMT earth_relief 02m grid; borders from Natural Earth 10m; the split within each country, and its trend, from M. Kumm, M. Taka and J. H. A. Guillaume, "Gridded global datasets for Gross Domestic Product and Human Development Index over 1990–2015," *Scientific Data* 5, 180004, 2018 (PPP); the projections from D. Murakami and Y. Yamagata, "Estimation of Gridded Population and GDP Scenarios with Spatially Explicit Statistical Downscaling," *Sustainability* 11(7), 2106, 2019 (SSP2, PPP at 2005 prices).

⁷ The center of gravity of the Indo-Pacific. What this thesis includes in the Indo-Pacific is North America (the United States, Canada and Mexico), Northeast Asia, Southeast Asia, South Asia, Oceania, the Gulf and the Arabian Sea, and East Africa and the Red Sea, 67 countries and territories in all. Russia, South America and southern Africa are not included. North America is counted whole rather than by its Pacific coast. The totals are this thesis's own aggregation from the country series of the IMF *World Economic Outlook* database, October 2025 (NGDPD, PPPSH, LP); at market exchange rates the share is 65.5 percent. The 1980 figure of 52.6 percent rests on the 49 of those 67 with data for that year. The largest single entry is the United States, and without it the figure falls to 51.7 percent; when government strategy documents write that the Indo-Pacific is 60 percent of world GDP, that figure also counts the United States inside the region. The center of gravity is from Danny Quah, "The Global Economy's Shifting Centre of Gravity," *Global Policy* 2(1), 2011, 3–9. The path after 2008 is an extrapolation of exponential trends fitted location by location, not a forecasting model.

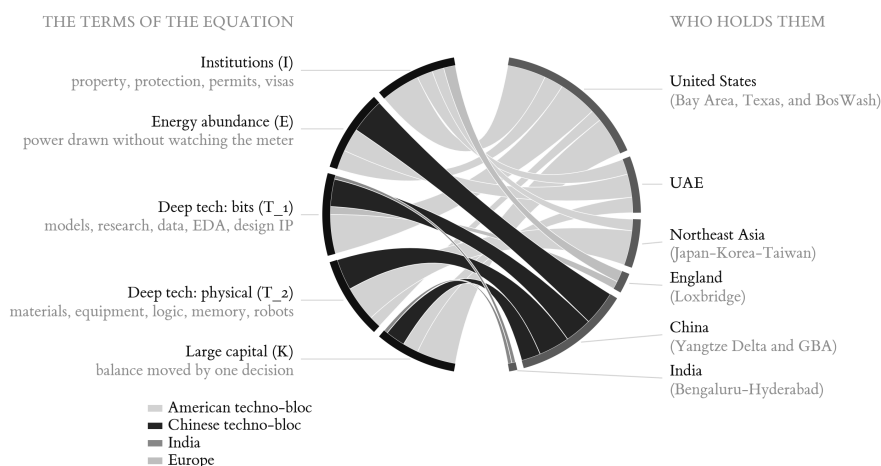
⁸ The penalty of the interior. Their development level is about 20 percent lower, and about half of that gap arises through trade. UN-OHRLLS, *The Development Economics of Landlockedness*, 2013 (also 43 days to export, more than twice a coastal developing country, and transport costs up to 45 percent higher). The population and export shares are from UNCTAD's 2025 statements. On growth: Ramesh C. Paudel, "Economic Growth in Developing Countries: Is Landlockedness Destiny?," ANU Working Papers in Trade and Development 2014/01 (2.1 to 3.2 percentage points a year lower). That paper's own conclusion is that the disadvantage can be mitigated by governance and trade openness.

K, large capital. A balance that can be moved, on a single decision, at a CAPEX of a different order.

Of the Chapter 1 equation, the labor input splits into deep tech and large capital, and deep tech splits into the two directions of Chapter 2. Labor input does, however, still have a third way of being filled: labor cheap enough that neither deep tech nor large capital need be bought. That way of filling it, by hands, still works widely on the present earth, and it tells in the construction speed of the Gulf states and in the manufacturing cost of China's two delta regions. But as 2.7 showed, the point where the price of the robot crosses under the price of those hands is already in sight. Cheap labor is an input that is working now, not an asset for the future.

Through whom can you reach for the term you lack? This chapter calls the bundle of those reaching relations a techno-bloc. There are at present only two techno-blocs that actually have the terms assembled. The American techno-bloc assembles them by division of labor; the Chinese techno-bloc tries to assemble them by vertical integration. The third candidate is India, which so far belongs to neither way of assembling them.

Figure 8-2 The terms of the equation, and where each is held



Schematic. The nine clusters are grouped here by country and region. Each term is taken as a whole and divided into the shares the author judges each place to carry; the widths are ordinal judgements, not measured shares. No place carries every term at world level: over eighty per cent of T₂ lies outside the United States, and China scores lower on institutions than on its other terms. Source: Original to this book.

⁹ The axis of a continent. Jared Diamond, *Guns, Germs, and Steel*, W. W. Norton, 1997, ch.10. For an empirical test of the axis hypothesis: David D. Laitin, Joachim Moortgat & Amanda Lea Robinson, "Geographic axes and the persistence of cultural diversity," *PNAS* 109(26), 2012, 10263–10268 (testing the implication that linguistic diversity persists more in areas oriented north to south, and finding support).

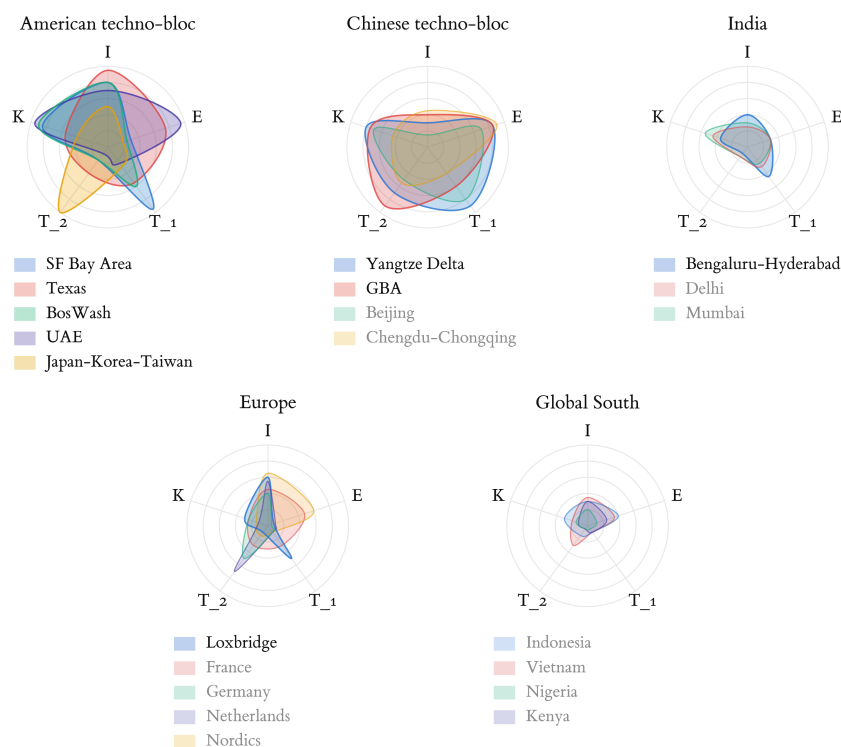
¹⁰ The place premium. Michael A. Clemens, Claudio E. Montenegro & Lant Pritchett, "The Place Premium: Bounding the Price Equivalent of Migration Barriers," *Review of Economics and Statistics* 101(2), 2019, 201–213. The ratios are estimated from wage data on two million workers in 43 countries for a male worker aged 35 to 39. Bounding selection on unobservables still leaves 3.95 times at the median. The wage data are centered on 1999, and these are cross-sectional comparisons of different people in different places rather than the realized gain of anyone who moved.

¹¹ The mobility transition. Wilbur Zelinsky, "The Hypothesis of the Mobility Transition," *Geographical Review* 61(2), 1971, 219–249. The pairing is from Table I, which lists the features of phase two.

¹² Urban scaling. Luis M. A. Bettencourt et al., "Growth, innovation, scaling, and the pace of life in cities," *PNAS* 104(17), 2007, 7301–7306, Table 1. The exponent on output is 1.15 for China, 1.26 for the EU and 1.13 for Germany; patents 1.27, total wages 1.12; length of electrical cable 0.87, road surface 0.83, gasoline stations 0.77. The 2.2 and 1.8 in the text are 1.15 and 0.87 applied to a doubling of population. The framework of expected income is John R. Harris & Michael P. Todaro, "Migration, Unemployment and Development: A Two-Sector Analysis," *American Economic Review* 60(1), 1970, 126–142.

¹³ What border enforcement does. Douglas S. Massey, Karen A. Pren & Jorge Durand, "Why Border Enforcement Backfired," *American Journal of Sociology* 121(5), 2016, 1557–1600. On visas: Mathias Czaika & Hein de Haas, "The Effect of Visas on Migration Processes," *International Migration Review* 51(4), 2017, 893–926 (38 destination countries and about 190 origin countries, 1973 to 2012; inflows fall and outflows fall further, so the net reduction is smaller than the reduction in inflows). On walls: Treb Allen, Cauê Dobbin & Melanie Morten, "Border Walls," NBER Working Paper

Figure 8-3 The five terms, region by region



Schematic. Each plate shows one region at the five terms of the Chapter 1 equation: institutions I, energy abundance E, deep tech in bits T₁, deep tech in the physical T₂, and large capital K. The scale is absolute and the same on every plate, so a small shape is a small shape, and the levels are consistent with the world shares drawn in figure 8-2. Names set in black are the nine clusters of this chapter; names in grey are placed for reference. Levels are ordinal judgements from the coefficients set out in this chapter, not measured values.
Source: Original to this book.

25267 (548 miles, 2.3 billion dollars, roughly 310,000 movements deterred).

¹⁴ Rural and urban fertility. Mathias Lerch, "Regional variations in the rural-urban fertility gradient in the global South," *PLOS ONE* 14(7), 2019, e0219624 (60 countries, 278 surveys; the rural total fertility rate reaches up to 2.0 times the urban rate in sub-Saharan Africa, about 1.5 times in Latin America and about 1.25 times in Asia).

¹⁵ Municipalities at risk of disappearing. Population Strategy Council, *Analysis Report on the Sustainability of Local Governments*, 24 April 2024, on the class where women aged 20 to 39 decline by 50 percent or more between 2020 and 2050. For a criticism of the measure: Keisuke Kondo, "Re-examining municipalities at risk of disappearing," RIETI Special Report, 7 May 2024 (over 1980 to 2020, 879 municipalities have already been through the same 50 percent decline). The classification should be read as a measure of the strength of the gradient, not as a forecast.

8.5 The American techno-bloc

The essence of the American techno-bloc is not that one country holds every term but that a division of labor holds. The chain does not function if any one link is pulled out. Not one of the five clusters is even across the terms: the Bay Area runs out along deep tech in bits, Japan-Korea-Taiwan along deep tech in the physical, and Texas, BosWash and the UAE along institutions, energy and capital. Yet laid over one another the five very nearly fill the pentagon. Not one of them is complete, and put together the gaps disappear.

The United States

The terms are most nearly complete here. It holds the world's strongest foundation models and computing resources and enormous capital, and Texas supplies abundant energy. On the physical side it holds a corner of the production equipment while falling short in manufacturing capacity of its own for leading-edge technology; what is missing arrives through alliance and capital. And on institutions it is strong on both faces of the pull, the one that calls in capital and the one that calls in people.

What holds implementation back is the speed at which gigawatts are connected to the grid and buildings are put up. This one thing depends on where you are, so computing resources and capital have to relocate to sites with room on the grid,

land that is easy to acquire and fast permitting. Inside the United States that is Texas. Outside it, it is the Gulf Cooperation Council (GCC) states.

The GCC

Its strength is capital, the construction capability underwritten by abundant energy and by location, and being a place where new rules and new businesses can be tried first: the one place where the three conditions of Chapter 1 are assembled in the present era. What the GCC carries is not a technological term but the flow of capital between terms, a node that belongs to every pole, meaning every camp, while trading with all of them.

There is one more input that does not show up well in the figures: the labor that goes into construction. The labor input of the Gulf states is still carried by large-scale, low-cost unskilled migration. Building costs there are low because people hired at the wage level of the sending country are building inside the price level of the receiving one. Neither Europe nor Northeast Asia has labor at that price, and neither can match the Gulf on construction capability.

In the Gulf states the soil for growing at home the deep tech that would automate cheap labor has still not formed. Both AI and the physical substrate are therefore handled by buying them with capital, and buying it keeps the labor input close to full. It is also the shortest-lived part of the Gulf states' advantage.

The UAE recognizes this weakness, and to create an ecosystem that produces deep tech it is widening the institutions that call in capital and people. The golden visa introduced in 2019 detaches residency from the employer and grants investors, researchers and professionals a self-contained ten-year right of stay; it is an institutional change aimed at the revolving-door structure itself.¹⁶ But a few years from enactment, neither the median length of stay nor whether a second generation stays on can yet be observed.

What counts in a place where new technology gets made is a particular stratum: frontier researchers, zero-to-one founders, deep engineers. What moves this stratum is less a tax rate than the density of the cluster itself: the probability of running into somebody smarter than you every week by accident, the confidence that if you fail the next job is within walking distance, the number of peers you can argue with at midnight. This is why the San Francisco Bay Area does not let go of the top tier of talent despite high taxes and a high cost of living. The foreign population of the Gulf states is still, in essence, a revolving door, and whether that stratum settles is what decides the coefficient on talent.

Japan-Korea-Taiwan

Japan carries chemicals and materials, machinery, industrial robots, semiconductor production equipment, automobiles, and the advanced industries that hang off them. Korea carries semiconductors and electronics, and shipbuilding. Taiwan carries the AI semiconductor itself. Only the three together constitute the physical-side supply chain of physical AI. The role of Japan-Korea-Taiwan is not the brain but the body, which is to say the very capacity to implement direction 1. But they

¹⁶ The UAE golden visa. Introduced in 2019. The indicators that measure its effect, meaning the median length of stay and the settlement rate of a second generation, cannot be observed a few years from enactment. This thesis reserves judgment on this point.

hold no large foundation model of their own. Their fate is therefore settled not by their own demand but by the demand of the stack they are built into.

Holding the upstream of that leading-edge physical supply chain is also a bargaining card, and as parties for the American techno-bloc to hand computing resources or orbital infrastructure to, Northeast Asia is at the top of the list. On the capital-calling side, institutions are high enough as well. Against that, the region is structurally set to run out of talent over the long run as its population falls, while the social inclusiveness on the side that calls in people is extremely low. And effective energy abundance is structurally low, dependent on imports, with a tight grid. For Northeast Asia, therefore, the implementation of next-generation energy, fusion among it, and the admission of high-skilled talent are not options but preconditions. If they go unfilled, the region remains one that holds the most complex and deepest technological supply chain in the world and cannot run it at home.

8.6 The Chinese techno-bloc

The Chinese techno-bloc holds an independent AI stack, assembled at home from the chip up to the foundation model. Its essence is not a division of labor but vertical integration.

What vertical integration assembles, and what it blocks

The advantage is speed of decision and resilience against a cut in supply; the defect is that if any one of the terms is structurally low, it becomes the bottleneck for the whole. What is actually behind is leading-edge logic alone, and it is behind because of sanctions. But the lag is not as large in computing capacity actually running as the generation gap in lithography would suggest. Separation of the supply chain from the American techno-bloc will proceed further, and technically the aim will be to complete a domestic stack inside a closed ecosystem.

Labor input has historically been carried by large volumes of hands. That is being replaced, fast, by the expansion of physical AI. Effective energy abundance, too, has been built at the largest scale in the world.

Against that, institutions are blocked on both faces, capital and people. On capital, direct investment coming in from abroad has turned down amid economic-security concerns, and local government finances are strained by shrinking land revenue.¹⁷ The side that calls in people is thinner still. On top of the difficulty of language and social integration, residency is not detached from the employer and the door to permanent settlement is effectively closed. There are constraints on taking assets out. No pronounced movement of the world's high-skilled talent into China is observed at present. And the wealth produced reaches households only thinly. With distribution thin, household income does not grow, labor supply exceeds demand, competition intensifies, youth unemployment stays high, births fall, demand shrinks, and oversupply deepens further.

¹⁷ Inward direct investment and local government finance. Actual inward direct investment, on the Ministry of Commerce's compilation, fell 27.1 percent year on year in 2024 to 826.2 billion yuan. Direct investment liabilities on a balance-of-payments basis turned to a full-year net outflow in 2023 for the first time since the series began, on the State Administration of Foreign Exchange's figures. Local government revenue from land-use-right transfers has shrunk by roughly 40 percent from its 2021 peak, on the Ministry of Finance's compilation. All three point the same way, but the cause cannot be attributed to economic security alone; the contribution

Not a country but a continent

The body of it is the coast. The Yangtze Delta has a population of about 240 million, more than half the European Union, and a GDP of about 4.65 trillion dollars, larger than Japan as a whole. The Pearl River Delta has a population of about 86 million and a GDP of about 2.1 trillion dollars. These two regions alone account for about 38 percent of China's GDP.⁴

The Yangtze Delta assembles, in balance, the AI model ecosystems of Hangzhou and Hefei, the machine manufacturing of Suzhou, and the capital and finance of Shanghai. The Pearl River Delta, on the axis of Shenzhen, is strong in the hardware and manufacture of physical AI. China's computation, manufacture and capital are gathered in these two.

Beijing holds an AI model ecosystem of its own, built around the link between industry and the universities, but its strength lies less in producing a pole than in turning out the people and administering the pole politically. What the two deltas want does not necessarily coincide with what Beijing wants. The regulatory tightening on Alibaba that began in 2020 is one instance; for a hardware engineer in Shenzhen or an AI engineer in Hangzhou, developing inside a stack closed off from the world is a condition set by policy, not one they chose. That fact puts a gradient of institutions across the clusters: the less directly central regulation reaches, the higher the score, if only slightly. Some of the people who graduate from Beijing's universities move to Hangzhou or Shenzhen, and some of those tired of the competition go to Chengdu.

Physical AI does not solve the contraction of demand

An oversupplied China has only three roads left: expand domestic demand, expand exports, or cut supply back. Japan took thirty years over that structural adjustment, and the exit came not from policy but from the side of population structure, in the form of a labor shortage. China is attempting to pass through these thirty years on a far lower GDP per capita than Japan's and on a far thinner accumulation. And labor, too, is in oversupply.

As derived in 2.7, demand is a function of population. Weighting toward supply becomes a self-reinforcing trap. Oversupply produces unemployment, unemployment kills births, falling births kill future demand, and vanished demand produces further oversupply.

Over the coming decades what happens in China is a selection: people vanish from the countryside, gather in the cities, and in the process winners and losers are settled fairly plainly region by region.

The three northeastern provinces. A process that can occur across China appears here first. Liaoning's birth rate in 2024 was 4.32 per thousand, well below China's national rate of 5.63 per thousand in 2025. The resident population of the three provinces fell by roughly 12.5 million over the ten years from 2014, and part of that is not death but outflow.¹⁸ Industry sank, the young left, the remaining population aged, births collapsed, housing stopped carrying a price, and public finance stopped turning. The population that disappeared did not disappear from China; it moved to

of the property market and of the interest rate differential is not separated out.

¹⁸ The three northeastern provinces. From each province's statistical communiqué. Net outflow in 2024 was 108,000 from Jilin and 49,000 from Liaoning. Hegang, in Heilongjiang, was the first provincial city in

the Yangtze Delta and the Pearl River Delta. Seen at the level of the country it is population decline; seen at the level of the cluster it is relocation.

The catch-up model stops working

China was fast to here not because it ran on its own but because it held a machine for catching up. Every time America changed the game, the state pointed the direction, poured in subsidy, took in IP from German and Japanese firms through joint ventures conditioned on technology transfer, and, on the large demand that population scale makes possible, put up the infrastructure first without waiting for short-term returns. The geniuses flowing back from the United States counted too. That combination produced explosive growth in a straight line.

That machine is coming apart in several places at once. China is now a target of Western economic security, so neither capital nor technology arrives as it did. The population is shrinking, so neither the return flow of geniuses nor that stampede of demand is easy to reproduce. As seen in Chapter 2, talent is an input factor of direction, so what must be asked is not the total but how large a layer such people emerge from. Researchers chosen from among some 20 million young people a year and researchers chosen once that pool has become 5 million a year are not the same. Before that, young people worn out by excessive competition have begun stepping off it. Population decline and deflation obscure the outlook, so investment thins; ageing swells social security, so subsidy thins. And once a country settles into stagnation it can no longer ignore the return. Being able to ignore it was its single greatest weapon. When America next changes the game, the stamina to follow will be smaller than it was.

The United States has evaded this problem by gathering geniuses from all over the world, but the premise for attracting talent is institutions and freedom. Closed institutions are not a design well suited to gathering talent from the world. On top of that, inside the US-China rivalry the American stack takes on the mission of rebuilding an independent set of manufacturing sites. The further that goes, the thinner the external demand Chinese manufacturing faces.

A wealthy China or a strong China

There is an exit. If energy is abundant, automation advances, and the wealth produced explodes, and if that wealth comes to be distributed naturally, then individuals become wealthy and the fertility rate reverses. China is now at the brink of deciding which it will make real first: the strengthening of supply capacity, or the wealth of the individual. The allocation of resources appears to lean toward strengthening supply capacity rather than household wealth. If not repeating the century of humiliation is something close to a national creed, then where this ends is a state that minimizes external dependence and has strong supply capacity. If that is so, the probability of failure is high. It is precisely because personal wealth becomes abundant that people reproduce their population and demand grows.

Having watched the population of my own country, Japan, fall throughout my life, I may be permitted this much: the least visible loss in a falling population is that the resource called youth, the part of the population that starts change, disappears from the society as a whole.

China to enter a state equivalent to fiscal reconstruction (reporting by *Nihon Keizai Shimbun*, 2021). Jilin's births in 2024 were 97,000. Heilongjiang has not published the disaggregated birth-count item for two years running; its age structure has those aged 0 to 14 at 8.4 percent of the total population and those aged 65 and over at 19.4 percent.

8.7 India, the third candidate

India converges not on an independent techno-bloc but on a regional great power with a sphere of influence inside the American techno-bloc. The conditions are assembled: the magnetism of three thousand years of civilizational accumulation, the world's largest population, the ease of integration into Western supply chains that comes with being English-speaking, competitiveness in IT services, pharmaceuticals and space, and the legitimacy of democracy. The diplomatic doctrine of strategic autonomy is likewise a declaration of intent not to be wholly subordinate to any pole.

But India's demographic dividend is correct as a statement about the country as a whole and already incorrect as a statement about clusters. According to the Sample Registration System of the Registrar General of India, the total fertility rate of the five southern states is 1.3 in Kerala, 1.3 in Tamil Nadu, 1.4 in Andhra Pradesh, 1.5 in Karnataka and 1.5 in Telangana. All India is 1.9. Bihar, by contrast, is 2.8 and Uttar Pradesh 2.6.¹⁹ The demographic dividend is in the north, and it is no longer in the south, which is the economic center.

Norway's total fertility rate in 2024 was 1.44, and the five southern states run from 1.3 to 1.5, which is the same level. Yet income per head across the five southern states is roughly 3,800 to 4,700 dollars. Norway's GDP per capita is 94,594 dollars.¹⁹ The same fertility rate, achieved on a twentieth of the income. Fertility decline arriving before wealth is not a forecast in southern India. It has already happened.

The reasons for reading India as never becoming an independent pole narrow to two: brain drain, and the absence of a semiconductor industry running from design through to manufacture. The first is that the layer producing talent is thick while its top keeps being drawn off to other clusters. The second is that stacking foundation-model technology without holding leading-edge physical technology leaves the ceiling set from outside.

There are therefore two branches: a road on which India aims at an AI stack of its own, and a road on which it settles inside the American stack. This thesis's reading leans to the latter. An independent stack demands a leading-edge physical supply chain, and that demands decades of accumulation, enormous capital, and above all the settling of talent. Convergence happens in the 2040s to the 2050s.

8.8 Europe

What comes out highest for Europe is leading-edge physical technology, pulled up by the Netherlands and Germany. The lithography machine, the machine tools and the industrial robots are real, and on this one axis Europe stands on the same ground as China's two deltas. But it does not reach Japan-Korea-Taiwan, and the reason is the depth of the supply-chain layers. Materials, production equipment, lithography, leading-edge logic, high-bandwidth memory, precision motors: the only region where all of them are assembled inside one place is Japan-Korea-Taiwan. Europe is strong at particular steps and does not hold every layer.

On effective energy abundance the contribution comes from the Nordics, with hydro and geothermal, and from France, with nuclear. On foundation models too,

¹⁹ India's fertility rates and incomes. Income per head is roughly 4,700 dollars in Karnataka, 4,500 in Tamil Nadu and 3,800 in Kerala. Registrar General of India, Sample Registration System (2024), cited by way of press reporting and not confirmed directly in that office's Statistical Report. The all-India figure of 1.9 is consistent with the World Bank's 2024 estimate of 1.96. Norway's GDP per capita is from the World Bank (2025).

sovereign-AI development has had a degree of success. Taken as a region, though, Europe is not on the same ground as Texas, the UAE or China's two deltas.

And there is a problem on the institutional side as well. Regulatory density is part of institutions, and it bears directly on both the preservation of returns and the speed of permitting. High industrial electricity, slow construction and heavy regulation are not separate problems but one term.

Loxbridge is BosWash at a smaller scale. Institutions, capital and research, with no energy and nothing physical: the same shape, differing only in size. What London and the Oxbridge corridor supply is the intelligence of basic research, and it is the party to which America grants top-level access first. But that intelligence is inclined to leak, and the most important AI research organization this corridor produced ended up under American capital, which is the same structure as the capital flight argued in Chapter 6. France, Germany, the Netherlands and the Nordics do not overlap even with Loxbridge. That Loxbridge is the region's only cluster follows from the same shape.

8.9 The global south

The twentieth-century development model is breaking from two sides at once. It ran in a single order. There is cheap labor; manufacturing agglomerates there; a middle class is born; a domestic consumption market forms; industry moves upmarket. Japan, Korea, Taiwan and China all climbed that staircase.

First, fertility decline arrives before wealth. Vietnam's total fertility rate has already fallen below 2, and urban areas are below 1.5. GDP per capita at that point is about 5,100 dollars. Korea and Taiwan met the same level of fertility at a stage past 10,000 dollars per head, and Vietnam is entering the same problem at roughly half that level. Indonesia too is at about 5,100 dollars per head, and urban Java is falling sharply.²⁰ The household Malthus mechanism applies directly. It used to be that wealth came first and the price structure throttled births afterwards. Now only the price structure is imported first. Urbanization pushes up housing and education costs, and the smartphone carries in the lifestyle of the advanced countries and the sense of opportunity cost at the same time. Income has not yet arrived, but the denominator of household Malthus arrives ahead of it.

Second, manufacturing does not come in the first place. The entrance to the twentieth-century sequence was cheap labor. But in a world where the robot is cheaper, cheap labor does not call in investment. As seen in Chapter 2, the automation of direction 1 has not stopped once in two hundred years. The largest reason Bangladesh or Ethiopia cannot become the next Vietnam is neither politics nor geography but that the labor-intensiveness of manufacturing is itself being lost.

The model of becoming wealthy by making use of a demographic dividend has a high probability of ending with Vietnam and Indonesia as the last generation, and even they sit on the near side of the trap. This is not the failure of one region but the end of the twentieth-century development model itself.

²⁰ Fertility decline before wealth. GDP per capita is from World Bank figures (2025). Urban fertility rates are from each country's statistical office and have not been confirmed against primary sources item by item.

The dividend and the institutions sit in different places

The demographic dividend lasts longest in sub-Saharan Africa, which can enjoy it from the 2030s into the 2060s. But it does not have the institutions and infrastructure to make use of it. Where the institutions are, fertility decline has already begun. In the global south plate of Figure 8–3, Indonesia, Vietnam, Nigeria and Kenya are still low on all five terms.

The more structural number is on the power side. In Nigeria, for a population of 237 million, average generation output stands at roughly 4,100 megawatts against 13,625 megawatts of grid-connected capacity, and the product of the three terms of Chapter 1 is structurally low.²¹ As for East Africa, outlooks have been offered for twenty years, and this thesis has not been able to confirm a record of grid connections that bears them out. The gap with Vietnam appears not as a difference of political system or of population but as a difference in state execution capacity, which is to say the capacity to connect gigawatts to the grid and put up buildings.

Head-count becomes the scarce factor

South America and sub-Saharan Africa, while settling the world's future in population, are placed outside the stack. As Figure 4–6 shows, a little over three tenths of the world's children are now born in sub-Saharan Africa. And as section 5.7 showed, the scarce thing is demand. The region where the demographic dividend lasts longest is also the only region where head-count is still rising during the half-century in which head-count is the scarce factor. What is missing is purchasing power, and purchasing power is a function of income, and income is a function of whether you are inside the stack. This is a region that is an object of investment and, once income arrives, the holder of the world's scarcest factor, head-count.

And if the cost of energy and automation really does collapse, the first and largest beneficiary ought to be not the frontier but the societies now made to pay the highest effective prices. The ethics of the cascade are settled by where that cheapness arrives first.

8.10 Orbit

What this thesis claims about orbit is not habitation but use. For the twenty-first century at least, habitation off Earth does not pay: life support is the use that demands energy, land and labor all at the highest unit price. The order begins with data centers in orbit; generation follows, then the automation and acceleration of transport, and production sites go up last. In the terms of the equation, an orbital infrastructure business is built out of T_2 and K , and what it sells is E .

Orbit beats the ground for two reasons and no more. Sunlight arrives unobstructed, and there is no queue for grid connection. The reliability of supply is if anything harsher than on the ground, but if what sets the speed is lead time, then a place with no queue is structurally strong.

And what runs short in orbit is not energy but the ability to get rid of heat. A vacuum has no medium to move heat into, so heat can only be radiated as infrared,

²¹ Nigeria. A currency devaluation in 2024 shrank its dollar GDP by about 47 percent and it lost its position as the largest economy in Africa. World Bank, *Technical Note 2: Igniting Economic Growth by Reforming Nigeria's Power Sector* (2020 data; more than 51 percent of installed capacity unavailable, annual losses from unstable electricity of 5 to 7 percent of GDP). Grid capacity and average output are from the Nigerian Electricity Regulatory Commission, *First Quarter 2026 Report*.

and the quantity radiated goes as the fourth power of absolute temperature.²² Extrapolate today's solid radiators from an ISS-class design and cooling one gigawatt of computation takes about a hundred thousand tonnes of radiator. At a hundred tonnes per launch for current heavy rockets, that is a thousand launches of radiator alone. The gigawatt-class orbital data center as currently conceived is a business of launching radiators rather than computers.

There are two ways to make the radiator lighter: raise the temperature, or stop using a panel. Raising the operating point from 20 to 120 degrees Celsius alone cuts the required area to under a third. The bottleneck of an orbital data center is not the performance of the semiconductor but the temperature at which the semiconductor can still run. The second way is to replace the solid panel with droplets of liquid, and a liquid-droplet radiator is reported to shed one gigawatt in about 2,200 tonnes, twenty launches. The distance between a hundred thousand tonnes and 2,200 is not a difference of cost but of possible and impossible. Europe's orbital data center programme ASCEND sets a timeline of 50 kilowatts demonstrated in 2031 and one gigawatt in 2050, but at 50 kilowatts heat is not yet a problem. The real test is the one in 2050.

Even once that is solved, energy investment on the ground does not become worthless the next day. If anyone loses, it is the cluster that invested earliest and heaviest in ground-based energy: the cheaper power in orbit becomes, the less that investment is worth. And at present the only holder of cheap access to orbit is the American techno-bloc. Does Wright's law fire first on reactors on the ground or on solar panels in orbit? The answer changes which cluster is ahead.

8.11 Division of labor beats vertical integration

Measure the nine clusters against the same equation and the answer that comes out is that no place carries every term at world level. No cluster is even across the terms, and the only place the pentagon fills is where five of them lie over one another. Extend the reading to here in a straight line and the relative advantage of the American techno-bloc accelerates from the middle of this century into its second half. The constraint comes off in the 2030s, it tells on price in the 2040s, and the design of distribution comes back as demand from the 2050s: the weight of this thesis's predictions sits on the later half, not the earlier one. China shrinks in relative terms under population decline and a closed ecosystem while keeping the strength of an overwhelming capacity to supply. India converges, Europe lacks technological autonomy, and the global south holds the head-count while staying outside the stack.

That reading holds because the division of labor has become harder to break than it was. The division of labor of the 1990s was an arbitrage on wage differences, so when wages moved the arrangement moved with them. What settles the division of labor now is sunlight, land, and a manufacturing lineage forty years in the accumulating, and none of those moves on price. To the extent that comparative advantage is fixed in geography and in time, the gap between the side that divides the terms among itself and the side that holds all of them within one country widens with time. So America, the UAE and Japan-Korea-Taiwan should be bound together more closely than they are. The bits are in America, the physical side is in

²² Waste heat in orbit. The figures in this section are not quotations from industry material but were recalculated by the author from the Stefan-Boltzmann law. Black-body radiation at 20 degrees Celsius is 419 watts per square meter on one side and 838 watts over two sides; at emissivity 0.85 it is 712 watts over two sides, giving roughly 1,400 square meters per megawatt. The International Space Station's measured performance is 70 kilowatts over 422 square meters, that is 166 watts per square meter, less than a fifth of the ideal value. The radiating area needed for one gigawatt is about 1.4 square kilometers on the ideal figure and about 6.0 on that measured one. Mass, extrapolated from the same design, is about 100 tonnes per megawatt for the solid radiator against about 10 tonnes for the computing racks. The 450 watts per kilogram for the liquid-droplet radiator is from an industry analysis, SpaceComputer, "Cooling for Orbital Compute: A Landscape Analysis", and the primary source for that value is unverified. This single point very nearly settles by itself whether gigawatt-scale computation in orbit succeeds or fails. ASCEND's time frame is from the European Space Agency's plan.

Japan–Korea–Taiwan, and energy, large capital and the speed of decision are in the UAE. The three already stand on the same side, but what joins them stops at the level of individual transactions. Put the three under one design and the five terms stand together for the first time. Putting them under one design also means bringing the unit of integration down from the country.

SDL's third axiom holds that all social and economic actors are resource integrators.²³ No actor holds every resource needed to create value. So value creation only comes about when what several actors offer is integrated by someone. And a resource can be integrated without being owned. What makes something a resource is that it can be drawn on, not that it is held. The individual actors who command the foundation models in the Bay Area, the individual actors who command lithography and leading-edge logic in Japan, Korea and Taiwan, the individual actors who command the gigawatts and the balance sheet in the UAE: what is needed among these three is not buying up one another's terms but a standing ability to reach another's term and combine it with one's own. Normann called this resource density, the best combination of resources that can be mobilized at a given time and place, made by unbundling resources and rebundling them, which is the general form of the division of labor.²³

²³ Resource integration and resource density. Robert F. Lusch & Stephen L. Vargo, *Service-Dominant Logic: Premises, Perspectives, Possibilities*, Cambridge University Press, 2014: chapter 6 (that no actor holds every resource it needs, and that a resource becomes one by being drawable on rather than by being owned) and chapter 9 (resource density, and the division of labor as its general form). The concept of resource density is from Richard Normann, *Reframing Business: When the Map Changes the Landscape*, John Wiley & Sons, 2001.

Chapter 9. On Investment

9.1 What to invest in

Which technologies and which institutions should capital go into, starting tomorrow, for the society on the far side of the constraint to arrive? The places map onto the terms of the equation in Chapter 1. Because it is a product, capital does most where the lowest term is. A second test: as section 5.7 showed, the only returns that can be relied on are the ones that do not depend on head-count, and businesses whose returns are proportional to the volume of demand sit in a structural headwind for the next half-century.

Raising the effective abundance of energy

- ⊙ **Fusion and SMRs.** Fusion at the top; conventional large reactors and small modular reactors (SMRs) built in series in a factory govern the transition. Renewables ride on top of firm power as a supplement and do not fill this term on their own. What sets the price is not discovery but volume. The cost curve follows Wright's law, and the contest turns on how many copies of one design get built. So what is worth buying is less the demonstration reactor itself than the production line and the supply chain that repeat one design after it. The effect on price arrives in the 2040s, and only capital that can wait that long belongs here.
- ⊙ **Grid infrastructure and heavy electrical equipment.** However much generation is added, nothing moves unless a new load can be connected to the grid. Three things are buyable: manufacturing capacity for transformers and switchgear, construction capacity for transmission lines and substations, and land already connected to the grid. The median period from application to entry into operation, for projects entering operation in the United States in 2025, was 61 months, and of the projects applied for between 2000 and 2020 only 13 percent by capacity reached operation at all.¹ A new interconnection right is no longer bought with time; it is bought with probability. The delivery time for large transformers went from under a year before the pandemic to a standard three years, with generator step-up transformers at 144 weeks.² And since neither electricity nor land can be exported, this is the one term that cannot be divided among partners, which makes the choice of site itself the investment decision.

¹ The interconnection queue in the United States. Lawrence Berkeley National Laboratory, *Queued Up: 2026 Edition* (of applications filed from 2000 to 2020, 13 percent by capacity reached operation and 75 percent were withdrawn). These are generation-side statistics and the load-side queue is outside their scope.

Raising labor input

- ⊙ **Physical AI.** Humanoids and automated plant. If mass production begins between the end of the 2020s and the early 2030s, the price of energy and the degree of automation move at the same time. Unit cost is already starting to ride Wright's law while the demonstration of quality lags, and the two advance separately. None of it can be planned on software's clock. Factories, parts, regulation, accidents and repairs all move in the time of the physical world.
- ⊙ **The leading-edge physical supply chain.** Materials, semiconductor production equipment, lithography, leading-edge logic, high-speed memory, industrial robots, precision motors. When humanoids enter volume production, what runs

² Transformer delivery times. U.S. Department of Energy, *Large Power Transformer Resilience Report*, 2024. The series in weeks is from Wood Mackenzie (high-voltage switchgear at 44 weeks). The floor is the same worldwide.

out first is not the finished machine but the capacity to make this layer. The only region where every layer sits together is Japan-Korea-Taiwan, and Japan-Korea-Taiwan is structurally short of energy. So investment in this layer means nothing unless it sits inside the same design as fusion and the grid.

- ⊙ **Implementation design for the Baumol sectors.** Housing, education, childcare and healthcare. Physical AI tells on population only when it reaches those four. The expected cost per child is made of them. Building the machines and designing where in a hospital, a school, a nursery or the production of housing to apply them are two different investments. The second is the thinner of the two.

Building institutions

- ⊙ **The institutions of distribution.** Capital accounts at birth, citizen-stake sovereign funds, and universal stakes in automation capital. This is not an ethical term but an engineering one: it decides whether a cluster is still working twenty years from now. Institutions can be made in law, so a decision moves them within a few years. Their effect, though, arrives twenty years later, which means institutionalizing them in the 2030s is buying the demand, the challengers and the tax base of the 2050s now.
- ⊙ **The speed of connection and permitting.** How good institutions are can be read off how long the procedures take.³ Britain in the 1840s held two things at once: incorporation by registration alone, and a statute that standardized the procedure for compulsory purchase. What sits in that position now is the interconnection queue, environmental assessment and building permits. The procedure can be shortened; the transformer's delivery time cannot.
- ⊙ **The infrastructure of measuring use-value.** When price stops being the device that rations scarcity, the job that remains is measurement. What is buyable is the technology that records the fact that a beneficiary felt a benefit, the contractual machinery that turns that record into payment, and the machinery that establishes the record was not faked. Without it, neither distribution nor co-creation has a unit to settle in. And once it arrives, the object of investment itself changes: what gets bought is not a share in a company but a claim on use-value that has not yet arrived. As Chapter 7 showed, this is the harder of the two, harder than energy.

Outside the equation

- ⊙ **Infrastructure outside the stack.** For the half-century from 2035, the only part of the world where head-count is rising is the part now placed outside the stack. The twentieth-century sequence in which cheap labor draws in manufacturing will not be replayed, so that head-count does not turn into purchasing power on its own. The constraint there is neither capability nor will but the cost of capital, the grid and the procedures. Investing to build the ability to buy and investing to sell are two different things: selling into a market where income has not yet arrived turns the sale into a loan.
- ⊙ **Orbit.** Computation in orbit beats the ground for two reasons: sunlight arrives unobstructed, and there is no connection queue. What stands in the way is not a law of physics but the engineering of getting rid of heat. So what is worth buying is semiconductors that run hot and a form of radiator to replace the solid

³ How E is measured. The effective price of industrial electricity comes from national published statistics (the UK Department for Energy Security and Net Zero's international comparison, Eurostat *nrg_pc_205*, US Energy Information Administration *Electric Power Monthly* Table 5.3). Reliability of supply is the series based on SAIDI/SAIFI that the World Bank's *Doing Business* Getting Electricity indicator used. For the lead time on a new grid connection alone, no published statistic exists that is bundled as an ex ante country-by-country screen. In the United States, Lawrence Berkeley National Laboratory's *Queued Up* publishes the period from application to commercial operation; in the United Kingdom NESO names a connection date; in the Netherlands TenneT publishes exhaustion by geography. China and the GCC publish nothing. This is the gap in information most unfavorable to the argument of this thesis.

panel. This is the rival to fusion and the grid, and at the same time the insurance on them.

One caution runs through all ten. Most of the returns named here exist because the constraint has not yet come off. Grid-connected land is dear because the procedure is slow, and faster permitting and connection thin it. Ground-based energy infrastructure is dear because orbit has not yet opened, and once orbit does open, whoever invested earliest and heaviest loses most.

9.2 How to invest

What I am proposing is a resource integrator that connects the terms held separately by actors in the United States, the UAE and Japan-Korea-Taiwan: a platform for co-creation that sits between a fund, a studio and a consortium.

As a venture fund it raises capital and puts it to work. As a studio it names, program by program, the theme to be taken on, and integrates the resources of the actors who gather under that theme to co-create concepts that do not yet exist. As a consortium it brings the energy, compute and regulatory parties into one contract. When one body does all three at once, what results can be called an ecosystem.

The studio works like this. It sets out one vision of a size no one reaches alone, such as building one fully automated manufacturing plant all the way through. The actors holding the component technologies that vision requires gather of their own accord to create value, lay their ideas over one another, conceive a range of products, build them, run them and test them. The faces change fluidly with the vision. The practice of putting people from different disciplines in one room and having them design together is not itself new.⁴ What changes here is the exit: it does not stop at a proposed concept but attaches gigawatts, compute and a site and carries the work through to the real thing. It is an ideathon run to completion.

The capital starts with the sovereign funds of the UAE, which are working to build domestic industry. It goes into the bits side, the foundation models and AI basic research of the San Francisco Bay Area, Texas and BosWash, and into the physical side, the actors around the leading-edge physical supply chain and the next generation of nuclear reactors in Japan-Korea-Taiwan. The counterparties are startups, large companies holding a technology, individuals holding the IP, and laboratories in universities and firms. Those invested in then choose, as the ground for their physical experiments, the UAE and Texas, where energy is cheap, a site can be had, permitting is fast and a regulatory sandbox is in place. This is how the UAE and Texas become the places on earth where implementation goes furthest.

For my own part I take the role on: as a strategic designer, who sets the vision, gathers the actors, prepares the conditions and designs the integration itself, and as a venture designer, who shapes each undertaking that stands up out of it.

9.3 The predictions

1. **Energy.** By the middle of the 2030s, in at least one major economic bloc, there will be a year in which new capital investment in nuclear fission exceeds new

⁴ The lineage of designing co-creation. Gathering people from different disciplines in one place to design together is something design education and design practice have built up over a long time, and the idea of drawing knowledge in from beyond the boundary of the organization has been set out as open innovation. Henry Chesbrough, *Open Business Models: How to Thrive in the New Innovation Landscape*, Harvard Business School Press, 2006. Eric von Hippel, *Democratizing Innovation*, MIT Press, 2006. What this section adds is not the method but the exit: attaching capital and physical conditions to the resource integration of section 8.11 and carrying it through to implementation.

capital investment in renewables. Fusion will achieve its first commercial delivery of power to the grid in the 2030s, but prices genuinely move only in the diffusion phase where the learning curve bites, which is to say the 2040s. The start-up of the first commercial reactor will be reported as the news that energy has become free, and that report will be wrong.

2. **The grid.** What settles the siting of gigawatt-scale projects shifts from the price of power to how long it takes to get connected to the grid. By the first half of the 2030s, investment flows will visibly separate between grid operators who can name a connection date and grid operators who cannot. Land already connected to the grid will be traded as an asset class of its own.
3. **Physical AI.** Mass production of humanoids and automated plant begins between the end of the 2020s and the early 2030s. Unit cost rides Wright's law, and the demonstration of quality lags behind it. What runs short first is the capacity to make industrial robots and precision motors.
4. **Population.** China's annual births will not come off their structurally declining path, whatever official support is put in place, because the population of potential mothers is already fixed. The first statistically significant reversal in fertility will be observed not in a country with generous family policy but in a country or a city that has structurally lowered the effective prices of energy and housing. The correlation between the size of a subsidy and the birth rate will weaken further over the coming decade.
5. **Labor.** The fall in youth employment relative to other age groups, in the occupations AI reaches most, will keep widening through the second half of the 2020s, and in at least one major economic bloc youth unemployment will become the central issue in politics. The first wave of solutions offered will be retraining, and it will not work, because tasks do not come back.
6. **Politics.** In the 2030s, in at least two major advanced economies, a wealth tax, or an income tax whose top rate is raised to the 70 percent level, will appear in the platform of a party in government. Where it is implemented, it will be rolled back within five years by the movement of capital and of people. The first half of this prediction has already begun, in New York and Seattle in 2025.
7. **Distribution.** In the same 2030s, institutions for making every citizen an investor, meaning capital accounts opened at birth, sovereign funds held as a national stake, and universal stakes in automation capital, will appear as mainstream policy in several countries. This is the beginning of what this thesis calls the update to capitalism. The first country to implement it in earnest buys itself the social stability of the 2050s.
8. **Demand.** By the second half of the 2030s, in at least one major country, the center of policy debate moves from a shortage of labor to a shortage of demand. Over the same years, the difference between returns that do not depend on head-count and returns proportional to the volume of demand comes to be treated as a distinction between asset classes.
9. **Measurement and money.** Money gives way first on the side of energy, not on the side that measures value in data. Exchange denominated in units of energy spreads outside regulated markets but does not displace a major currency within the first half of this century. In the 2030s the first commercial cases will appear in which a direct record of use-value, rather than the optimization of

advertising and recommendation, settles what is paid. The divergence between GDP and welfare is acknowledged from inside official statistics, as a satellite account.

10. **The unit of value creation.** On the bits side, assembly and dispersal without employment settles in as one of the forms in which major AI products are built. On the physical side the firm survives as a unit. The average headcount of a company falls in software and AI and does not fall in materials, equipment and power.
11. **Clusters and orbit.** Texas and the UAE stay near the top over the long run, Japan-Korea-Taiwan holds its place as a technological chokepoint, and Europe produces no cluster other than Loxbridge. Computation in orbit reaches a demonstration at the 50-kilowatt scale in 2031, but at that scale heat is not yet a test. Whether a gigawatt-scale system stands by 2050 turns on whether semiconductors that run hot or liquid-droplet radiators are demonstrated.

9.4 What would falsify this

Writing the predictions down is not enough. What would make me admit that one of them was wrong has to be written down to the same precision. The purpose of this table is not to protect the predictions but to make sure a wrong one can be thrown away for certain.⁵

What breaks in the skeleton of this thesis when one of them is falsified? If energy is lost, the timeline of Chapter 5 slides backwards, but the identification of the constraint survives. If the grid is lost, the identification itself, that the constraint sits at connection, is wrong. If China's births are lost, the central proposition of Chapter 4 is itself wrong. If labor is lost, the break between task and income, which is the premise of Chapter 6, was never happening in the first place. If politics and distribution are lost, the design argument of Chapter 6 was necessary and yet politically impossible. If measurement and the unit of value creation are lost, the direction of Chapter 7 survives but its timing slides a long way back. The one whose failure would reach furthest is China's births. The one that most needs to hold is distribution. Some of these will be wrong, but the way they go wrong is the information. I intend to keep updating this reading by collision with reality.

⁵ On the tracking indicators in the falsification table. Every series listed in the Source column of the table is a published statistic updated annually or quarterly. Rows whose date of judgment has passed are kept, with the result of the judgment appended.

#	The prediction	Falsification condition (observe this and the prediction is lost)	Tracking indicator	Source	Date of judgment
I	Energy	By the end of 2035, no year has appeared, in any major economic bloc, in which new capital investment in nuclear exceeded new capital investment in renewables	Annual new capital investment by generation type	IEA <i>World Energy Investment</i> (annual)	2036
I'	Fusion	Commercial delivery of power to the grid is not achieved even once by the end of 2040, or the levelized cost of fusion reaches the level of existing sources by the end of 2040	Delivery to a commercial grid; LCOE	Company disclosures, IEA, IAEA	2040

#	The prediction	Falsification condition (observe this and the prediction is lost)	Tracking indicator	Source	Date of judgment
II	The grid	Regions that rank at the top on the three indicators of effective industrial electricity price, reliability and interconnection lead time are not chosen, by 2035, as sites for gigawatt-scale new computing and manufacturing	Siting decisions for new data centers and advanced fabs	Company investment disclosures, grid operators' queues	2035
III	Physical AI	By the end of 2032, no producer has reached an annual output of 10,000 humanoids	Units shipped	Company disclosures, IFR	2033
IV	China's births	Annual births in 2032 exceed 7 million	Annual births	National Bureau of Statistics of China, annual communiqué	Early 2033
IV'	Reversal in the birth rate	A significant reversal in the birth rate is observed in a country centered on cash transfers before it appears in one that lowered the effective prices of energy and housing	TFR; real prices of housing and electricity	National statistical offices, World Bank, OECD	2035
V	Labor	The fall in youth employment relative to other age groups, in occupations AI reaches most, levels off in the second half of the 2020s and does not widen thereafter	Employment series by age and AI exposure	Stanford Digital Economy Lab, national employment statistics	2030
V'	Retraining	A large-scale retraining program restores the employment rate of the target cohort to a statistically significant degree	RCT or quasi-experimental evaluation of the program	National labor ministries, peer-reviewed papers	2035
VI	Politics	By the end of 2040, a wealth tax or a top marginal rate at the 70 percent level has not appeared in the platforms of parties in government in two or more major advanced economies	Party platforms and legislative status	National parliamentary records, OECD tax database	2040
VI'	Capital flight	In a country that implements it, neither a net outflow of the wealthy nor a fall in revenue from that tax is observed five years after it takes effect	Departure filings by top taxpayers; revenue from that tax	National finance ministries and statistical offices	Effect + 5 years
VII	Distribution	By the end of 2040, none of capital accounts opened at birth, funds held as a national stake, or universal stakes in automation capital has been institutionalized as mainstream policy in a major country	The fact of legislation and institutionalization	National legislative records	2040
VIII	Demand	By the end of 2040, a shortage of labor still ranks above a shortage of demand in the policy documents of major countries	The subject of medium-term plans and white papers	National government documents, OECD <i>Economic Surveys</i>	2040
IX	Measurement	By the end of 2040, not one commercial case has appeared in which payment is settled on a direct measurement of use-value	The fact of the pricing method	Company disclosures, regulatory approvals	2040
IX'	Units denominated in energy	By the end of 2040, no price denominated in energy consumed or in emission allowances is used to settle a commercial transaction outside a regulated market	The fact of denomination and settlement	Company disclosures, exchanges and registries	2040

#	The prediction	Falsification condition (observe this and the prediction is lost)	Tracking indicator	Source	Date of judgment
X	The unit of value creation	By the end of 2040, the average headcount of companies in software and AI has not fallen	The distribution of company size	National statistical offices, industry statistics	2040
XI	Clusters and orbit	By the end of 2040, neither Texas nor the UAE is among the leading sites for gigawatt-scale computing and manufacturing; or no orbital demonstration at the 50-kilowatt scale by the end of 2032	Capacity of new sites; demonstrated power in orbit	Company investment disclosures, ESA	2040
XI'	Orbit	Computation at the 10-megawatt scale is demonstrated in orbit by the end of 2040	The fact of the demonstration	European Space Agency, company disclosures	2040
Zero	The framework itself	Within a window of ten to thirty years, the quality of institutions is shown to move systematically with resource endowment (if institutions are settled from the resource side on this horizon, the first term of the equation is not an independent term)	A panel of institutional indicators and resource endowment over the same period	World Governance Indicators, resource endowment data	2040
Zero'	Automation cannot compensate	In regions where energy prices are high, investment in automation offsets the price rise and the output share of energy-intensive industry is maintained	Output by industry, robot density, electricity prices	IFR, Eurostat, national statistical offices	2032

9.5 What is not established here

Four things this thesis asserts and does not establish.

The link between energy prices, technological structure and the birth rate is set out as a mechanism, not measured as one; settling it takes data on energy prices, generation mix, real wages and fertility for many advanced economies over many years. The decomposition of the price system is qualitative. What is missing is an estimate, across housing, education, childcare and healthcare, of how much each cost driver contributes to the household Malthus, energy and non-automated labor above all. The account of urban growth rests on historical cases rather than on a model in which cheap energy, cheap labor and abundant land endogenously settle infrastructure investment and agglomeration. And the effect of automation on hours of work and on fertility behavior is argued rather than modeled. What shorter hours, redistributed income and reduced anxiety about the future do to the decision to marry and to have a child belongs in a household macro model this thesis does not build.

9.6 Past this, people decide

This thesis has restated, from one layer down, things that are usually discussed as questions of values or of policy: energy, computing capacity, technology, the price system, the household constraint. What geology and physics settle is whether the raw material of energy is there at all. What turns it into energy that is very nearly free, or leaves it in the ground, is institutions. Past that, people decide.

If this reading is right, we become the first generation in human history to design the far side of the constraint rather than the inside of it.

I intend to spend my working life on making that co-creation happen. It is not a thing one person can assemble, so the design goes out first.

Let us meet and talk.

About the Author

Hi, I'm Arata Hoshino, a strategic and venture designer based in the United Arab Emirates. Although raised in Japan, I spent my teens reading about Silicon Valley technology, longtermism, techno-libertarianism, and the ideas of Buckminster Fuller.

I did BEng Mechanical and Energy Engineering at Waseda University; MSc Media Design at Keio University, with research work in service design and venture design; CEMS MIM (MSc International Management) at HEC Paris and the National University of Singapore. After that, I worked in VC and strategy consulting in San Francisco and Singapore. From 2024 to 2026, I advised UAE ministries on economic strategy at a policy advisory firm spun out of the OECD and Harvard.

I am grateful to Kei Irizawa, a classmate since junior high, for reviewing the data and the structure of the argument. The errors that remain are mine.

You can reach me by email. A summary is on LinkedIn, the scope of each project is in my portfolio (PDF), and where I stand is on vision paper (PDF).

Appendix. Objections and Limits

Chapter 1: Objections and Limits

The body carries only the bones of the argument. Anticipated objections and how they are handled are gathered here. Where an objection attaches to one passage, the heading gives the section. No figures are placed here.

Why the equation takes this form (1.2)

There is a reason not to collapse it into a single efficiency term. Write competitiveness as something like “energy times institutional efficiency” and everything unobservable ends up inside the second term. That is the same structure as the Solow residual in growth accounting. Moses Abramovitz called that residual “a measure of our ignorance” (“Resource and Output Trends in the United States since 1870,” *American Economic Review* 46(2), 1956, 5–23). An equation that merely names a residual predicts nothing.

A product of three terms is a different object. Institutions are observable through net inflows of talent, the depth of capital markets, and permitting lead times; effective energy abundance through the level of prices and, for the expectation of persistence, through the price and tenor of long-term power purchase agreements and forward curves; labor input through robot density, computing resources, and access to chokepoint technologies. Each is independently observable. None of the three is a residual.

Capital and infrastructure are not placed as terms in order to avoid circularity. Capital is what institutions and energy call in, and physical infrastructure is what gets built as the product of the three. Line up the result on the side of the cause and the equation explains itself. Defining the K in $L \approx T \times K$ as a balance is for the same reason. What determines how much can be built this period is capital accumulated through prior periods, not capital created this period. That the result thickens the next period’s balance is not denied. The opening and closing of windows in 1.3 is exactly that circulation.

With that said, this equation should be read as an enumeration of necessary conditions rather than a decomposition of causes. The claim is that if any one of the three is missing, nothing gets built. It is not the claim that if all three are present, something necessarily does.

How much of this structure belongs to prior literature (1.2)

The general form of “resources are not given but become resources through institutions” is Zimmermann’s functional theory of resources: resources are not, they become (Erich W. Zimmermann, *World Resources and Industries*, Harper & Brothers, 1933; rev. ed. 1951, pp.814–815). That resource dependence is an endogenous outcome of institutions is Brunnschweiler and Bulte (*Journal of Environmental Economics and Management* 55(3), 2008, 248–264); the path from cheap energy to automation and competitiveness is Allen (Robert C. Allen, *The British Industrial Revolution in Global Perspective*, Cambridge University Press, 2009); the causal identification of failures of conversion is Allcott et al. (*American Economic Review* 106(3), 2016, 587–624). The view that technological chokepoints open and close according to alliance and network position is Farrell and Newman (*International Security* 44(1), 2019, 42–79).

The difference here is one point. The existing literature places institutions as a moderator; this thesis places institutions on the side that calls in capital and talent, which are the mediating variables. That is a choice of formulation rather than a discovery, and it is very nearly the whole of what this thesis can claim as its own. Writing institutions as two functions, the preservation of returns and the attraction of talent, follows Acemoglu and Johnson, who separated property rights institutions from contracting institutions and concluded that what matters for long-run growth and investment is the former, not the latter (Daron Acemoglu & Simon Johnson, “Unbundling Institutions,” *Journal of Political Economy* 113(5), 2005, 949–995).

Zimmermann is cited via a citation database and the pages of the original have not been checked. Brunnschweiler and Bulte rest on the publisher's abstract and indexing records; the full text has not been obtained.

Labor input L is used in two senses (1.2 and 1.3)

The L of 1.2 is automation, defined as the removal of humans from labor input by machines together with productivity gains from AI and computing. But the cases offered in 1.3 as instances of all three conditions being present are Britain in the 1840s and the UAE today, and in both of them what is cheap is human labor. Child labor and migrant labor, not automation.

It follows that there is not a single case in this chapter that demonstrates the L of the equation. And the decomposition $L \approx T \times K$ applies only to the present; Britain in the 1840s had neither technology access nor large capital.

This thesis puts the two into the same term because L is taken broadly, as input for getting work done without spending human hands, and that breadth derives from the definition of terms in 1.1. The line at the end of 1.3, that the abundance the nineteenth century bought with human pain the twenty-first buys back with machines, is a declaration that the two are different realizations of the same term. But a declaration is not a demonstration. This is a weak part of the thesis and is marked as such.

What was never conceived cannot be observed (1.2)

Section 1.2 says the constraint binds not only what can be built but what can be conceived. That claim, by its nature, cannot be directly observed. The distribution of business plans that were never written is recorded nowhere.

There are two indirect supports. The first is the discontinuous turnover in the objects of investment immediately after a constraint loosens. Railways in 1840s Britain and computing substrate in the 2020s Gulf both saw an asset class of a scale that had not previously existed stand up within a few years of the loosening. The second is that in other countries of the same period, where the same technology was known, investment at the same scale did not occur. Availability of technology does not explain it; it has to be read as a difference on the side of the premises.

But these two are suggestive, not probative. Since there is no way to observe the counterfactual, this is the hardest claim in the thesis to test. It is marked as such.

Are the three conditions only judgeable after the fact (1.3)

What the body offers are cases where all three were present and something was built, and cases where one was missing and nothing was. It does not offer a case where all three were present and nothing was built. Section 1.3 is therefore an illustration of the equation's application rather than a test of it. What corresponds to a test is placed in the falsification table of Chapter 9.

The direction of causation (1.2)

Correcting for type I error from overfitted VARs, no genuine causal effect from energy consumption to growth is found across the literature as a whole; what is robustly detected is only the direction from output to energy consumption (Stephan B. Bruns, Christian Gross & David I. Stern, "Is There Really Granger Causality Between Energy Use and Output?" *The Energy Journal* 35(4), 2014, 97–120). This thesis does not dismiss that result.

What it relies on is Stern and Kander's conditional formulation: energy constrains growth when it is scarce, while in non-scarce regimes labor-augmenting technical change dominates (David I. Stern & Astrid Kander, "The Role of Energy in the Industrial Revolution and Modern Economic Growth," *The Energy Journal* 33(3), 2012, 125–152). What this thesis treats is a regime in which scarcity returns, and econometric results from a non-scarce period cannot be extrapolated into it directly. This corresponds to one line of the falsification table in Chapter 9.

Chapter 2: Objections and Limits

The body carries only the bones of the argument. Anticipated objections and how they are handled are gathered here. Where an objection attaches to one passage, the heading gives the section.

Is the law of competition only judgeable after the fact (2.3)

This is the heaviest objection. Does “the more the essence of the problem is information processing, the more direction 2 wins” not amount to identifying the essence after seeing which side won? Shopping looked like a physical problem until Amazon won.

The decomposition by mass in the body is written to answer this. Counting the mass that must be moved at each step can be done before the outcome is known. But the granularity of the decomposition is itself a judgment. Cut shopping into search, comparison, ordering, payment, and delivery, or into going to the shop, choosing, and carrying home, and the proportion of zero-mass steps changes. This law is therefore best read as a tool of classification rather than a tool of prediction. It does not tell you which side will win; it points to the axis along which the outcome will be decided. The same limit written about the three conditions in Chapter 1 applies here.

Is the division into two directions itself arbitrary (2.1 and 2.2)

The division into matter configuration and predictive models is neither exclusive nor exhaustive. Which is the production of energy itself? Which is the building of institutions? This thesis uses the division not as an ontological classification but as a working partition covering only what can be measured on a single scale, convergence cost. Energy production can be handled as a special case of matter configuration; institutions cannot. That is why institutions are erected as a separate term in Chapter 1. The two directions do not cover the whole of civilization, and no claim is made that they do.

The proviso that two layers suffice, against the rest of this thesis (2.1)

Section 2.1 collapses five layers into two. Yet Chapter 1 makes chokepoint technology T half of labor input, and the latter half of this thesis argues that holding a particular layer is decisive. Is it a contradiction to fold layers away and then call them decisive?

It is not a contradiction, but the risk in how it is written is acknowledged. Two layers is the resolution appropriate to asking where things are headed; five layers is the resolution appropriate to asking where things are. At the former, intermediate layers turn over; at the latter, the location of the intermediate layers is everything. The same object is being viewed at different resolutions according to the question.

Does this depend on Moravec’s paradox (2.6)

It does not, and that is set down here. The claim in the body is that the human brain is not an organ designed for computation, and the general understanding of evolutionary biology is sufficient for that. Moravec is cited only as corroboration. And the present scene cited as corroboration, that white-collar work is being eroded first, is equally well explained by the availability of training data. Which of the two is operating cannot be separated at present, and the argument of that section is written so as to survive Moravec being wrong.

The objection that “energy is only 6 percent of GDP”

American energy expenditure runs between 5.6 and 6.7 percent of nominal GDP (EIA, “U.S. energy spending increased by more than 20% in 2022,” *Today in Energy*, August 28, 2024). But a small share for an input that enters the production function of nearly every good, and whose short-run elasticity of substitution is extremely low, is a statement about the size of the damage under a supply shock. The GDP lost to a single day of widespread blackout runs to many tens of times that day’s energy expenditure.

The residual in growth accounting accounted for roughly nine tenths of growth in Solow's measurement (Robert M. Solow, "Technical Change and the Aggregate Production Function," *Review of Economics and Statistics* 39(3), 1957, 312–320). Ayres and Warr showed that part of it is improvement in thermodynamic efficiency (Robert U. Ayres & Benjamin Warr, *The Economic Growth Engine*, Edward Elgar, 2009), but the most rigorous replication gives roughly 25 percent of the increase in UK GDP from 1971 to 2013, and this thesis claims no more than that. The bulk of the residual has not disappeared. A quarter of it has been given a name.

Chapter 3: Objections and Limits

Degrowth as the opposing school (3.1)

The diagnosis in this chapter and those after it stands on the premise that the expansion of energy is desirable. There is a school that opposes that premise head-on: degrowth. This thesis does not take that position, but it does not ignore it either.

Van den Bergh criticized both growth maximalism and degrowth, and proposed a policy framework neutral with respect to growth, which he called a-growth (Jeroen C. J. M. van den Bergh, "Environment versus growth — A criticism of 'degrowth' and a plea for 'a-growth'," *Ecological Economics* 70(5), 2011, 881–890). The argument is that the criterion of judgment should be placed on the social and environmental objectives one wants to achieve, not on the growth rate itself.

The claim of this thesis answers that criticism in one particular form. What this thesis defends is not the growth rate but the breadth of the options a household can choose from. What increases when the effective price of energy falls is not, in the first instance, the quantity consumed. It is the number of options that can be made to work. Having a child, owning a home, starting a business. What degrowth criticizes and what this thesis is trying to restore are not the same object.

The relation between the three variables and the equation of Chapter 1 (3.6)

The three variables raised in 3.6, automation, energy cost, and the speed at which research and development gets implemented, are a different cut from the product of three terms in section 1.2 of Chapter 1. The former are observational indicators for diagnosis; the latter is a definition of the capacity to build. Automation corresponds to L and energy cost to E, but the speed of implementation corresponds directly to none of the three terms. This mismatch has not been resolved. The speed of implementation is best read as a secondary quantity that appears as a composite of institutions, capital and energy, but it has not yet been put into a form in which that can be written down.

On the arithmetic of the eight billion robots (3.6)

The calculation at the end of 3.6 is an order-of-magnitude sketch meant to show the scale of energy demand, not an estimate. The energy equivalent to the metabolism of one human being and the energy consumed by one piece of automated plant are not, in the first place, things that can be compared on the same scale. What the calculation asserts is one thing only: that the number of consumers being added bites at the scale of the population. It does not depend on the precision of the multiple.

Chapter 4: Objections and Limits

The edge of the framework

Two edges of the framework, and only two.

The Nordics. In the countries with the most generous family policy in the world, the TFR has collapsed since the 2010s¹⁴. Family policy transfers childcare and leave, but it does not reach housing, the largest term in C_child, or the expected value of the opportunity cost of a career. On top of that, the decision to have a child rests not on current prices but on expected

lifetime prices. A generation that grew up in an era of low growth discounts the sustainability of policy however generous that policy is. An expected price system is also a price. Even so, this case marks the edge of this thesis's explanatory power. Price explains the conditions under which fertility falls. It does not explain what makes it rise.

¹⁴ The Nordics. Julia Hellstrand, Jessica Nisén, Vitor Miranda, Peter Fallesen, Lars Dommermuth & Mikko Myrskylä, "Not Just Later, but Fewer: Novel Trends in Cohort Fertility in the Nordic Countries," *Demography* 58(4), 2021.

Israel. The TFR is around 2.9, and this is not a phenomenon confined to the ultra-Orthodox: the secular population holds at around 2¹⁵. And this in a high-price society. Religion, community and a national ethos can overwrite price signals. But so far this is the only country that has overwritten price.

¹⁵ Israel. Israel Central Bureau of Statistics, *Statistical Abstract of Israel 2025*, Table 2.28. The 2024 TFR for the whole population is 2.87 (Jews 3.07, Muslims 2.74, Christians 1.61). By religiosity: Haredi 6.38, religious 3.77, traditional 2.80, secular 1.98 (three-year centered moving average, 2020–2022). The secular figure is "around 2" but has lately fallen slightly below replacement, drifting down from the 2.1 to 2.3 of a decade ago. The religiosity categories are based on self-report in social surveys.

Korean fertility has recovered in recent years (4.3)

Korea's TFR bottomed at 0.72 in 2023 and has recovered for two consecutive years, to 0.75 in 2024 and 0.80 in 2025. Seoul has risen from 0.55 to 0.63 (birth statistics from Statistics Korea and the National Data Office). This can be read as an achievement of policy. This thesis does not read it that way because an explanation in terms of rebound and of people bringing decisions forward holds equally well, and not enough time has passed to separate the two. What this thesis uses as evidence is only the spending side; it does not use the reversal side. If that judgment turns out to be wrong, the central proposition of Chapter 4 is itself weakened. The row marked IV' in the falsification table in Chapter 9 corresponds to this.

The Chinese estimate depends on TFR, a quantity that is not published (4.4)

China's National Bureau of Statistics publishes births but does not publish a total fertility rate. The 0.93 this thesis uses is a private estimate; the *China Population and Employment Statistics Yearbook* gives 1.14 for 2024 and the United Nations estimate is about 1.01, so the sources diverge widely. What is robust in this thesis's estimate is the population base, the part that says every woman who gives birth in 2050 has been born by 2032 at the latest, not the part that depends on the level of the TFR. The absolute number of births should therefore be read with a range around it, and what this thesis is claiming is not the level but the shape.

On the residual the price account cannot explain (4.5)

The study by Kearney and colleagues treated in the body shows that economic variables explain only a small part of the fall in American fertility. This thesis's response is as written in the body, but it should also be said that the response is not complete. The objection that a shift in priorities is a thing to be explained is only finished once it shows what explains that shift. This thesis has not yet shown it. The first and the fourth of the four things Chapter 9 asserts without establishing correspond to this gap.

And there is a price to be paid for an argument that separates the regime from the marginal. The distinction protects this thesis's claim from data inside the regime, but at the same time it forfeits any support from data inside the regime. No record exists of a state with the constraint removed, so what is left is only the three cases of periods when the constraint loosened. Three cases are too few to establish a law. That is why this thesis writes this part as a conditional prediction rather than as a law.

Chapter 5: Objections and Limits

The rebound effect is not a general law (5.2)

Whether the size of the rebound effect across an entire economy, meaning a backfire above 100 percent, is a general law is still disputed in the literature. This thesis does not settle that dispute. What it rests on is not a general law but a single point: that in domains where suppressed demand is large, there is no end to the places a cost collapse can be absorbed into. In a world that contains the cities that could not be built, the children that could not be had and the science that was never implemented, all seen in Part II, a collapse in cost does not show up as a contraction in spending. Read with that limitation removed, as “efficiency always increases total consumption,” it is a stronger claim than this thesis makes.

The assumptions behind the valley-of-demand calculation (5.7)

The assumptions of the calculation. The World Bank’s 2025 population structure by five-year band and by sex was interpolated to single years of age, smoothed, and run through the cohort component method. Age-specific fertility was set as a skew-normal distribution with a mean of 29.0, a standard deviation of 5.2 and a skewness of 1.2, scaled by the total fertility rate. Mortality was fixed using single-year survival rates of the Gompertz-Makeham form, calibrated to match life expectancy in the region concerned. Migration was set to zero. This calculation is a model, not a forecast. Its purpose is to show that even under the most optimistic assumptions the valley of demand lasts half a century.

The strongest objection to this model is migration. Because migration substantially governs the actual path of the working-age population in high-income countries, the assumption of zero migration works in the direction of making the valley of demand look deeper. What lets the claim of this section survive nonetheless is that the claim concerns timing rather than level. The year of the floor does not move outside the range 2085 to 2090 even when the life expectancy assumption is shifted four years either way. And migration sums to zero for the world as a whole: to the extent it fills the high-income valley, it deepens the valley somewhere else.

What this thesis is not confident about on the timeline (5.6)

The reservations set out in 5.6 are every one of them about the speed of removal, not about the identification of the constraint. There is exactly one case in which that distinction breaks down: the case in which capital markets break first and AI investment bends badly. Removal then slips back by a decade or more, and the investment judgments in this thesis need more than added years of waiting. Of the ten priorities this thesis sets out, the judgment about returns that do not depend on head-count is the one most directly affected.

Chapter 6: Objections and Limits

If AI does not deliver explosive growth (6.3)

Daron Acemoglu’s cautious estimate puts AI’s contribution to total factor productivity over the coming decade at 0.66 percent at most, and under 0.53 percent on a more careful reckoning, with a contribution to GDP of about 1 percent ⁹. This is the strongest academic downgrade of the abundance argument of this thesis.

⁹ The estimate on the downgrading side. Daron Acemoglu, “The Simple Macroeconomics of AI,” NBER WP 32487, 2024 (peer-reviewed version *Economic Policy* 40(121), 2025, 13–58).

But the estimate does not weaken the distributive argument of this thesis. The reverse. The same paper simultaneously predicts a widening gap between capital income and labor income. If the pie does not expand quickly while the share going to capital rises, the relative position of people who earn their income from tasks gets worse rather than better. The distributive argument of this thesis therefore does not presuppose explosive growth driven by AI. The necessary condition is

not unlimited plenty but whether ownership can be widened faster than the returns to the means of production concentrate in a few hands. The more the cascade of Chapter 5 is delayed, the more urgent the design of Chapter 6 becomes.

The break between task and income is still only a correlation (6.2)

The Stanford work on the relative fall in youth employment is qualified by the authors themselves as correlation rather than causation³. The numbers move as the data is updated. Over the same period there is also the observation that use of AI itself leans slightly toward augmenting people⁴. This thesis treats the two as compatible because it has moved the question from “how many people has AI made redundant” to “do the entry-level jobs survive, who owns the output, and does the fruit of productivity accrue to wages or to capital.” Employment statistics cannot falsify this claim. What would falsify it is the fifth row of the falsification table in Chapter 9, the case in which the relative fall levels off.

The defeat of the wealth tax does not entail the dispersal of ownership (6.3 to 6.4)

The fact that taxing wealth country by country has not historically been sustained is not a proof that taxing wealth is wrong. It is no more than the fact that it was hard to sustain under the condition of mobility. Zucman’s proposal for international coordination is an attempt to change precisely that condition⁷. This thesis presses for the dispersal of ownership not because taxation fails, but because taxation and the dispersal of ownership solve different problems. Taxation settles how much is distributed; ownership settles who it reaches. Even if the first succeeds, the problem of who it reaches remains so long as the way in through tasks stays shut.

How to read the results of the cash-transfer experiments (6.3)

The OpenResearch and Finnish experiments are both about transfers of limited duration⁸. A transfer known to end in three years and capital income premised on lasting a lifetime do not mean the same thing to the behavior of the recipient. The conclusion of this thesis, that a cheque does not move the structure, therefore reaches one step beyond what the experiments showed. That step is, for now, an extrapolation by logic and not a demonstration.

Chapter 7: Objections and Limits

Four reservations about the argument that use-value becomes measurable (7.5 and 7.6)

First, service-dominant logic does not support this conclusion. The original work has a side to it that defends money. Money is a device for learning relative exchange-values quickly on a common scale, and without exchange an actor has little information on exchange-value at all, as chapter 4 of the book states. Whether direct measurement by data can replace that learning function depends on whether measurement is cheap and, just as much, on whether it is trusted.

Second, the current state of the technology is a long way off. Invasive brain-machine interfaces are at the clinical-trial stage, and what has reached practical use non-invasively is at the level of reading fragments of intention from muscle potentials. Smart contact lenses are at the prototype stage. And the quantification of subjective experience such as pleasure and satisfaction still depends, for now, on proxies of its own: dopamine response, electrodermal response.

Third, a problem of principle remains there. If we depend on proxies, we may simply have replaced one proxy, price, with another, biometric data. And what the fourth axiom says is that an external party cannot assess the beneficiary’s value. Brain activity measured from outside is precisely an assessment from outside. This argument holds only where measurement stops standing proxy and records the determination that happens inside the person’s own experience. Including the inverted qualia problem, the possibility that the same stimulus is felt differently by different people, this is a question that remains on the side of philosophy rather than engineering.

Fourth, what can be measured gets manufactured. An institution that pays for measured pleasure simultaneously creates the incentive to manufacture measured pleasure. Price was a coarse measure, but it was hard to fake at scale. There is no

guarantee that biometric data has the same property. Being that much easier to handle, it may be that much easier to manipulate. Just how measured pleasure would be manufactured, though, I cannot yet picture. If anyone can, I would like to hear it.

Where SDL ends and this thesis begins

This chapter uses SDL's axioms and vocabulary as a foundation. That the line is drawn by who performs an act rather than by what the act consists of (7.3), that the producer-consumer division has no explanatory power, that money is a claim on future competence: all of this is SDL's claim and not a contribution of this thesis.

There are three things SDL does not say. First, it does not support the retirement of money (the first reservation above). Second, it does not claim that the weight of the uncounted side will rise. SDL's claim is that the classification is wrong, not a forecast that the outside of the classification grows in size. The argument about weight in 7.1 and 7.4 belongs to this thesis. Third, SDL does not treat the lifting of the constraint, the collapse in energy prices, or universal capital ownership. Every step from the axioms to the conclusions of this thesis is this thesis's own. Blur that distinction and an established theory looks as though it were guaranteeing an unestablished claim.

The limits of GDP are not in themselves a new point (7.4)

The observation about the uncounted economy was formally set out by the Stiglitz-Sen-Fitoussi Commission in 2009 ¹¹, and the satellite accounts for household production have a series running back to 1965 ⁸. Go further back and the critique of the classification itself was written by Bastiat in 1848 ⁶. What is new here is not the limits of measurement themselves but the direction: that those limits may be lifted by technology. The first half is an introduction to existing knowledge, not a contribution of this thesis.

Chapter 8: Objections and Limits

Figures 8-2 and 8-3 are not measured values

Both are original schematics to this thesis. The thickness of the ribbons in Figure 8-2 is the author's ordinal judgment about how much of a term a place carries in the world, not a measured share. The levels in Figure 8-3 are likewise ordinal judgments. The two figures do not claim quantities. They claim an ordering. Where the ranking could change places, I have said so at each point in the body. That several shapes laid over one another fill the pentagon in Figure 8-3 carries one further assumption: that the best value for each term inside a group is actually available to the rest of it, which is to say that the chain holds. That assumption does not last forever.

The five terms carry no axis for cheap labor

As 8.4 says, labor input is still filled in part by labor cheap enough that neither deep tech nor large capital need be bought. This chapter does not make that a term because the input does not accumulate and the point at which it is replaced is in sight. Figures 8-2 and 8-3 therefore understate the part of the Gulf states' construction cost and China's two delta regions' manufacturing cost that this input carries. If the replacement runs slower than this chapter reads it, the present position of both is stronger than the figures show.

Can China get out of oversupply on demand from outside? (8.6)

The idea of absorbing the supply into Central Asia, Southeast Asia, Pakistan and sub-Saharan Africa has three ceilings. The scale does not match: the whole GDP of sub-Saharan Africa is about two trillion dollars, a little over a tenth of China's, and taking all of those markets still does not fill the hole in the domestic demand of 1.4 billion people. Demand is purchasing power, and purchasing power is a function of income; to sell into markets where the income has not yet arrived is to lend to

them, and that Belt and Road lending has shrunk from its 2016 peak with renegotiations running ever since is the cost of this strategy appearing on the collection side. If the buyer cannot pay, what you hold is not demand but a receivable. And as 8.9 shows, an inflow of cheap goods holds back the start of that country's own manufacturing: the strategy thins, in the act of selling, the middle class that was to become the future demand. The idea is not itself wrong; the time frames do not match.

This chapter does not treat geopolitics

The contest of power between states, alliances, deterrence and the succession of hegemony are not treated. The arrangement of clusters in this chapter can therefore be overwritten by changes in power. This thesis deliberately leaves blank a subject that requires an independent book of its own.

Whether coastal concentration is still increasing is disputed (8.2)

The paper by Kumm and colleagues cited in the body shows that the band within 100 kilometers of the coast and below 100 meters produces 42 percent of world GDP, and concludes at the same time that the common belief in a continuing accumulation of people and wealth along coasts is not supported by its results. The population share of that band is close to flat from 1900 to 2010. In the tighter bands of 10 to 50 kilometers, concentration is observed. What this thesis rests on is the level, not the trend.

On geographic determinism (8.2)

Acemoglu, Johnson and Robinson showed that among regions colonized by Europeans the places that were rich in 1500 are the poor ones now, and argued that geography cannot explain this reversal. They also write, however, that relative prosperity in 1500 may itself have been geographic. What this thesis attributes to geography is initial conditions and the cost of transport, not the quality of institutions. That distinction is why institutions stand as an independent term in the equation of Chapter 1.

Urban scaling laws depend on where the boundary is drawn (8.3)

Arcaute and colleagues showed that redrawing city boundaries thousands of ways for England and Wales returns most indicators to linearity and leaves the non-linear exponents unstable. What this thesis uses is the direction, that the physical quantity per head falls as density rises, and not the values of the exponents.

This is not a claim that restricting migration does nothing (8.3)

Walls and border enforcement do reduce inflows. The estimated effects are small and the costs are high. The claim here is not that restriction fails, but that what restriction changes is mainly the form of the flow rather than its direction.